# CLEVRER Video Annotations

This document describes short physics videos from the CLEVRER benchmark. Each video is 128 frames (frame 0 to frame 127, about 5 seconds) showing solid objects sliding on a flat floor and colliding.

Every video has its own chapter titled 'Video <number>'. Questions refer to a video by that number.

Each chapter has three sections. 'Objects' lists every object with its color, material (metal or rubber) and shape (cube, sphere or cylinder). 'Events' lists, in time order, when objects enter the scene, exit the scene and collide with each other. 'Motion log' gives sampled positions (x, y) and velocities (vx, vy) on the floor plane for objects that are visible in the video.

An object is stationary at a frame if its speed is below 0.1; otherwise it is moving. Only what is visible during the 128 frames is recorded; nothing after the video ends is described.

Videos in this document: Video 10146, Video 10170, Video 10193, Video 10892, Video 10937, Video 11544, Video 11716, Video 11851, Video 12082, Video 12123, Video 12564, Video 12821, Video 13177, Video 13265, Video 13461, Video 13549, Video 13894, Video 14082, Video 14195, Video 14981.

# Video 10146

## Video 10146: Objects

Object 0: purple metal cube (color=purple, material=metal, shape=cube)

Object 1: yellow metal cylinder (color=yellow, material=metal, shape=cylinder)

Object 2: cyan metal sphere (color=cyan, material=metal, shape=sphere)

Object 3: yellow metal sphere (color=yellow, material=metal, shape=sphere)

Object 4: purple rubber cylinder (color=purple, material=rubber, shape=cylinder)

Object 5: blue metal cylinder (color=blue, material=metal, shape=cylinder)

## Video 10146: Events

Frame 0: Object 2 (cyan metal sphere) is in view when the video begins.

Frame 0: Object 4 (purple rubber cylinder) is in view when the video begins.

Frame 0: Object 5 (blue metal cylinder) is in view when the video begins.

Frame 12: Object 1 (yellow metal cylinder) enters the scene.

Frame 14: Object 0 (purple metal cube) enters the scene.

Frame 14: Object 3 (yellow metal sphere) enters the scene.

Frame 47: Object 0 (purple metal cube) collides with Object 1 (yellow metal cylinder).

Frame 67: Object 0 (purple metal cube) collides with Object 2 (cyan metal sphere).

Frame 118: Object 1 (yellow metal cylinder) collides with Object 3 (yellow metal sphere).

Frame 127: The video ends.

## Video 10146: Motion log

| frame | object | x | y | vx | vy | speed |
|---|---|---|---|---|---|---|
| 0 | 2 cyan metal sphere | 0.21 | -1.17 | 0.00 | 0.00 | 0.00 |
| 0 | 4 purple rubber cylinder | -1.18 | -0.91 | 0.00 | 0.00 | 0.00 |
| 0 | 5 blue metal cylinder | 1.08 | 1.32 | 0.00 | 0.00 | 0.00 |
| 8 | 2 cyan metal sphere | 0.21 | -1.17 | -0.00 | -0.00 | 0.00 |
| 8 | 4 purple rubber cylinder | -1.18 | -0.91 | -0.00 | -0.00 | 0.00 |
| 8 | 5 blue metal cylinder | 1.08 | 1.32 | -0.00 | 0.00 | 0.00 |
| 12 | 1 yellow metal cylinder | 0.03 | 2.48 | 0.33 | -2.48 | 2.50 |
| 12 | 2 cyan metal sphere | 0.21 | -1.17 | -0.00 | -0.00 | 0.00 |
| 12 | 4 purple rubber cylinder | -1.18 | -0.91 | -0.00 | -0.00 | 0.00 |
| 12 | 5 blue metal cylinder | 1.08 | 1.32 | -0.00 | 0.00 | 0.00 |
| 14 | 0 purple metal cube | 1.28 | -2.04 | -1.02 | 2.35 | 2.56 |
| 14 | 1 yellow metal cylinder | 0.05 | 2.34 | 0.33 | -2.44 | 2.46 |
| 14 | 2 cyan metal sphere | 0.21 | -1.17 | 0.00 | -0.00 | 0.00 |
| 14 | 3 yellow metal sphere | -6.41 | 2.09 | 2.65 | -0.68 | 2.73 |
| 14 | 4 purple rubber cylinder | -1.18 | -0.91 | -0.00 | -0.00 | 0.00 |
| 14 | 5 blue metal cylinder | 1.08 | 1.32 | -0.00 | 0.00 | 0.00 |
| 16 | 0 purple metal cube | 1.22 | -1.91 | -1.01 | 2.32 | 2.53 |
| 16 | 1 yellow metal cylinder | 0.07 | 2.20 | 0.32 | -2.41 | 2.43 |
| 16 | 2 cyan metal sphere | 0.21 | -1.17 | 0.00 | -0.00 | 0.00 |
| 16 | 3 yellow metal sphere | -6.26 | 2.05 | 2.62 | -0.67 | 2.71 |
| 16 | 4 purple rubber cylinder | -1.18 | -0.91 | -0.00 | -0.00 | 0.00 |
| 16 | 5 blue metal cylinder | 1.08 | 1.32 | 0.00 | 0.00 | 0.00 |

| frame | object | x | y | vx | vy | speed |
|---|---|---|---|---|---|---|
| 24 | 0 purple metal cube | 1.00 | -1.41 | -0.95 | 2.19 | 2.38 |
| 24 | 1 yellow metal cylinder | 0.14 | 1.68 | 0.30 | -2.27 | 2.29 |
| 24 | 2 cyan metal sphere | 0.21 | -1.17 | 0.00 | -0.00 | 0.00 |
| 24 | 3 yellow metal sphere | -5.69 | 1.90 | 2.54 | -0.65 | 2.62 |
| 24 | 4 purple rubber cylinder | -1.18 | -0.91 | -0.00 | -0.00 | 0.00 |
| 24 | 5 blue metal cylinder | 1.08 | 1.32 | 0.00 | -0.00 | 0.00 |
| 32 | 0 purple metal cube | 0.80 | -0.93 | -0.89 | 2.06 | 2.25 |
| 32 | 1 yellow metal cylinder | 0.20 | 1.19 | 0.28 | -2.14 | 2.16 |
| 32 | 2 cyan metal sphere | 0.21 | -1.17 | 0.00 | -0.00 | 0.00 |
| 32 | 3 yellow metal sphere | -5.13 | 1.76 | 2.46 | -0.63 | 2.54 |
| 32 | 4 purple rubber cylinder | -1.18 | -0.91 | -0.00 | -0.00 | 0.00 |
| 32 | 5 blue metal cylinder | 1.08 | 1.32 | 0.00 | -0.00 | 0.00 |
| 40 | 0 purple metal cube | 0.60 | -0.48 | -0.84 | 1.95 | 2.12 |
| 40 | 1 yellow metal cylinder | 0.26 | 0.72 | 0.26 | -2.02 | 2.04 |
| 40 | 2 cyan metal sphere | 0.21 | -1.17 | 0.00 | -0.00 | 0.00 |
| 40 | 3 yellow metal sphere | -4.58 | 1.62 | 2.38 | -0.61 | 2.46 |
| 40 | 4 purple rubber cylinder | -1.18 | -0.91 | -0.00 | -0.00 | 0.00 |
| 40 | 5 blue metal cylinder | 1.08 | 1.32 | 0.00 | -0.00 | 0.00 |
| 47 | 0 purple metal cube | 0.44 | -0.11 | -0.80 | 1.85 | 2.02 |
| 47 | 1 yellow metal cylinder | 0.31 | 0.34 | 0.25 | -1.92 | 1.93 |
| 47 | 2 cyan metal sphere | 0.21 | -1.17 | 0.00 | -0.00 | 0.00 |
| 47 | 3 yellow metal sphere | -4.12 | 1.50 | 2.32 | -0.59 | 2.39 |
| 47 | 4 purple rubber cylinder | -1.18 | -0.91 | -0.00 | -0.00 | 0.00 |
| 47 | 5 blue metal cylinder | 1.08 | 1.32 | 0.00 | -0.00 | 0.00 |
| 48 | 0 purple metal cube | 0.43 | -0.14 | -0.46 | -1.12 | 1.21 |
| 48 | 1 yellow metal cylinder | 0.32 | 0.34 | 0.02 | 0.12 | 0.13 |
| 48 | 2 cyan metal sphere | 0.21 | -1.17 | 0.00 | -0.00 | 0.00 |
| 48 | 3 yellow metal sphere | -4.06 | 1.48 | 2.31 | -0.59 | 2.39 |
| 48 | 4 purple rubber cylinder | -1.18 | -0.91 | -0.00 | -0.00 | 0.00 |
| 48 | 5 blue metal cylinder | 1.08 | 1.32 | -0.00 | -0.00 | 0.00 |
| 56 | 0 purple metal cube | 0.33 | -0.38 | -0.44 | -1.08 | 1.17 |
| 56 | 1 yellow metal cylinder | 0.32 | 0.36 | 0.03 | 0.08 | 0.09 |
| 56 | 2 cyan metal sphere | 0.21 | -1.17 | 0.00 | -0.00 | 0.00 |
| 56 | 3 yellow metal sphere | -3.55 | 1.35 | 2.24 | -0.57 | 2.31 |
| 56 | 4 purple rubber cylinder | -1.18 | -0.91 | 0.00 | -0.00 | 0.00 |
| 56 | 5 blue metal cylinder | 1.08 | 1.32 | 0.00 | 0.00 | 0.00 |
| 64 | 0 purple metal cube | 0.23 | -0.62 | -0.41 | -1.01 | 1.09 |
| 64 | 1 yellow metal cylinder | 0.32 | 0.37 | 0.00 | 0.02 | 0.02 |
| 64 | 2 cyan metal sphere | 0.21 | -1.17 | 0.00 | -0.00 | 0.00 |
| 64 | 3 yellow metal sphere | -3.05 | 1.23 | 2.18 | -0.56 | 2.25 |
| 64 | 4 purple rubber cylinder | -1.18 | -0.91 | 0.00 | -0.00 | 0.00 |
| 64 | 5 blue metal cylinder | 1.08 | 1.32 | 0.00 | -0.00 | 0.00 |
| 67 | 0 purple metal cube | 0.20 | -0.70 | -0.40 | -0.97 | 1.05 |
| 67 | 1 yellow metal cylinder | 0.32 | 0.37 | -0.00 | -0.00 | 0.00 |
| 67 | 2 cyan metal sphere | 0.21 | -1.17 | -0.00 | -0.02 | 0.02 |
| 67 | 3 yellow metal sphere | -2.87 | 1.18 | 2.15 | -0.55 | 2.22 |
| 67 | 4 purple rubber cylinder | -1.18 | -0.91 | -0.00 | 0.00 | 0.00 |
| 67 | 5 blue metal cylinder | 1.08 | 1.32 | -0.00 | 0.00 | 0.00 |
| 72 | 0 purple metal cube | 0.14 | -0.78 | -0.37 | -0.54 | 0.65 |
| 72 | 1 yellow metal cylinder | 0.32 | 0.37 | 0.00 | -0.00 | 0.00 |
| 72 | 2 cyan metal sphere | 0.20 | -1.24 | -0.02 | -0.44 | 0.44 |
| 72 | 3 yellow metal sphere | -2.57 | 1.10 | 2.12 | -0.54 | 2.18 |
| 72 | 4 purple rubber cylinder | -1.18 | -0.91 | 0.00 | -0.00 | 0.00 |
| 72 | 5 blue metal cylinder | 1.08 | 1.32 | 0.00 | -0.00 | 0.00 |
| 80 | 0 purple metal cube | 0.07 | -0.89 | -0.33 | -0.48 | 0.59 |
| 80 | 1 yellow metal cylinder | 0.32 | 0.37 | 0.00 | -0.00 | 0.00 |
| 80 | 2 cyan metal sphere | 0.20 | -1.33 | -0.02 | -0.43 | 0.43 |

| frame | object | x | y | vx | vy | speed |
|---|---|---|---|---|---|---|
| 80 | 3 yellow metal sphere | -2.11 | 0.98 | 2.06 | -0.53 | 2.12 |
| 80 | 4 purple rubber cylinder | -1.18 | -0.91 | -0.00 | -0.00 | 0.00 |
| 80 | 5 blue metal cylinder | 1.08 | 1.32 | 0.00 | -0.00 | 0.00 |
| 88 | 0 purple metal cube | -0.00 | -1.00 | -0.29 | -0.43 | 0.52 |
| 88 | 1 yellow metal cylinder | 0.32 | 0.37 | 0.00 | -0.00 | 0.00 |
| 88 | 2 cyan metal sphere | 0.20 | -1.43 | -0.02 | -0.43 | 0.43 |
| 88 | 3 yellow metal sphere | -1.65 | 0.87 | 2.00 | -0.51 | 2.07 |
| 88 | 4 purple rubber cylinder | -1.18 | -0.91 | -0.00 | -0.00 | 0.00 |
| 88 | 5 blue metal cylinder | 1.08 | 1.32 | 0.00 | -0.00 | 0.00 |
| 96 | 0 purple metal cube | -0.06 | -1.09 | -0.25 | -0.37 | 0.45 |
| 96 | 1 yellow metal cylinder | 0.32 | 0.37 | 0.00 | -0.00 | 0.00 |
| 96 | 2 cyan metal sphere | 0.19 | -1.53 | -0.02 | -0.42 | 0.42 |
| 96 | 3 yellow metal sphere | -1.21 | 0.75 | 1.95 | -0.50 | 2.01 |
| 96 | 4 purple rubber cylinder | -1.18 | -0.91 | -0.00 | -0.00 | 0.00 |
| 96 | 5 blue metal cylinder | 1.08 | 1.32 | 0.00 | -0.00 | 0.00 |
| 104 | 0 purple metal cube | -0.11 | -1.16 | -0.21 | -0.32 | 0.38 |
| 104 | 1 yellow metal cylinder | 0.32 | 0.37 | 0.00 | -0.00 | 0.00 |
| 104 | 2 cyan metal sphere | 0.19 | -1.62 | -0.02 | -0.42 | 0.42 |
| 104 | 3 yellow metal sphere | -0.78 | 0.64 | 1.90 | -0.49 | 1.96 |
| 104 | 4 purple rubber cylinder | -1.18 | -0.91 | -0.00 | -0.00 | 0.00 |
| 104 | 5 blue metal cylinder | 1.08 | 1.32 | 0.00 | -0.00 | 0.00 |
| 112 | 0 purple metal cube | -0.16 | -1.23 | -0.17 | -0.26 | 0.31 |
| 112 | 1 yellow metal cylinder | 0.32 | 0.37 | 0.00 | -0.00 | 0.00 |
| 112 | 2 cyan metal sphere | 0.19 | -1.71 | -0.02 | -0.41 | 0.41 |
| 112 | 3 yellow metal sphere | -0.36 | 0.54 | 1.85 | -0.47 | 1.91 |
| 112 | 4 purple rubber cylinder | -1.18 | -0.91 | -0.00 | -0.00 | 0.00 |
| 112 | 5 blue metal cylinder | 1.08 | 1.32 | 0.00 | -0.00 | 0.00 |
| 118 | 0 purple metal cube | -0.18 | -1.27 | -0.14 | -0.22 | 0.26 |
| 118 | 1 yellow metal cylinder | 0.33 | 0.37 | 0.97 | -0.20 | 0.99 |
| 118 | 2 cyan metal sphere | 0.18 | -1.78 | -0.02 | -0.41 | 0.41 |
| 118 | 3 yellow metal sphere | -0.07 | 0.46 | 0.05 | -0.10 | 0.11 |
| 118 | 4 purple rubber cylinder | -1.18 | -0.91 | 0.00 | -0.00 | 0.00 |
| 118 | 5 blue metal cylinder | 1.08 | 1.32 | 0.00 | -0.00 | 0.00 |
| 120 | 0 purple metal cube | -0.19 | -1.28 | -0.13 | -0.20 | 0.24 |
| 120 | 1 yellow metal cylinder | 0.39 | 0.36 | 0.95 | -0.20 | 0.97 |
| 120 | 2 cyan metal sphere | 0.18 | -1.80 | -0.02 | -0.41 | 0.41 |
| 120 | 3 yellow metal sphere | -0.07 | 0.45 | 0.05 | -0.10 | 0.11 |
| 120 | 4 purple rubber cylinder | -1.18 | -0.91 | 0.00 | -0.00 | 0.00 |
| 120 | 5 blue metal cylinder | 1.08 | 1.32 | 0.00 | -0.00 | 0.00 |
| 127 | 0 purple metal cube | -0.21 | -1.31 | -0.10 | -0.16 | 0.18 |
| 127 | 1 yellow metal cylinder | 0.56 | 0.32 | 0.88 | -0.18 | 0.89 |
| 127 | 2 cyan metal sphere | 0.18 | -1.88 | -0.02 | -0.40 | 0.40 |
| 127 | 3 yellow metal sphere | -0.06 | 0.43 | 0.05 | -0.10 | 0.11 |
| 127 | 4 purple rubber cylinder | -1.18 | -0.91 | -0.00 | -0.00 | 0.00 |
| 127 | 5 blue metal cylinder | 1.08 | 1.32 | -0.00 | 0.00 | 0.00 |

# Video 10170

## Video 10170: Objects

Object 0: cyan metal cylinder (color=cyan, material=metal, shape=cylinder)

Object 1: green rubber cube (color=green, material=rubber, shape=cube)

Object 2: brown metal cylinder (color=brown, material=metal, shape=cylinder)

Object 3: blue metal sphere (color=blue, material=metal, shape=sphere)

## Video 10170: Events

Frame 0: Object 1 (green rubber cube) is in view when the video begins.

Frame 0: Object 3 (blue metal sphere) is in view when the video begins.

Frame 18: Object 0 (cyan metal cylinder) enters the scene.

Frame 42: Object 2 (brown metal cylinder) enters the scene.

Frame 59: Object 0 (cyan metal cylinder) collides with Object 1 (green rubber cube).

Frame 127: The video ends.

## Video 10170: Motion log

| frame | object | x | y | vx | vy | speed |
|---|---|---|---|---|---|---|
| 0 | 1 green rubber cube | -0.35 | 0.14 | 0.00 | 0.00 | 0.00 |
| 0 | 3 blue metal sphere | 1.12 | 0.87 | 0.00 | 0.00 | 0.00 |
| 8 | 1 green rubber cube | -0.35 | 0.14 | 0.00 | -0.00 | 0.00 |
| 8 | 3 blue metal sphere | 1.12 | 0.87 | 0.00 | -0.00 | 0.00 |
| 16 | 1 green rubber cube | -0.35 | 0.14 | 0.00 | -0.00 | 0.00 |
| 16 | 3 blue metal sphere | 1.12 | 0.87 | 0.00 | -0.00 | 0.00 |
| 18 | 0 cyan metal cylinder | 2.53 | -0.37 | -2.43 | 0.51 | 2.48 |
| 18 | 1 green rubber cube | -0.35 | 0.14 | 0.00 | -0.00 | 0.00 |
| 18 | 3 blue metal sphere | 1.12 | 0.87 | 0.00 | -0.00 | 0.00 |
| 24 | 0 cyan metal cylinder | 2.13 | -0.29 | -2.33 | 0.49 | 2.38 |
| 24 | 1 green rubber cube | -0.35 | 0.14 | 0.00 | -0.00 | 0.00 |
| 24 | 3 blue metal sphere | 1.12 | 0.87 | 0.00 | -0.00 | 0.00 |
| 32 | 0 cyan metal cylinder | 1.62 | -0.18 | -2.20 | 0.46 | 2.24 |
| 32 | 1 green rubber cube | -0.35 | 0.14 | -0.00 | 0.00 | 0.00 |
| 32 | 3 blue metal sphere | 1.12 | 0.87 | 0.00 | -0.00 | 0.00 |
| 40 | 0 cyan metal cylinder | 1.15 | -0.08 | -2.07 | 0.43 | 2.12 |
| 40 | 1 green rubber cube | -0.35 | 0.14 | 0.00 | -0.00 | 0.00 |
| 40 | 3 blue metal sphere | 1.12 | 0.87 | 0.00 | -0.00 | 0.00 |
| 42 | 0 cyan metal cylinder | 1.03 | -0.06 | -2.04 | 0.43 | 2.08 |
| 42 | 1 green rubber cube | -0.35 | 0.14 | 0.00 | -0.00 | 0.00 |
| 42 | 2 brown metal cylinder | -6.64 | 0.60 | 2.11 | -0.01 | 2.11 |
| 42 | 3 blue metal sphere | 1.12 | 0.87 | 0.00 | -0.00 | 0.00 |
| 48 | 0 cyan metal cylinder | 0.70 | 0.01 | -1.95 | 0.41 | 1.99 |
| 48 | 1 green rubber cube | -0.35 | 0.14 | 0.00 | -0.00 | 0.00 |
| 48 | 2 brown metal cylinder | -6.30 | 0.60 | 2.02 | -0.01 | 2.02 |
| 48 | 3 blue metal sphere | 1.12 | 0.87 | 0.00 | -0.00 | 0.00 |
| 56 | 0 cyan metal cylinder | 0.27 | 0.10 | -1.84 | 0.38 | 1.88 |
| 56 | 1 green rubber cube | -0.35 | 0.14 | -0.00 | 0.00 | 0.00 |
| 56 | 2 brown metal cylinder | -5.86 | 0.60 | 1.90 | -0.01 | 1.90 |
| 56 | 3 blue metal sphere | 1.12 | 0.87 | 0.00 | -0.00 | 0.00 |
| 59 | 0 cyan metal cylinder | 0.12 | 0.13 | -1.80 | 0.37 | 1.83 |

| frame | object | x | y | vx | vy | speed |
|---|---|---|---|---|---|---|
| 59 | 1 green rubber cube | -0.35 | 0.14 | -0.00 | 0.00 | 0.00 |
| 59 | 2 brown metal cylinder | -5.70 | 0.59 | 1.86 | -0.01 | 1.86 |
| 59 | 3 blue metal sphere | 1.12 | 0.87 | 0.00 | -0.00 | 0.00 |
| 64 | 0 cyan metal cylinder | -0.02 | 0.16 | -0.92 | 0.17 | 0.93 |
| 64 | 1 green rubber cube | -0.50 | 0.17 | -1.09 | 0.26 | 1.12 |
| 64 | 2 brown metal cylinder | -5.44 | 0.59 | 1.79 | -0.01 | 1.79 |
| 64 | 3 blue metal sphere | 1.12 | 0.87 | 0.00 | -0.00 | 0.00 |
| 72 | 0 cyan metal cylinder | -0.21 | 0.19 | -0.84 | 0.15 | 0.85 |
| 72 | 1 green rubber cube | -0.73 | 0.23 | -1.01 | 0.24 | 1.04 |
| 72 | 2 brown metal cylinder | -5.06 | 0.59 | 1.68 | -0.01 | 1.68 |
| 72 | 3 blue metal sphere | 1.12 | 0.87 | 0.00 | -0.00 | 0.00 |
| 80 | 0 cyan metal cylinder | -0.39 | 0.23 | -0.76 | 0.14 | 0.77 |
| 80 | 1 green rubber cube | -0.95 | 0.28 | -0.93 | 0.22 | 0.95 |
| 80 | 2 brown metal cylinder | -4.69 | 0.59 | 1.57 | -0.01 | 1.57 |
| 80 | 3 blue metal sphere | 1.12 | 0.87 | 0.00 | -0.00 | 0.00 |
| 88 | 0 cyan metal cylinder | -0.55 | 0.26 | -0.68 | 0.12 | 0.69 |
| 88 | 1 green rubber cube | -1.15 | 0.33 | -0.85 | 0.20 | 0.87 |
| 88 | 2 brown metal cylinder | -4.35 | 0.59 | 1.47 | -0.01 | 1.47 |
| 88 | 3 blue metal sphere | 1.12 | 0.87 | 0.00 | -0.00 | 0.00 |
| 96 | 0 cyan metal cylinder | -0.70 | 0.28 | -0.61 | 0.11 | 0.62 |
| 96 | 1 green rubber cube | -1.33 | 0.37 | -0.77 | 0.18 | 0.79 |
| 96 | 2 brown metal cylinder | -4.03 | 0.58 | 1.38 | -0.01 | 1.38 |
| 96 | 3 blue metal sphere | 1.12 | 0.87 | 0.00 | -0.00 | 0.00 |
| 104 | 0 cyan metal cylinder | -0.83 | 0.30 | -0.54 | 0.10 | 0.55 |
| 104 | 1 green rubber cube | -1.49 | 0.41 | -0.70 | 0.16 | 0.72 |
| 104 | 2 brown metal cylinder | -3.74 | 0.58 | 1.28 | -0.01 | 1.28 |
| 104 | 3 blue metal sphere | 1.12 | 0.87 | 0.00 | -0.00 | 0.00 |
| 112 | 0 cyan metal cylinder | -0.94 | 0.32 | -0.47 | 0.08 | 0.47 |
| 112 | 1 green rubber cube | -1.64 | 0.45 | -0.62 | 0.15 | 0.64 |
| 112 | 2 brown metal cylinder | -3.46 | 0.58 | 1.19 | -0.01 | 1.19 |
| 112 | 3 blue metal sphere | 1.12 | 0.87 | 0.00 | -0.00 | 0.00 |
| 120 | 0 cyan metal cylinder | -1.03 | 0.34 | -0.40 | 0.07 | 0.40 |
| 120 | 1 green rubber cube | -1.77 | 0.48 | -0.55 | 0.13 | 0.57 |
| 120 | 2 brown metal cylinder | -3.20 | 0.58 | 1.10 | -0.01 | 1.10 |
| 120 | 3 blue metal sphere | 1.12 | 0.87 | 0.00 | -0.00 | 0.00 |
| 127 | 0 cyan metal cylinder | -1.11 | 0.35 | -0.34 | 0.06 | 0.34 |
| 127 | 1 green rubber cube | -1.88 | 0.50 | -0.49 | 0.11 | 0.50 |
| 127 | 2 brown metal cylinder | -2.99 | 0.58 | 1.03 | -0.01 | 1.03 |
| 127 | 3 blue metal sphere | 1.12 | 0.87 | 0.00 | -0.00 | 0.00 |

# Video 10193

## Video 10193: Objects

Object 0: brown rubber cube (color=brown, material=rubber, shape=cube)

Object 1: blue rubber cylinder (color=blue, material=rubber, shape=cylinder)

Object 2: yellow metal cylinder (color=yellow, material=metal, shape=cylinder)

Object 3: yellow rubber cube (color=yellow, material=rubber, shape=cube)

Object 4: yellow metal cube (color=yellow, material=metal, shape=cube)

Object 5: green rubber sphere (color=green, material=rubber, shape=sphere)

## Video 10193: Events

Frame 0: Object 1 (blue rubber cylinder) is in view when the video begins.

Frame 0: Object 4 (yellow metal cube) is in view when the video begins.

Frame 0: Object 5 (green rubber sphere) is in view when the video begins.

Frame 17: Object 0 (brown rubber cube) enters the scene.

Frame 17: Object 2 (yellow metal cylinder) enters the scene.

Frame 33: Object 3 (yellow rubber cube) enters the scene.

Frame 40: Object 0 (brown rubber cube) collides with Object 1 (blue rubber cylinder).

Frame 111: Object 0 (brown rubber cube) collides with Object 2 (yellow metal cylinder).

Frame 127: The video ends.

## Video 10193: Motion log

| frame | object | x | y | vx | vy | speed |
|---|---|---|---|---|---|---|
| 0 | 1 blue rubber cylinder | -0.59 | -1.33 | 0.00 | 0.00 | 0.00 |
| 0 | 4 yellow metal cube | 0.42 | 0.84 | 0.00 | 0.00 | 0.00 |
| 0 | 5 green rubber sphere | 1.00 | -0.30 | 0.00 | 0.00 | 0.00 |
| 8 | 1 blue rubber cylinder | -0.59 | -1.33 | 0.00 | 0.00 | 0.00 |
| 8 | 4 yellow metal cube | 0.42 | 0.84 | -0.00 | 0.00 | 0.00 |
| 8 | 5 green rubber sphere | 1.00 | -0.30 | 0.00 | -0.00 | 0.00 |
| 16 | 1 blue rubber cylinder | -0.59 | -1.33 | 0.00 | -0.00 | 0.00 |
| 16 | 4 yellow metal cube | 0.42 | 0.84 | 0.00 | 0.00 | 0.00 |
| 16 | 5 green rubber sphere | 1.00 | -0.30 | 0.00 | -0.00 | 0.00 |
| 17 | 0 brown rubber cube | 1.18 | -2.07 | -2.27 | 1.30 | 2.61 |
| 17 | 1 blue rubber cylinder | -0.59 | -1.33 | 0.00 | -0.00 | 0.00 |
| 17 | 2 yellow metal cylinder | -6.60 | -1.47 | 2.33 | 0.75 | 2.45 |
| 17 | 4 yellow metal cube | 0.42 | 0.84 | 0.00 | 0.00 | 0.00 |
| 17 | 5 green rubber sphere | 1.00 | -0.30 | 0.00 | -0.00 | 0.00 |
| 24 | 0 brown rubber cube | 0.74 | -1.82 | -2.15 | 1.23 | 2.48 |
| 24 | 1 blue rubber cylinder | -0.59 | -1.33 | -0.00 | -0.00 | 0.00 |
| 24 | 2 yellow metal cylinder | -6.15 | -1.33 | 2.22 | 0.71 | 2.33 |
| 24 | 4 yellow metal cube | 0.42 | 0.84 | -0.00 | 0.00 | 0.00 |
| 24 | 5 green rubber sphere | 1.00 | -0.30 | 0.00 | -0.00 | 0.00 |
| 32 | 0 brown rubber cube | 0.27 | -1.55 | -2.03 | 1.16 | 2.34 |
| 32 | 1 blue rubber cylinder | -0.59 | -1.33 | -0.00 | -0.00 | 0.00 |
| 32 | 2 yellow metal cylinder | -5.67 | -1.17 | 2.09 | 0.67 | 2.20 |
| 32 | 4 yellow metal cube | 0.42 | 0.84 | -0.00 | 0.00 | 0.00 |

| frame | object | x | y | vx | vy | speed |
|---|---|---|---|---|---|---|
| 32 | 5 green rubber sphere | 1.00 | -0.30 | 0.00 | -0.00 | 0.00 |
| 33 | 0 brown rubber cube | 0.22 | -1.52 | -2.02 | 1.15 | 2.33 |
| 33 | 1 blue rubber cylinder | -0.59 | -1.33 | -0.00 | -0.00 | 0.00 |
| 33 | 2 yellow metal cylinder | -5.61 | -1.15 | 2.08 | 0.67 | 2.18 |
| 33 | 3 yellow rubber cube | -6.65 | -1.90 | 2.09 | 0.89 | 2.28 |
| 33 | 4 yellow metal cube | 0.42 | 0.84 | -0.00 | 0.00 | 0.00 |
| 33 | 5 green rubber sphere | 1.00 | -0.30 | 0.00 | -0.00 | 0.00 |
| 40 | 0 brown rubber cube | -0.16 | -1.31 | -1.14 | 0.88 | 1.44 |
| 40 | 1 blue rubber cylinder | -0.60 | -1.33 | -0.98 | 0.27 | 1.02 |
| 40 | 2 yellow metal cylinder | -5.21 | -1.03 | 1.97 | 0.63 | 2.07 |
| 40 | 3 yellow rubber cube | -6.25 | -1.73 | 1.99 | 0.84 | 2.16 |
| 40 | 4 yellow metal cube | 0.42 | 0.84 | -0.00 | 0.00 | 0.00 |
| 40 | 5 green rubber sphere | 1.00 | -0.30 | 0.00 | -0.00 | 0.00 |
| 48 | 0 brown rubber cube | -0.40 | -1.11 | -1.07 | 0.83 | 1.35 |
| 48 | 1 blue rubber cylinder | -0.81 | -1.27 | -0.91 | 0.25 | 0.94 |
| 48 | 2 yellow metal cylinder | -4.78 | -0.89 | 1.86 | 0.60 | 1.95 |
| 48 | 3 yellow rubber cube | -5.82 | -1.55 | 1.88 | 0.79 | 2.04 |
| 48 | 4 yellow metal cube | 0.42 | 0.84 | -0.00 | 0.00 | 0.00 |
| 48 | 5 green rubber sphere | 1.00 | -0.30 | 0.00 | -0.00 | 0.00 |
| 56 | 0 brown rubber cube | -0.63 | -0.94 | -0.99 | 0.77 | 1.26 |
| 56 | 1 blue rubber cylinder | -1.01 | -1.22 | -0.83 | 0.23 | 0.86 |
| 56 | 2 yellow metal cylinder | -4.38 | -0.76 | 1.75 | 0.56 | 1.84 |
| 56 | 3 yellow rubber cube | -5.41 | -1.38 | 1.77 | 0.75 | 1.92 |
| 56 | 4 yellow metal cube | 0.42 | 0.84 | -0.00 | 0.00 | 0.00 |
| 56 | 5 green rubber sphere | 1.00 | -0.30 | 0.00 | -0.00 | 0.00 |
| 64 | 0 brown rubber cube | -0.85 | -0.77 | -0.93 | 0.72 | 1.17 |
| 64 | 1 blue rubber cylinder | -1.19 | -1.17 | -0.75 | 0.21 | 0.78 |
| 64 | 2 yellow metal cylinder | -4.00 | -0.64 | 1.65 | 0.52 | 1.73 |
| 64 | 3 yellow rubber cube | -5.03 | -1.21 | 1.67 | 0.70 | 1.81 |
| 64 | 4 yellow metal cube | 0.42 | 0.84 | -0.00 | 0.00 | 0.00 |
| 64 | 5 green rubber sphere | 1.00 | -0.30 | 0.00 | -0.00 | 0.00 |
| 72 | 0 brown rubber cube | -1.05 | -0.61 | -0.86 | 0.66 | 1.09 |
| 72 | 1 blue rubber cylinder | -1.35 | -1.13 | -0.68 | 0.18 | 0.70 |
| 72 | 2 yellow metal cylinder | -3.64 | -0.53 | 1.54 | 0.49 | 1.62 |
| 72 | 3 yellow rubber cube | -4.66 | -1.06 | 1.57 | 0.66 | 1.70 |
| 72 | 4 yellow metal cube | 0.42 | 0.84 | -0.00 | 0.00 | 0.00 |
| 72 | 5 green rubber sphere | 1.00 | -0.30 | 0.00 | -0.00 | 0.00 |
| 80 | 0 brown rubber cube | -1.23 | -0.47 | -0.79 | 0.61 | 1.00 |
| 80 | 1 blue rubber cylinder | -1.49 | -1.09 | -0.61 | 0.16 | 0.63 |
| 80 | 2 yellow metal cylinder | -3.31 | -0.42 | 1.45 | 0.46 | 1.52 |
| 80 | 3 yellow rubber cube | -4.32 | -0.92 | 1.47 | 0.61 | 1.60 |
| 80 | 4 yellow metal cube | 0.42 | 0.84 | -0.00 | 0.00 | 0.00 |
| 80 | 5 green rubber sphere | 1.00 | -0.30 | 0.00 | -0.00 | 0.00 |
| 88 | 0 brown rubber cube | -1.40 | -0.34 | -0.73 | 0.56 | 0.92 |
| 88 | 1 blue rubber cylinder | -1.62 | -1.05 | -0.54 | 0.14 | 0.55 |
| 88 | 2 yellow metal cylinder | -2.99 | -0.32 | 1.35 | 0.43 | 1.42 |
| 88 | 3 yellow rubber cube | -4.00 | -0.79 | 1.38 | 0.57 | 1.50 |
| 88 | 4 yellow metal cube | 0.42 | 0.84 | -0.00 | 0.00 | 0.00 |
| 88 | 5 green rubber sphere | 1.00 | -0.30 | 0.00 | -0.00 | 0.00 |
| 96 | 0 brown rubber cube | -1.56 | -0.22 | -0.66 | 0.51 | 0.84 |
| 96 | 1 blue rubber cylinder | -1.73 | -1.02 | -0.47 | 0.12 | 0.48 |
| 96 | 2 yellow metal cylinder | -2.70 | -0.23 | 1.26 | 0.40 | 1.33 |
| 96 | 3 yellow rubber cube | -3.70 | -0.66 | 1.29 | 0.53 | 1.40 |
| 96 | 4 yellow metal cube | 0.42 | 0.84 | -0.00 | 0.00 | 0.00 |
| 96 | 5 green rubber sphere | 1.00 | -0.30 | 0.00 | -0.00 | 0.00 |
| 104 | 0 brown rubber cube | -1.70 | -0.11 | -0.60 | 0.47 | 0.76 |
| 104 | 1 blue rubber cylinder | -1.83 | -1.00 | -0.40 | 0.11 | 0.41 |

| frame | object | x | y | vx | vy | speed |
|---|---|---|---|---|---|---|
| 104 | 2 yellow metal cylinder | -2.43 | -0.14 | 1.18 | 0.37 | 1.23 |
| 104 | 3 yellow rubber cube | -3.42 | -0.55 | 1.21 | 0.50 | 1.31 |
| 104 | 4 yellow metal cube | 0.42 | 0.84 | 0.00 | 0.00 | 0.00 |
| 104 | 5 green rubber sphere | 1.00 | -0.30 | 0.00 | -0.00 | 0.00 |
| 111 | 0 brown rubber cube | -1.81 | -0.02 | 0.59 | 0.35 | 0.69 |
| 111 | 1 blue rubber cylinder | -1.90 | -0.98 | -0.34 | 0.09 | 0.35 |
| 111 | 2 yellow metal cylinder | -2.21 | -0.07 | 0.08 | 0.39 | 0.40 |
| 111 | 3 yellow rubber cube | -3.19 | -0.46 | 1.14 | 0.46 | 1.23 |
| 111 | 4 yellow metal cube | 0.42 | 0.84 | -0.00 | 0.00 | 0.00 |
| 111 | 5 green rubber sphere | 1.00 | -0.30 | 0.00 | -0.00 | 0.00 |
| 112 | 0 brown rubber cube | -1.79 | -0.01 | 0.59 | 0.35 | 0.69 |
| 112 | 1 blue rubber cylinder | -1.91 | -0.98 | -0.33 | 0.09 | 0.34 |
| 112 | 2 yellow metal cylinder | -2.21 | -0.06 | 0.08 | 0.39 | 0.40 |
| 112 | 3 yellow rubber cube | -3.16 | -0.44 | 1.13 | 0.46 | 1.22 |
| 112 | 4 yellow metal cube | 0.42 | 0.84 | -0.00 | 0.00 | 0.00 |
| 112 | 5 green rubber sphere | 1.00 | -0.30 | 0.00 | -0.00 | 0.00 |
| 120 | 0 brown rubber cube | -1.66 | 0.06 | 0.53 | 0.33 | 0.63 |
| 120 | 1 blue rubber cylinder | -1.97 | -0.96 | -0.26 | 0.07 | 0.27 |
| 120 | 2 yellow metal cylinder | -2.19 | 0.02 | 0.07 | 0.32 | 0.33 |
| 120 | 3 yellow rubber cube | -2.92 | -0.34 | 1.05 | 0.42 | 1.13 |
| 120 | 4 yellow metal cube | 0.42 | 0.84 | -0.00 | -0.00 | 0.00 |
| 120 | 5 green rubber sphere | 1.00 | -0.30 | 0.00 | -0.00 | 0.00 |
| 127 | 0 brown rubber cube | -1.56 | 0.12 | 0.48 | 0.29 | 0.56 |
| 127 | 1 blue rubber cylinder | -2.02 | -0.95 | -0.21 | 0.05 | 0.21 |
| 127 | 2 yellow metal cylinder | -2.18 | 0.08 | 0.05 | 0.27 | 0.27 |
| 127 | 3 yellow rubber cube | -2.72 | -0.26 | 0.98 | 0.40 | 1.05 |
| 127 | 4 yellow metal cube | 0.42 | 0.84 | -0.00 | -0.00 | 0.00 |
| 127 | 5 green rubber sphere | 1.00 | -0.30 | 0.00 | -0.00 | 0.00 |

# Video 10892

## Video 10892: Objects

Object 0: brown metal cube (color=brown, material=metal, shape=cube)

Object 1: yellow rubber sphere (color=yellow, material=rubber, shape=sphere)

Object 2: brown rubber cylinder (color=brown, material=rubber, shape=cylinder)

Object 3: red rubber cube (color=red, material=rubber, shape=cube)

Object 4: red rubber sphere (color=red, material=rubber, shape=sphere)

Object 5: purple metal cube (color=purple, material=metal, shape=cube)

## Video 10892: Events

Frame 0: Object 1 (yellow rubber sphere) is in view when the video begins.

Frame 0: Object 4 (red rubber sphere) is in view when the video begins.

Frame 0: Object 5 (purple metal cube) is in view when the video begins.

Frame 7: Object 3 (red rubber cube) enters the scene.

Frame 8: Object 0 (brown metal cube) enters the scene.

Frame 41: Object 0 (brown metal cube) collides with Object 1 (yellow rubber sphere).

Frame 61: Object 2 (brown rubber cylinder) enters the scene.

Frame 75: Object 0 (brown metal cube) collides with Object 2 (brown rubber cylinder).

Frame 110: Object 1 (yellow rubber sphere) collides with Object 3 (red rubber cube).

Frame 127: The video ends.

## Video 10892: Motion log

| frame | object | x | y | vx | vy | speed |
|---|---|---|---|---|---|---|
| 0 | 1 yellow rubber sphere | 0.01 | 0.91 | 0.00 | 0.00 | 0.00 |
| 0 | 4 red rubber sphere | 0.39 | -0.85 | 0.00 | 0.00 | 0.00 |
| 0 | 5 purple metal cube | -2.01 | -0.83 | 0.00 | 0.00 | 0.00 |
| 7 | 1 yellow rubber sphere | 0.01 | 0.91 | -0.00 | -0.00 | 0.00 |
| 7 | 3 red rubber cube | -6.65 | 4.88 | 2.19 | -1.75 | 2.80 |
| 7 | 4 red rubber sphere | 0.39 | -0.85 | 0.00 | -0.00 | 0.00 |
| 7 | 5 purple metal cube | -2.01 | -0.83 | -0.00 | 0.00 | 0.00 |
| 8 | 0 brown metal cube | 2.49 | 0.01 | -2.53 | 0.94 | 2.70 |
| 8 | 1 yellow rubber sphere | 0.01 | 0.91 | -0.00 | -0.00 | 0.00 |
| 8 | 3 red rubber cube | -6.59 | 4.83 | 2.17 | -1.73 | 2.78 |
| 8 | 4 red rubber sphere | 0.39 | -0.85 | 0.00 | -0.00 | 0.00 |
| 8 | 5 purple metal cube | -2.01 | -0.83 | -0.00 | 0.00 | 0.00 |
| 16 | 0 brown metal cube | 1.95 | 0.22 | -2.39 | 0.89 | 2.55 |
| 16 | 1 yellow rubber sphere | 0.01 | 0.91 | -0.00 | -0.00 | 0.00 |
| 16 | 3 red rubber cube | -6.12 | 4.45 | 2.05 | -1.64 | 2.62 |
| 16 | 4 red rubber sphere | 0.39 | -0.85 | 0.00 | -0.00 | 0.00 |
| 16 | 5 purple metal cube | -2.01 | -0.83 | -0.00 | 0.00 | 0.00 |
| 24 | 0 brown metal cube | 1.43 | 0.41 | -2.25 | 0.84 | 2.40 |
| 24 | 1 yellow rubber sphere | 0.01 | 0.91 | -0.00 | -0.00 | 0.00 |
| 24 | 3 red rubber cube | -5.67 | 4.10 | 1.94 | -1.55 | 2.48 |
| 24 | 4 red rubber sphere | 0.39 | -0.85 | 0.00 | -0.00 | 0.00 |
| 24 | 5 purple metal cube | -2.01 | -0.83 | -0.00 | 0.00 | 0.00 |

| frame | object | x | y | vx | vy | speed |
|---|---|---|---|---|---|---|
| 32 | 0 brown metal cube | 0.94 | 0.59 | -2.12 | 0.79 | 2.27 |
| 32 | 1 yellow rubber sphere | 0.01 | 0.91 | -0.00 | -0.00 | 0.00 |
| 32 | 3 red rubber cube | -5.25 | 3.76 | 1.83 | -1.46 | 2.34 |
| 32 | 4 red rubber sphere | 0.39 | -0.85 | 0.00 | -0.00 | 0.00 |
| 32 | 5 purple metal cube | -2.01 | -0.83 | -0.00 | 0.00 | 0.00 |
| 40 | 0 brown metal cube | 0.47 | 0.77 | -2.00 | 0.75 | 2.14 |
| 40 | 1 yellow rubber sphere | 0.01 | 0.91 | -0.00 | -0.00 | 0.00 |
| 40 | 3 red rubber cube | -4.85 | 3.44 | 1.73 | -1.38 | 2.21 |
| 40 | 4 red rubber sphere | 0.39 | -0.85 | 0.00 | -0.00 | 0.00 |
| 40 | 5 purple metal cube | -2.01 | -0.83 | -0.00 | 0.00 | 0.00 |
| 41 | 0 brown metal cube | 0.42 | 0.79 | -1.99 | 0.74 | 2.12 |
| 41 | 1 yellow rubber sphere | 0.01 | 0.91 | -0.00 | -0.00 | 0.00 |
| 41 | 3 red rubber cube | -4.81 | 3.40 | 1.72 | -1.37 | 2.19 |
| 41 | 4 red rubber sphere | 0.39 | -0.85 | 0.00 | -0.00 | 0.00 |
| 41 | 5 purple metal cube | -2.01 | -0.83 | -0.00 | 0.00 | 0.00 |
| 48 | 0 brown metal cube | 0.30 | 0.90 | -0.36 | 0.49 | 0.61 |
| 48 | 1 yellow rubber sphere | -0.16 | 0.93 | -1.02 | 0.14 | 1.03 |
| 48 | 3 red rubber cube | -4.48 | 3.14 | 1.63 | -1.30 | 2.08 |
| 48 | 4 red rubber sphere | 0.39 | -0.85 | 0.00 | -0.00 | 0.00 |
| 48 | 5 purple metal cube | -2.01 | -0.83 | -0.00 | 0.00 | 0.00 |
| 56 | 0 brown metal cube | 0.22 | 1.00 | -0.32 | 0.44 | 0.55 |
| 56 | 1 yellow rubber sphere | -0.39 | 0.97 | -1.00 | 0.14 | 1.01 |
| 56 | 3 red rubber cube | -4.12 | 2.86 | 1.54 | -1.22 | 1.96 |
| 56 | 4 red rubber sphere | 0.39 | -0.85 | 0.00 | -0.00 | 0.00 |
| 56 | 5 purple metal cube | -2.01 | -0.83 | -0.00 | -0.00 | 0.00 |
| 61 | 0 brown metal cube | 0.18 | 1.06 | -0.30 | 0.40 | 0.50 |
| 61 | 1 yellow rubber sphere | -0.53 | 0.98 | -0.99 | 0.14 | 1.00 |
| 61 | 2 brown rubber cylinder | 0.63 | 2.22 | -0.93 | -1.57 | 1.83 |
| 61 | 3 red rubber cube | -3.91 | 2.69 | 1.48 | -1.18 | 1.89 |
| 61 | 4 red rubber sphere | 0.39 | -0.85 | 0.00 | -0.00 | 0.00 |
| 61 | 5 purple metal cube | -2.01 | -0.83 | -0.00 | 0.00 | 0.00 |
| 64 | 0 brown metal cube | 0.16 | 1.09 | -0.28 | 0.38 | 0.48 |
| 64 | 1 yellow rubber sphere | -0.61 | 1.00 | -0.99 | 0.14 | 1.00 |
| 64 | 2 brown rubber cylinder | 0.55 | 2.09 | -0.91 | -1.54 | 1.78 |
| 64 | 3 red rubber cube | -3.79 | 2.60 | 1.45 | -1.15 | 1.85 |
| 64 | 4 red rubber sphere | 0.39 | -0.85 | 0.00 | -0.00 | 0.00 |
| 64 | 5 purple metal cube | -2.01 | -0.83 | -0.00 | 0.00 | 0.00 |
| 72 | 0 brown metal cube | 0.10 | 1.17 | -0.24 | 0.33 | 0.40 |
| 72 | 1 yellow rubber sphere | -0.83 | 1.03 | -0.97 | 0.13 | 0.98 |
| 72 | 2 brown rubber cylinder | 0.36 | 1.76 | -0.85 | -1.44 | 1.68 |
| 72 | 3 red rubber cube | -3.48 | 2.35 | 1.36 | -1.08 | 1.74 |
| 72 | 4 red rubber sphere | 0.39 | -0.85 | 0.00 | -0.00 | 0.00 |
| 72 | 5 purple metal cube | -2.01 | -0.83 | -0.00 | 0.00 | 0.00 |
| 75 | 0 brown metal cube | 0.08 | 1.20 | -0.22 | 0.31 | 0.38 |
| 75 | 1 yellow rubber sphere | -0.91 | 1.04 | -0.96 | 0.13 | 0.97 |
| 75 | 2 brown rubber cylinder | 0.28 | 1.64 | -0.83 | -1.41 | 1.64 |
| 75 | 3 red rubber cube | -3.36 | 2.26 | 1.33 | -1.06 | 1.70 |
| 75 | 4 red rubber sphere | 0.39 | -0.85 | 0.00 | -0.00 | 0.00 |
| 75 | 5 purple metal cube | -2.01 | -0.83 | 0.00 | 0.00 | 0.00 |
| 80 | 0 brown metal cube | -0.04 | 1.07 | -0.88 | -0.98 | 1.31 |
| 80 | 1 yellow rubber sphere | -1.05 | 1.06 | -0.95 | 0.13 | 0.96 |
| 80 | 2 brown rubber cylinder | 0.23 | 1.56 | -0.37 | -0.53 | 0.65 |
| 80 | 3 red rubber cube | -3.18 | 2.11 | 1.28 | -1.01 | 1.63 |
| 80 | 4 red rubber sphere | 0.39 | -0.85 | 0.00 | -0.00 | 0.00 |
| 80 | 5 purple metal cube | -2.01 | -0.83 | -0.00 | 0.00 | 0.00 |
| 88 | 0 brown metal cube | -0.23 | 0.86 | -0.82 | -0.91 | 1.23 |
| 88 | 1 yellow rubber sphere | -1.26 | 1.09 | -0.93 | 0.13 | 0.94 |

| frame | object | x | y | vx | vy | speed |
|---|---|---|---|---|---|---|
| 88 | 2 brown rubber cylinder | 0.15 | 1.44 | -0.33 | -0.47 | 0.57 |
| 88 | 3 red rubber cube | -2.90 | 1.89 | 1.20 | -0.95 | 1.53 |
| 88 | 4 red rubber sphere | 0.39 | -0.85 | 0.00 | -0.00 | 0.00 |
| 88 | 5 purple metal cube | -2.01 | -0.83 | -0.00 | 0.00 | 0.00 |
| 96 | 0 brown metal cube | -0.41 | 0.66 | -0.76 | -0.85 | 1.14 |
| 96 | 1 yellow rubber sphere | -1.47 | 1.11 | -0.92 | 0.13 | 0.93 |
| 96 | 2 brown rubber cylinder | 0.08 | 1.35 | -0.28 | -0.41 | 0.50 |
| 96 | 3 red rubber cube | -2.64 | 1.69 | 1.12 | -0.89 | 1.43 |
| 96 | 4 red rubber sphere | 0.39 | -0.85 | 0.00 | -0.00 | 0.00 |
| 96 | 5 purple metal cube | -2.01 | -0.83 | -0.00 | 0.00 | 0.00 |
| 104 | 0 brown metal cube | -0.57 | 0.48 | -0.70 | -0.78 | 1.05 |
| 104 | 1 yellow rubber sphere | -1.67 | 1.14 | -0.90 | 0.12 | 0.91 |
| 104 | 2 brown rubber cylinder | 0.02 | 1.26 | -0.24 | -0.35 | 0.43 |
| 104 | 3 red rubber cube | -2.40 | 1.49 | 1.05 | -0.83 | 1.34 |
| 104 | 4 red rubber sphere | 0.39 | -0.85 | 0.00 | -0.00 | 0.00 |
| 104 | 5 purple metal cube | -2.01 | -0.83 | -0.00 | 0.00 | 0.00 |
| 110 | 0 brown metal cube | -0.69 | 0.35 | -0.66 | -0.74 | 0.99 |
| 110 | 1 yellow rubber sphere | -1.81 | 1.16 | -0.22 | -0.25 | 0.33 |
| 110 | 2 brown rubber cylinder | -0.02 | 1.20 | -0.21 | -0.31 | 0.38 |
| 110 | 3 red rubber cube | -2.24 | 1.36 | 0.12 | -0.30 | 0.32 |
| 110 | 4 red rubber sphere | 0.39 | -0.85 | 0.00 | -0.00 | 0.00 |
| 110 | 5 purple metal cube | -2.01 | -0.83 | -0.00 | 0.00 | 0.00 |
| 112 | 0 brown metal cube | -0.72 | 0.31 | -0.65 | -0.72 | 0.97 |
| 112 | 1 yellow rubber sphere | -1.82 | 1.14 | -0.13 | -0.31 | 0.33 |
| 112 | 2 brown rubber cylinder | -0.03 | 1.19 | -0.20 | -0.30 | 0.36 |
| 112 | 3 red rubber cube | -2.24 | 1.35 | 0.00 | -0.22 | 0.22 |
| 112 | 4 red rubber sphere | 0.39 | -0.85 | 0.00 | -0.00 | 0.00 |
| 112 | 5 purple metal cube | -2.01 | -0.83 | -0.00 | 0.00 | 0.00 |
| 120 | 0 brown metal cube | -0.86 | 0.16 | -0.59 | -0.66 | 0.89 |
| 120 | 1 yellow rubber sphere | -1.85 | 1.07 | -0.12 | -0.30 | 0.33 |
| 120 | 2 brown rubber cylinder | -0.07 | 1.13 | -0.16 | -0.24 | 0.29 |
| 120 | 3 red rubber cube | -2.24 | 1.31 | -0.01 | -0.16 | 0.16 |
| 120 | 4 red rubber sphere | 0.39 | -0.85 | 0.00 | -0.00 | 0.00 |
| 120 | 5 purple metal cube | -2.01 | -0.83 | -0.00 | 0.00 | 0.00 |
| 127 | 0 brown metal cube | -0.97 | 0.03 | -0.55 | -0.61 | 0.82 |
| 127 | 1 yellow rubber sphere | -1.87 | 1.02 | -0.12 | -0.30 | 0.32 |
| 127 | 2 brown rubber cylinder | -0.10 | 1.08 | -0.13 | -0.19 | 0.23 |
| 127 | 3 red rubber cube | -2.24 | 1.28 | -0.00 | -0.10 | 0.10 |
| 127 | 4 red rubber sphere | 0.39 | -0.85 | 0.00 | -0.00 | 0.00 |
| 127 | 5 purple metal cube | -2.01 | -0.83 | -0.00 | 0.00 | 0.00 |

# Video 10937

## Video 10937: Objects

Object 0: brown metal cylinder (color=brown, material=metal, shape=cylinder)

Object 1: red rubber sphere (color=red, material=rubber, shape=sphere)

Object 2: purple rubber cube (color=purple, material=rubber, shape=cube)

Object 3: green metal cylinder (color=green, material=metal, shape=cylinder)

Object 4: cyan metal cylinder (color=cyan, material=metal, shape=cylinder)

## Video 10937: Events

Frame 0: Object 0 (brown metal cylinder) is in view when the video begins.

Frame 0: Object 3 (green metal cylinder) is in view when the video begins.

Frame 0: Object 4 (cyan metal cylinder) is in view when the video begins.

Frame 5: Object 2 (purple rubber cube) enters the scene.

Frame 13: Object 1 (red rubber sphere) enters the scene.

Frame 41: Object 0 (brown metal cylinder) collides with Object 1 (red rubber sphere).

Frame 59: Object 1 (red rubber sphere) collides with Object 3 (green metal cylinder).

Frame 105: Object 0 (brown metal cylinder) collides with Object 2 (purple rubber cube).

Frame 127: The video ends.

## Video 10937: Motion log

| frame | object | x | y | vx | vy | speed |
|---|---|---|---|---|---|---|
| 0 | 0 brown metal cylinder | -2.95 | -2.03 | 2.52 | 1.28 | 2.83 |
| 0 | 3 green metal cylinder | 0.99 | -0.13 | 0.00 | 0.00 | 0.00 |
| 0 | 4 cyan metal cylinder | -2.49 | 1.49 | 0.00 | 0.00 | 0.00 |
| 5 | 0 brown metal cylinder | -2.60 | -1.85 | 2.44 | 1.23 | 2.73 |
| 5 | 2 purple rubber cube | -4.06 | -4.15 | 1.21 | 2.47 | 2.75 |
| 5 | 3 green metal cylinder | 0.99 | -0.13 | -0.00 | -0.00 | 0.00 |
| 5 | 4 cyan metal cylinder | -2.49 | 1.49 | 0.00 | 0.00 | 0.00 |
| 8 | 0 brown metal cylinder | -2.40 | -1.75 | 2.38 | 1.20 | 2.67 |
| 8 | 2 purple rubber cube | -3.96 | -3.94 | 1.18 | 2.42 | 2.69 |
| 8 | 3 green metal cylinder | 0.99 | -0.13 | -0.00 | 0.00 | 0.00 |
| 8 | 4 cyan metal cylinder | -2.49 | 1.49 | -0.00 | 0.00 | 0.00 |
| 13 | 0 brown metal cylinder | -2.07 | -1.58 | 2.30 | 1.16 | 2.58 |
| 13 | 1 red rubber sphere | 1.58 | -1.79 | -2.22 | 1.34 | 2.60 |
| 13 | 2 purple rubber cube | -3.80 | -3.61 | 1.14 | 2.33 | 2.60 |
| 13 | 3 green metal cylinder | 0.99 | -0.13 | -0.00 | 0.00 | 0.00 |
| 13 | 4 cyan metal cylinder | -2.49 | 1.49 | -0.00 | -0.00 | 0.00 |
| 16 | 0 brown metal cylinder | -1.88 | -1.49 | 2.25 | 1.14 | 2.52 |
| 16 | 1 red rubber sphere | 1.39 | -1.67 | -2.20 | 1.33 | 2.57 |
| 16 | 2 purple rubber cube | -3.70 | -3.42 | 1.12 | 2.28 | 2.54 |
| 16 | 3 green metal cylinder | 0.99 | -0.13 | -0.00 | 0.00 | 0.00 |
| 16 | 4 cyan metal cylinder | -2.49 | 1.49 | -0.00 | -0.00 | 0.00 |
| 24 | 0 brown metal cylinder | -1.39 | -1.24 | 2.12 | 1.08 | 2.38 |
| 24 | 1 red rubber sphere | 0.91 | -1.38 | -2.13 | 1.29 | 2.49 |
| 24 | 2 purple rubber cube | -3.46 | -2.92 | 1.05 | 2.15 | 2.40 |
| 24 | 3 green metal cylinder | 0.99 | -0.13 | 0.00 | 0.00 | 0.00 |

| frame | object | x | y | vx | vy | speed |
|---|---|---|---|---|---|---|
| 24 | 4 cyan metal cylinder | -2.49 | 1.49 | 0.00 | -0.00 | 0.00 |
| 32 | 0 brown metal cylinder | -0.93 | -1.01 | 2.00 | 1.01 | 2.25 |
| 32 | 1 red rubber sphere | 0.44 | -1.10 | -2.06 | 1.25 | 2.41 |
| 32 | 2 purple rubber cube | -3.23 | -2.45 | 0.99 | 2.03 | 2.26 |
| 32 | 3 green metal cylinder | 0.99 | -0.13 | 0.00 | 0.00 | 0.00 |
| 32 | 4 cyan metal cylinder | -2.49 | 1.49 | 0.00 | -0.00 | 0.00 |
| 40 | 0 brown metal cylinder | -0.50 | -0.79 | 1.89 | 0.95 | 2.12 |
| 40 | 1 red rubber sphere | -0.02 | -0.82 | -2.00 | 1.21 | 2.34 |
| 40 | 2 purple rubber cube | -3.02 | -2.01 | 0.94 | 1.92 | 2.13 |
| 40 | 3 green metal cylinder | 0.99 | -0.13 | 0.00 | 0.00 | 0.00 |
| 40 | 4 cyan metal cylinder | -2.49 | 1.49 | 0.00 | -0.00 | 0.00 |
| 41 | 0 brown metal cylinder | -0.46 | -0.76 | -0.62 | 1.13 | 1.29 |
| 41 | 1 red rubber sphere | -0.05 | -0.79 | 1.35 | 0.96 | 1.66 |
| 41 | 2 purple rubber cube | -2.99 | -1.96 | 0.93 | 1.90 | 2.12 |
| 41 | 3 green metal cylinder | 0.99 | -0.13 | 0.00 | 0.00 | 0.00 |
| 41 | 4 cyan metal cylinder | -2.49 | 1.49 | 0.00 | -0.00 | 0.00 |
| 48 | 0 brown metal cylinder | -0.58 | -0.54 | -0.58 | 1.06 | 1.21 |
| 48 | 1 red rubber sphere | 0.22 | -0.60 | 1.32 | 0.94 | 1.62 |
| 48 | 2 purple rubber cube | -2.81 | -1.59 | 0.88 | 1.81 | 2.01 |
| 48 | 3 green metal cylinder | 0.99 | -0.13 | 0.00 | 0.00 | 0.00 |
| 48 | 4 cyan metal cylinder | -2.49 | 1.49 | 0.00 | -0.00 | 0.00 |
| 56 | 0 brown metal cylinder | -0.71 | -0.32 | -0.54 | 0.98 | 1.12 |
| 56 | 1 red rubber sphere | 0.51 | -0.40 | 1.29 | 0.92 | 1.59 |
| 56 | 2 purple rubber cube | -2.62 | -1.20 | 0.83 | 1.70 | 1.89 |
| 56 | 3 green metal cylinder | 0.99 | -0.13 | 0.00 | 0.00 | 0.00 |
| 56 | 4 cyan metal cylinder | -2.49 | 1.49 | 0.00 | -0.00 | 0.00 |
| 59 | 0 brown metal cylinder | -0.75 | -0.23 | -0.52 | 0.95 | 1.09 |
| 59 | 1 red rubber sphere | 0.62 | -0.32 | 1.28 | 0.91 | 1.57 |
| 59 | 2 purple rubber cube | -2.55 | -1.06 | 0.81 | 1.66 | 1.85 |
| 59 | 3 green metal cylinder | 0.99 | -0.13 | 0.00 | -0.00 | 0.00 |
| 59 | 4 cyan metal cylinder | -2.49 | 1.49 | 0.00 | -0.00 | 0.00 |
| 64 | 0 brown metal cylinder | -0.82 | -0.10 | -0.50 | 0.91 | 1.03 |
| 64 | 1 red rubber sphere | 0.63 | -0.27 | 0.03 | 0.28 | 0.28 |
| 64 | 2 purple rubber cube | -2.44 | -0.83 | 0.78 | 1.60 | 1.78 |
| 64 | 3 green metal cylinder | 1.09 | -0.08 | 0.77 | 0.39 | 0.86 |
| 64 | 4 cyan metal cylinder | -2.49 | 1.49 | 0.00 | -0.00 | 0.00 |
| 72 | 0 brown metal cylinder | -0.93 | 0.09 | -0.46 | 0.83 | 0.95 |
| 72 | 1 red rubber sphere | 0.64 | -0.21 | 0.03 | 0.28 | 0.28 |
| 72 | 2 purple rubber cube | -2.27 | -0.48 | 0.73 | 1.50 | 1.67 |
| 72 | 3 green metal cylinder | 1.26 | 0.01 | 0.70 | 0.35 | 0.79 |
| 72 | 4 cyan metal cylinder | -2.49 | 1.49 | 0.00 | -0.00 | 0.00 |
| 80 | 0 brown metal cylinder | -1.03 | 0.27 | -0.42 | 0.76 | 0.87 |
| 80 | 1 red rubber sphere | 0.64 | -0.15 | 0.03 | 0.27 | 0.28 |
| 80 | 2 purple rubber cube | -2.11 | -0.16 | 0.69 | 1.41 | 1.57 |
| 80 | 3 green metal cylinder | 1.41 | 0.08 | 0.63 | 0.32 | 0.71 |
| 80 | 4 cyan metal cylinder | -2.49 | 1.49 | 0.00 | -0.00 | 0.00 |
| 88 | 0 brown metal cylinder | -1.12 | 0.43 | -0.38 | 0.69 | 0.79 |
| 88 | 1 red rubber sphere | 0.65 | -0.09 | 0.03 | 0.27 | 0.27 |
| 88 | 2 purple rubber cube | -1.96 | 0.15 | 0.64 | 1.32 | 1.47 |
| 88 | 3 green metal cylinder | 1.54 | 0.15 | 0.57 | 0.28 | 0.63 |
| 88 | 4 cyan metal cylinder | -2.49 | 1.49 | 0.00 | -0.00 | 0.00 |
| 96 | 0 brown metal cylinder | -1.20 | 0.58 | -0.34 | 0.63 | 0.71 |
| 96 | 1 red rubber sphere | 0.65 | -0.03 | 0.03 | 0.27 | 0.27 |
| 96 | 2 purple rubber cube | -1.82 | 0.43 | 0.60 | 1.23 | 1.37 |
| 96 | 3 green metal cylinder | 1.66 | 0.21 | 0.50 | 0.25 | 0.56 |
| 96 | 4 cyan metal cylinder | -2.49 | 1.49 | 0.00 | -0.00 | 0.00 |
| 104 | 0 brown metal cylinder | -1.27 | 0.71 | -0.30 | 0.56 | 0.64 |

| frame | object | x | y | vx | vy | speed |
|---|---|---|---|---|---|---|
| 104 | 1 red rubber sphere | 0.66 | 0.03 | 0.03 | 0.27 | 0.27 |
| 104 | 2 purple rubber cube | -1.69 | 0.70 | 0.56 | 1.15 | 1.28 |
| 104 | 3 green metal cylinder | 1.76 | 0.26 | 0.44 | 0.22 | 0.49 |
| 104 | 4 cyan metal cylinder | -2.49 | 1.49 | 0.00 | -0.00 | 0.00 |
| 105 | 0 brown metal cylinder | -1.27 | 0.73 | 0.04 | 0.51 | 0.51 |
| 105 | 1 red rubber sphere | 0.66 | 0.04 | 0.03 | 0.27 | 0.27 |
| 105 | 2 purple rubber cube | -1.68 | 0.73 | 0.26 | 1.17 | 1.20 |
| 105 | 3 green metal cylinder | 1.78 | 0.27 | 0.43 | 0.21 | 0.48 |
| 105 | 4 cyan metal cylinder | -2.49 | 1.49 | 0.00 | -0.00 | 0.00 |
| 112 | 0 brown metal cylinder | -1.25 | 0.82 | 0.13 | 0.44 | 0.46 |
| 112 | 1 red rubber sphere | 0.66 | 0.09 | 0.03 | 0.26 | 0.26 |
| 112 | 2 purple rubber cube | -1.65 | 0.96 | 0.16 | 1.11 | 1.12 |
| 112 | 3 green metal cylinder | 1.85 | 0.31 | 0.37 | 0.18 | 0.42 |
| 112 | 4 cyan metal cylinder | -2.49 | 1.49 | 0.00 | -0.00 | 0.00 |
| 120 | 0 brown metal cylinder | -1.22 | 0.91 | 0.11 | 0.37 | 0.39 |
| 120 | 1 red rubber sphere | 0.67 | 0.15 | 0.02 | 0.26 | 0.26 |
| 120 | 2 purple rubber cube | -1.61 | 1.19 | 0.15 | 1.02 | 1.03 |
| 120 | 3 green metal cylinder | 1.93 | 0.34 | 0.31 | 0.15 | 0.35 |
| 120 | 4 cyan metal cylinder | -2.49 | 1.49 | 0.00 | -0.00 | 0.00 |
| 127 | 0 brown metal cylinder | -1.20 | 0.98 | 0.09 | 0.31 | 0.32 |
| 127 | 1 red rubber sphere | 0.67 | 0.20 | 0.02 | 0.26 | 0.26 |
| 127 | 2 purple rubber cube | -1.58 | 1.39 | 0.14 | 0.95 | 0.96 |
| 127 | 3 green metal cylinder | 1.99 | 0.37 | 0.26 | 0.12 | 0.29 |
| 127 | 4 cyan metal cylinder | -2.49 | 1.49 | 0.00 | -0.00 | 0.00 |

# Video 11544

## Video 11544: Objects

Object 0: gray rubber cube (color=gray, material=rubber, shape=cube)

Object 1: blue rubber cube (color=blue, material=rubber, shape=cube)

Object 2: yellow metal cylinder (color=yellow, material=metal, shape=cylinder)

Object 3: gray metal sphere (color=gray, material=metal, shape=sphere)

Object 4: yellow rubber cylinder (color=yellow, material=rubber, shape=cylinder)

## Video 11544: Events

Frame 0: Object 0 (gray rubber cube) is in view when the video begins.

Frame 0: Object 1 (blue rubber cube) is in view when the video begins.

Frame 0: Object 4 (yellow rubber cylinder) is in view when the video begins.

Frame 23: Object 0 (gray rubber cube) collides with Object 1 (blue rubber cube).

Frame 52: Object 2 (yellow metal cylinder) enters the scene.

Frame 92: Object 1 (blue rubber cube) exits the scene.

Frame 118: Object 3 (gray metal sphere) enters the scene.

Frame 127: The video ends.

## Video 11544: Motion log

| frame | object | x | y | vx | vy | speed |
|---|---|---|---|---|---|---|
| 0 | 0 gray rubber cube | -3.55 | 0.63 | 1.89 | -2.16 | 2.87 |
| 0 | 1 blue rubber cube | -2.21 | -1.07 | 0.00 | 0.00 | 0.00 |
| 0 | 4 yellow rubber cylinder | -0.43 | 0.00 | 0.00 | 0.00 | 0.00 |
| 8 | 0 gray rubber cube | -3.14 | 0.16 | 1.78 | -2.04 | 2.71 |
| 8 | 1 blue rubber cube | -2.21 | -1.07 | 0.00 | -0.00 | 0.00 |
| 8 | 4 yellow rubber cylinder | -0.43 | 0.00 | -0.00 | -0.00 | 0.00 |
| 16 | 0 gray rubber cube | -2.75 | -0.29 | 1.68 | -1.93 | 2.56 |
| 16 | 1 blue rubber cube | -2.21 | -1.07 | 0.00 | -0.00 | 0.00 |
| 16 | 4 yellow rubber cylinder | -0.43 | 0.00 | -0.00 | -0.00 | 0.00 |
| 23 | 0 gray rubber cube | -2.43 | -0.65 | 1.30 | -1.17 | 1.74 |
| 23 | 1 blue rubber cube | -2.21 | -1.07 | 0.30 | -0.65 | 0.72 |
| 23 | 4 yellow rubber cylinder | -0.43 | 0.00 | 0.00 | -0.00 | 0.00 |
| 24 | 0 gray rubber cube | -2.40 | -0.67 | 0.95 | -0.45 | 1.05 |
| 24 | 1 blue rubber cube | -2.19 | -1.11 | 0.62 | -1.35 | 1.49 |
| 24 | 4 yellow rubber cylinder | -0.43 | 0.00 | 0.00 | -0.00 | 0.00 |
| 32 | 0 gray rubber cube | -2.20 | -0.76 | 0.88 | -0.41 | 0.97 |
| 32 | 1 blue rubber cube | -2.06 | -1.41 | 0.58 | -1.27 | 1.40 |
| 32 | 4 yellow rubber cylinder | -0.43 | 0.00 | -0.00 | -0.00 | 0.00 |
| 40 | 0 gray rubber cube | -2.01 | -0.85 | 0.81 | -0.38 | 0.89 |
| 40 | 1 blue rubber cube | -1.93 | -1.68 | 0.54 | -1.19 | 1.30 |
| 40 | 4 yellow rubber cylinder | -0.43 | 0.00 | 0.00 | -0.00 | 0.00 |
| 48 | 0 gray rubber cube | -1.83 | -0.93 | 0.73 | -0.34 | 0.81 |
| 48 | 1 blue rubber cube | -1.81 | -1.94 | 0.51 | -1.10 | 1.21 |
| 48 | 4 yellow rubber cylinder | -0.43 | 0.00 | 0.00 | -0.00 | 0.00 |
| 52 | 0 gray rubber cube | -1.75 | -0.97 | 0.70 | -0.33 | 0.77 |
| 52 | 1 blue rubber cube | -1.76 | -2.06 | 0.49 | -1.06 | 1.17 |

| frame | object | x | y | vx | vy | speed |
|---|---|---|---|---|---|---|
| 52 | 2 yellow metal cylinder | -3.66 | -3.92 | 1.43 | 1.36 | 1.98 |
| 52 | 4 yellow rubber cylinder | -0.43 | 0.00 | 0.00 | -0.00 | 0.00 |
| 56 | 0 gray rubber cube | -1.68 | -1.00 | 0.66 | -0.31 | 0.73 |
| 56 | 1 blue rubber cube | -1.71 | -2.17 | 0.47 | -1.02 | 1.13 |
| 56 | 2 yellow metal cylinder | -3.50 | -3.77 | 1.39 | 1.32 | 1.92 |
| 56 | 4 yellow rubber cylinder | -0.43 | 0.00 | 0.00 | -0.00 | 0.00 |
| 64 | 0 gray rubber cube | -1.54 | -1.07 | 0.60 | -0.28 | 0.66 |
| 64 | 1 blue rubber cube | -1.61 | -2.40 | 0.43 | -0.95 | 1.04 |
| 64 | 2 yellow metal cylinder | -3.20 | -3.49 | 1.31 | 1.24 | 1.80 |
| 64 | 4 yellow rubber cylinder | -0.43 | 0.00 | 0.00 | -0.00 | 0.00 |
| 72 | 0 gray rubber cube | -1.41 | -1.13 | 0.53 | -0.24 | 0.58 |
| 72 | 1 blue rubber cube | -1.51 | -2.60 | 0.40 | -0.87 | 0.96 |
| 72 | 2 yellow metal cylinder | -2.91 | -3.22 | 1.23 | 1.17 | 1.70 |
| 72 | 4 yellow rubber cylinder | -0.43 | 0.00 | 0.00 | -0.00 | 0.00 |
| 80 | 0 gray rubber cube | -1.30 | -1.18 | 0.46 | -0.21 | 0.51 |
| 80 | 1 blue rubber cube | -1.43 | -2.79 | 0.36 | -0.80 | 0.88 |
| 80 | 2 yellow metal cylinder | -2.65 | -2.97 | 1.15 | 1.09 | 1.59 |
| 80 | 4 yellow rubber cylinder | -0.43 | 0.00 | -0.00 | -0.00 | 0.00 |
| 88 | 0 gray rubber cube | -1.20 | -1.22 | 0.40 | -0.18 | 0.44 |
| 88 | 1 blue rubber cube | -1.35 | -2.96 | 0.33 | -0.73 | 0.80 |
| 88 | 2 yellow metal cylinder | -2.40 | -2.73 | 1.08 | 1.03 | 1.49 |
| 88 | 4 yellow rubber cylinder | -0.43 | 0.00 | 0.00 | -0.00 | 0.00 |
| 92 | 0 gray rubber cube | -1.16 | -1.24 | 0.37 | -0.17 | 0.40 |
| 92 | 2 yellow metal cylinder | -2.28 | -2.62 | 1.05 | 0.99 | 1.44 |
| 92 | 4 yellow rubber cylinder | -0.43 | 0.00 | -0.00 | -0.00 | 0.00 |
| 96 | 0 gray rubber cube | -1.12 | -1.26 | 0.33 | -0.15 | 0.37 |
| 96 | 2 yellow metal cylinder | -2.16 | -2.51 | 1.01 | 0.96 | 1.39 |
| 96 | 4 yellow rubber cylinder | -0.43 | 0.00 | 0.00 | -0.00 | 0.00 |
| 104 | 0 gray rubber cube | -1.05 | -1.29 | 0.27 | -0.12 | 0.30 |
| 104 | 2 yellow metal cylinder | -1.94 | -2.30 | 0.94 | 0.89 | 1.30 |
| 104 | 4 yellow rubber cylinder | -0.43 | 0.00 | 0.00 | -0.00 | 0.00 |
| 112 | 0 gray rubber cube | -1.00 | -1.31 | 0.21 | -0.09 | 0.23 |
| 112 | 2 yellow metal cylinder | -1.74 | -2.11 | 0.88 | 0.83 | 1.21 |
| 112 | 4 yellow rubber cylinder | -0.43 | 0.00 | -0.00 | -0.00 | 0.00 |
| 118 | 0 gray rubber cube | -0.97 | -1.33 | 0.16 | -0.07 | 0.18 |
| 118 | 2 yellow metal cylinder | -1.60 | -1.97 | 0.83 | 0.79 | 1.14 |
| 118 | 3 gray metal sphere | 0.76 | -2.14 | -1.55 | 0.95 | 1.82 |
| 118 | 4 yellow rubber cylinder | -0.43 | 0.00 | -0.00 | -0.00 | 0.00 |
| 120 | 0 gray rubber cube | -0.96 | -1.33 | 0.15 | -0.07 | 0.16 |
| 120 | 2 yellow metal cylinder | -1.55 | -1.93 | 0.81 | 0.77 | 1.12 |
| 120 | 3 gray metal sphere | 0.67 | -2.09 | -1.54 | 0.94 | 1.81 |
| 120 | 4 yellow rubber cylinder | -0.43 | 0.00 | 0.00 | -0.00 | 0.00 |
| 127 | 0 gray rubber cube | -0.94 | -1.34 | 0.09 | -0.04 | 0.10 |
| 127 | 2 yellow metal cylinder | -1.40 | -1.78 | 0.76 | 0.72 | 1.05 |
| 127 | 3 gray metal sphere | 0.37 | -1.90 | -1.51 | 0.92 | 1.77 |
| 127 | 4 yellow rubber cylinder | -0.43 | 0.00 | -0.00 | -0.00 | 0.00 |

# Video 11716

## Video 11716: Objects

Object 0: purple metal sphere (color=purple, material=metal, shape=sphere)

Object 1: cyan rubber cube (color=cyan, material=rubber, shape=cube)

Object 2: yellow rubber cylinder (color=yellow, material=rubber, shape=cylinder)

Object 3: purple rubber sphere (color=purple, material=rubber, shape=sphere)

## Video 11716: Events

Frame 0: Object 1 (cyan rubber cube) is in view when the video begins.

Frame 8: Object 0 (purple metal sphere) enters the scene.

Frame 19: Object 2 (yellow rubber cylinder) enters the scene.

Frame 35: Object 0 (purple metal sphere) collides with Object 1 (cyan rubber cube).

Frame 90: Object 3 (purple rubber sphere) enters the scene.

Frame 118: Object 0 (purple metal sphere) collides with Object 2 (yellow rubber cylinder).

Frame 127: The video ends.

## Video 11716: Motion log

| frame | object | x | y | vx | vy | speed |
|---|---|---|---|---|---|---|
| 0 | 1 cyan rubber cube | 0.05 | -0.42 | 0.00 | 0.00 | 0.00 |
| 8 | 0 purple metal sphere | 2.40 | -0.93 | -2.75 | 0.52 | 2.79 |
| 8 | 1 cyan rubber cube | 0.05 | -0.42 | 0.00 | -0.00 | 0.00 |
| 16 | 0 purple metal sphere | 1.79 | -0.82 | -2.66 | 0.50 | 2.70 |
| 16 | 1 cyan rubber cube | 0.05 | -0.42 | 0.00 | -0.00 | 0.00 |
| 19 | 0 purple metal sphere | 1.57 | -0.77 | -2.62 | 0.50 | 2.67 |
| 19 | 1 cyan rubber cube | 0.05 | -0.42 | 0.00 | -0.00 | 0.00 |
| 19 | 2 yellow rubber cylinder | -6.62 | -2.13 | 2.25 | 0.90 | 2.42 |
| 24 | 0 purple metal sphere | 1.21 | -0.70 | -2.57 | 0.49 | 2.61 |
| 24 | 1 cyan rubber cube | 0.05 | -0.42 | 0.00 | -0.00 | 0.00 |
| 24 | 2 yellow rubber cylinder | -6.31 | -2.00 | 2.17 | 0.87 | 2.34 |
| 32 | 0 purple metal sphere | 0.64 | -0.60 | -2.49 | 0.47 | 2.53 |
| 32 | 1 cyan rubber cube | 0.05 | -0.42 | 0.00 | -0.00 | 0.00 |
| 32 | 2 yellow rubber cylinder | -5.84 | -1.81 | 2.04 | 0.82 | 2.20 |
| 35 | 0 purple metal sphere | 0.44 | -0.56 | -2.46 | 0.46 | 2.50 |
| 35 | 1 cyan rubber cube | 0.05 | -0.42 | 0.00 | -0.00 | 0.00 |
| 35 | 2 yellow rubber cylinder | -5.67 | -1.74 | 2.00 | 0.81 | 2.16 |
| 40 | 0 purple metal sphere | 0.28 | -0.54 | -1.08 | 0.12 | 1.09 |
| 40 | 1 cyan rubber cube | -0.17 | -0.37 | -1.56 | 0.39 | 1.60 |
| 40 | 2 yellow rubber cylinder | -5.40 | -1.63 | 1.93 | 0.78 | 2.08 |
| 48 | 0 purple metal sphere | 0.04 | -0.52 | -1.06 | 0.12 | 1.07 |
| 48 | 1 cyan rubber cube | -0.51 | -0.28 | -1.46 | 0.37 | 1.50 |
| 48 | 2 yellow rubber cylinder | -4.98 | -1.46 | 1.81 | 0.74 | 1.96 |
| 56 | 0 purple metal sphere | -0.19 | -0.49 | -1.04 | 0.11 | 1.05 |
| 56 | 1 cyan rubber cube | -0.82 | -0.20 | -1.36 | 0.34 | 1.41 |
| 56 | 2 yellow rubber cylinder | -4.58 | -1.30 | 1.71 | 0.69 | 1.84 |
| 64 | 0 purple metal sphere | -0.42 | -0.47 | -1.02 | 0.11 | 1.03 |
| 64 | 1 cyan rubber cube | -1.12 | -0.13 | -1.27 | 0.31 | 1.31 |
| 64 | 2 yellow rubber cylinder | -4.21 | -1.15 | 1.60 | 0.65 | 1.73 |

| frame | object | x | y | vx | vy | speed |
|---|---|---|---|---|---|---|
| 72 | 0 purple metal sphere | -0.65 | -0.44 | -1.01 | 0.11 | 1.01 |
| 72 | 1 cyan rubber cube | -1.39 | -0.06 | -1.19 | 0.29 | 1.22 |
| 72 | 2 yellow rubber cylinder | -3.87 | -1.01 | 1.51 | 0.61 | 1.63 |
| 80 | 0 purple metal sphere | -0.87 | -0.42 | -0.99 | 0.11 | 0.99 |
| 80 | 1 cyan rubber cube | -1.65 | 0.00 | -1.10 | 0.27 | 1.13 |
| 80 | 2 yellow rubber cylinder | -3.54 | -0.88 | 1.41 | 0.57 | 1.52 |
| 88 | 0 purple metal sphere | -1.09 | -0.39 | -0.97 | 0.11 | 0.98 |
| 88 | 1 cyan rubber cube | -1.89 | 0.06 | -1.02 | 0.25 | 1.05 |
| 88 | 2 yellow rubber cylinder | -3.23 | -0.76 | 1.32 | 0.53 | 1.43 |
| 90 | 0 purple metal sphere | -1.15 | -0.39 | -0.97 | 0.10 | 0.97 |
| 90 | 1 cyan rubber cube | -1.94 | 0.07 | -1.00 | 0.24 | 1.03 |
| 90 | 2 yellow rubber cylinder | -3.16 | -0.73 | 1.30 | 0.52 | 1.40 |
| 90 | 3 purple rubber sphere | 0.02 | 2.42 | -1.06 | -1.66 | 1.97 |
| 96 | 0 purple metal sphere | -1.31 | -0.37 | -0.95 | 0.10 | 0.96 |
| 96 | 1 cyan rubber cube | -2.10 | 0.11 | -0.94 | 0.22 | 0.96 |
| 96 | 2 yellow rubber cylinder | -2.95 | -0.64 | 1.23 | 0.50 | 1.33 |
| 96 | 3 purple rubber sphere | -0.15 | 2.14 | -1.04 | -1.63 | 1.93 |
| 104 | 0 purple metal sphere | -1.52 | -0.35 | -0.94 | 0.10 | 0.94 |
| 104 | 1 cyan rubber cube | -2.31 | 0.16 | -0.86 | 0.20 | 0.88 |
| 104 | 2 yellow rubber cylinder | -2.68 | -0.53 | 1.15 | 0.46 | 1.24 |
| 104 | 3 purple rubber sphere | -0.38 | 1.78 | -1.01 | -1.59 | 1.88 |
| 112 | 0 purple metal sphere | -1.73 | -0.32 | -0.92 | 0.10 | 0.93 |
| 112 | 1 cyan rubber cube | -2.49 | 0.20 | -0.78 | 0.18 | 0.80 |
| 112 | 2 yellow rubber cylinder | -2.43 | -0.43 | 1.07 | 0.43 | 1.15 |
| 112 | 3 purple rubber sphere | -0.61 | 1.43 | -0.99 | -1.55 | 1.84 |
| 118 | 0 purple metal sphere | -1.87 | -0.31 | 0.37 | 0.28 | 0.46 |
| 118 | 1 cyan rubber cube | -2.62 | 0.23 | -0.73 | 0.17 | 0.75 |
| 118 | 2 yellow rubber cylinder | -2.28 | -0.37 | -0.56 | 0.18 | 0.59 |
| 118 | 3 purple rubber sphere | -0.77 | 1.17 | -0.97 | -1.52 | 1.80 |
| 120 | 0 purple metal sphere | -1.85 | -0.29 | 0.36 | 0.28 | 0.46 |
| 120 | 1 cyan rubber cube | -2.66 | 0.24 | -0.71 | 0.16 | 0.73 |
| 120 | 2 yellow rubber cylinder | -2.31 | -0.36 | -0.55 | 0.17 | 0.57 |
| 120 | 3 purple rubber sphere | -0.82 | 1.09 | -0.96 | -1.51 | 1.79 |
| 127 | 0 purple metal sphere | -1.78 | -0.23 | 0.36 | 0.28 | 0.46 |
| 127 | 1 cyan rubber cube | -2.79 | 0.27 | -0.64 | 0.15 | 0.66 |
| 127 | 2 yellow rubber cylinder | -2.41 | -0.32 | -0.48 | 0.15 | 0.51 |
| 127 | 3 purple rubber sphere | -1.01 | 0.79 | -0.94 | -1.48 | 1.75 |

# Video 11851

## Video 11851: Objects

Object 0: brown metal cylinder (color=brown, material=metal, shape=cylinder)

Object 1: brown rubber sphere (color=brown, material=rubber, shape=sphere)

Object 2: red rubber sphere (color=red, material=rubber, shape=sphere)

Object 3: purple metal cube (color=purple, material=metal, shape=cube)

Object 4: purple metal sphere (color=purple, material=metal, shape=sphere)

## Video 11851: Events

Frame 0: Object 2 (red rubber sphere) is in view when the video begins.

Frame 0: Object 4 (purple metal sphere) is in view when the video begins.

Frame 12: Object 3 (purple metal cube) enters the scene.

Frame 13: Object 0 (brown metal cylinder) enters the scene.

Frame 31: Object 1 (brown rubber sphere) enters the scene.

Frame 51: Object 0 (brown metal cylinder) collides with Object 1 (brown rubber sphere).

Frame 69: Object 1 (brown rubber sphere) collides with Object 2 (red rubber sphere).

Frame 127: The video ends.

## Video 11851: Motion log

| frame | object | x | y | vx | vy | speed |
|---|---|---|---|---|---|---|
| 0 | 2 red rubber sphere | 0.16 | 1.27 | 0.00 | 0.00 | 0.00 |
| 0 | 4 purple metal sphere | 0.21 | -1.23 | 0.00 | 0.00 | 0.00 |
| 8 | 2 red rubber sphere | 0.16 | 1.27 | 0.00 | 0.00 | 0.00 |
| 8 | 4 purple metal sphere | 0.21 | -1.23 | -0.00 | 0.00 | 0.00 |
| 12 | 2 red rubber sphere | 0.16 | 1.27 | 0.00 | -0.00 | 0.00 |
| 12 | 3 purple metal cube | -6.66 | 2.65 | 2.28 | -1.11 | 2.53 |
| 12 | 4 purple metal sphere | 0.21 | -1.23 | -0.00 | 0.00 | 0.00 |
| 13 | 0 brown metal cylinder | 1.70 | -1.78 | -0.89 | 2.32 | 2.48 |
| 13 | 2 red rubber sphere | 0.16 | 1.27 | 0.00 | 0.00 | 0.00 |
| 13 | 3 purple metal cube | -6.60 | 2.62 | 2.26 | -1.10 | 2.51 |
| 13 | 4 purple metal sphere | 0.21 | -1.23 | -0.00 | 0.00 | 0.00 |
| 16 | 0 brown metal cylinder | 1.62 | -1.59 | -0.87 | 2.27 | 2.43 |
| 16 | 2 red rubber sphere | 0.16 | 1.27 | -0.00 | -0.00 | 0.00 |
| 16 | 3 purple metal cube | -6.41 | 2.53 | 2.21 | -1.07 | 2.46 |
| 16 | 4 purple metal sphere | 0.21 | -1.23 | 0.00 | 0.00 | 0.00 |
| 24 | 0 brown metal cylinder | 1.43 | -1.10 | -0.82 | 2.14 | 2.29 |
| 24 | 2 red rubber sphere | 0.16 | 1.27 | 0.00 | -0.00 | 0.00 |
| 24 | 3 purple metal cube | -5.93 | 2.29 | 2.09 | -1.01 | 2.32 |
| 24 | 4 purple metal sphere | 0.21 | -1.23 | 0.00 | -0.00 | 0.00 |
| 31 | 0 brown metal cylinder | 1.28 | -0.69 | -0.78 | 2.03 | 2.18 |
| 31 | 1 brown rubber sphere | 1.95 | 1.67 | -1.81 | -1.77 | 2.53 |
| 31 | 2 red rubber sphere | 0.16 | 1.27 | -0.00 | -0.00 | 0.00 |
| 31 | 3 purple metal cube | -5.53 | 2.10 | 1.99 | -0.96 | 2.21 |
| 31 | 4 purple metal sphere | 0.21 | -1.23 | 0.00 | -0.00 | 0.00 |
| 32 | 0 brown metal cylinder | 1.25 | -0.63 | -0.77 | 2.02 | 2.16 |
| 32 | 1 brown rubber sphere | 1.90 | 1.62 | -1.80 | -1.76 | 2.52 |

| frame | object | x | y | vx | vy | speed |
|---|---|---|---|---|---|---|
| 32 | 2 red rubber sphere | 0.16 | 1.27 | -0.00 | -0.00 | 0.00 |
| 32 | 3 purple metal cube | -5.47 | 2.07 | 1.97 | -0.95 | 2.19 |
| 32 | 4 purple metal sphere | 0.21 | -1.23 | 0.00 | -0.00 | 0.00 |
| 40 | 0 brown metal cylinder | 1.09 | -0.19 | -0.73 | 1.91 | 2.04 |
| 40 | 1 brown rubber sphere | 1.51 | 1.24 | -1.75 | -1.71 | 2.45 |
| 40 | 2 red rubber sphere | 0.16 | 1.27 | 0.00 | -0.00 | 0.00 |
| 40 | 3 purple metal cube | -5.05 | 1.86 | 1.86 | -0.90 | 2.07 |
| 40 | 4 purple metal sphere | 0.21 | -1.23 | 0.00 | -0.00 | 0.00 |
| 48 | 0 brown metal cylinder | 0.93 | 0.22 | -0.68 | 1.80 | 1.92 |
| 48 | 1 brown rubber sphere | 1.12 | 0.86 | -1.70 | -1.66 | 2.37 |
| 48 | 2 red rubber sphere | 0.16 | 1.27 | -0.00 | -0.00 | 0.00 |
| 48 | 3 purple metal cube | -4.64 | 1.67 | 1.75 | -0.84 | 1.95 |
| 48 | 4 purple metal sphere | 0.21 | -1.23 | 0.00 | -0.00 | 0.00 |
| 51 | 0 brown metal cylinder | 0.87 | 0.35 | -1.27 | -0.17 | 1.28 |
| 51 | 1 brown rubber sphere | 0.99 | 0.75 | -0.95 | 0.66 | 1.16 |
| 51 | 2 red rubber sphere | 0.16 | 1.27 | 0.00 | -0.00 | 0.00 |
| 51 | 3 purple metal cube | -4.50 | 1.60 | 1.72 | -0.82 | 1.90 |
| 51 | 4 purple metal sphere | 0.21 | -1.23 | 0.00 | -0.00 | 0.00 |
| 56 | 0 brown metal cylinder | 0.69 | 0.33 | -1.22 | -0.16 | 1.23 |
| 56 | 1 brown rubber sphere | 0.85 | 0.84 | -0.94 | 0.65 | 1.14 |
| 56 | 2 red rubber sphere | 0.16 | 1.27 | 0.00 | -0.00 | 0.00 |
| 56 | 3 purple metal cube | -4.26 | 1.49 | 1.65 | -0.79 | 1.83 |
| 56 | 4 purple metal sphere | 0.21 | -1.23 | 0.00 | -0.00 | 0.00 |
| 64 | 0 brown metal cylinder | 0.43 | 0.29 | -1.13 | -0.15 | 1.14 |
| 64 | 1 brown rubber sphere | 0.65 | 0.98 | -0.92 | 0.64 | 1.12 |
| 64 | 2 red rubber sphere | 0.16 | 1.27 | 0.00 | -0.00 | 0.00 |
| 64 | 3 purple metal cube | -3.90 | 1.31 | 1.56 | -0.74 | 1.72 |
| 64 | 4 purple metal sphere | 0.21 | -1.23 | 0.00 | -0.00 | 0.00 |
| 69 | 0 brown metal cylinder | 0.28 | 0.27 | -1.07 | -0.14 | 1.08 |
| 69 | 1 brown rubber sphere | 0.52 | 1.07 | -0.85 | 0.60 | 1.03 |
| 69 | 2 red rubber sphere | 0.16 | 1.27 | -0.05 | 0.03 | 0.05 |
| 69 | 3 purple metal cube | -3.69 | 1.21 | 1.50 | -0.72 | 1.66 |
| 69 | 4 purple metal sphere | 0.21 | -1.23 | 0.00 | -0.00 | 0.00 |
| 72 | 0 brown metal cylinder | 0.19 | 0.26 | -1.04 | -0.14 | 1.05 |
| 72 | 1 brown rubber sphere | 0.49 | 1.10 | -0.36 | 0.32 | 0.48 |
| 72 | 2 red rubber sphere | 0.13 | 1.29 | -0.41 | 0.23 | 0.47 |
| 72 | 3 purple metal cube | -3.56 | 1.15 | 1.46 | -0.70 | 1.62 |
| 72 | 4 purple metal sphere | 0.21 | -1.23 | 0.00 | -0.00 | 0.00 |
| 80 | 0 brown metal cylinder | -0.04 | 0.23 | -0.96 | -0.13 | 0.97 |
| 80 | 1 brown rubber sphere | 0.41 | 1.17 | -0.35 | 0.32 | 0.47 |
| 80 | 2 red rubber sphere | 0.04 | 1.34 | -0.40 | 0.23 | 0.46 |
| 80 | 3 purple metal cube | -3.25 | 1.00 | 1.37 | -0.65 | 1.52 |
| 80 | 4 purple metal sphere | 0.21 | -1.23 | 0.00 | -0.00 | 0.00 |
| 88 | 0 brown metal cylinder | -0.24 | 0.20 | -0.88 | -0.12 | 0.88 |
| 88 | 1 brown rubber sphere | 0.33 | 1.24 | -0.35 | 0.31 | 0.47 |
| 88 | 2 red rubber sphere | -0.05 | 1.39 | -0.40 | 0.22 | 0.46 |
| 88 | 3 purple metal cube | -2.95 | 0.86 | 1.28 | -0.61 | 1.42 |
| 88 | 4 purple metal sphere | 0.21 | -1.23 | 0.00 | -0.00 | 0.00 |
| 96 | 0 brown metal cylinder | -0.43 | 0.18 | -0.80 | -0.11 | 0.81 |
| 96 | 1 brown rubber sphere | 0.25 | 1.31 | -0.34 | 0.31 | 0.46 |
| 96 | 2 red rubber sphere | -0.14 | 1.44 | -0.39 | 0.22 | 0.45 |
| 96 | 3 purple metal cube | -2.67 | 0.73 | 1.20 | -0.57 | 1.33 |
| 96 | 4 purple metal sphere | 0.21 | -1.23 | 0.00 | -0.00 | 0.00 |
| 104 | 0 brown metal cylinder | -0.60 | 0.15 | -0.72 | -0.09 | 0.73 |
| 104 | 1 brown rubber sphere | 0.17 | 1.38 | -0.34 | 0.30 | 0.46 |
| 104 | 2 red rubber sphere | -0.23 | 1.48 | -0.39 | 0.22 | 0.44 |
| 104 | 3 purple metal cube | -2.41 | 0.61 | 1.12 | -0.53 | 1.24 |

| frame | object | x | y | vx | vy | speed |
|---|---|---|---|---|---|---|
| 104 | 4 purple metal sphere | 0.21 | -1.23 | 0.00 | -0.00 | 0.00 |
| 112 | 0 brown metal cylinder | -0.75 | 0.13 | -0.65 | -0.08 | 0.65 |
| 112 | 1 brown rubber sphere | 0.10 | 1.45 | -0.33 | 0.30 | 0.45 |
| 112 | 2 red rubber sphere | -0.31 | 1.53 | -0.38 | 0.21 | 0.44 |
| 112 | 3 purple metal cube | -2.17 | 0.49 | 1.04 | -0.49 | 1.15 |
| 112 | 4 purple metal sphere | 0.21 | -1.23 | 0.00 | -0.00 | 0.00 |
| 120 | 0 brown metal cylinder | -0.89 | 0.12 | -0.57 | -0.07 | 0.58 |
| 120 | 1 brown rubber sphere | 0.03 | 1.51 | -0.33 | 0.30 | 0.44 |
| 120 | 2 red rubber sphere | -0.40 | 1.58 | -0.38 | 0.21 | 0.43 |
| 120 | 3 purple metal cube | -1.94 | 0.39 | 0.96 | -0.45 | 1.06 |
| 120 | 4 purple metal sphere | 0.21 | -1.23 | 0.00 | -0.00 | 0.00 |
| 127 | 0 brown metal cylinder | -0.99 | 0.10 | -0.51 | -0.07 | 0.51 |
| 127 | 1 brown rubber sphere | -0.04 | 1.57 | -0.33 | 0.29 | 0.44 |
| 127 | 2 red rubber sphere | -0.47 | 1.62 | -0.37 | 0.21 | 0.43 |
| 127 | 3 purple metal cube | -1.76 | 0.30 | 0.90 | -0.42 | 0.99 |
| 127 | 4 purple metal sphere | 0.21 | -1.23 | 0.00 | -0.00 | 0.00 |

# Video 12082

## Video 12082: Objects

Object 0: yellow rubber cylinder (color=yellow, material=rubber, shape=cylinder)

Object 1: cyan metal cylinder (color=cyan, material=metal, shape=cylinder)

Object 2: yellow metal sphere (color=yellow, material=metal, shape=sphere)

Object 3: brown metal sphere (color=brown, material=metal, shape=sphere)

Object 4: cyan rubber sphere (color=cyan, material=rubber, shape=sphere)

Object 5: cyan metal sphere (color=cyan, material=metal, shape=sphere)

## Video 12082: Events

Frame 0: Object 0 (yellow rubber cylinder) is in view when the video begins.

Frame 0: Object 1 (cyan metal cylinder) is in view when the video begins.

Frame 0: Object 3 (brown metal sphere) is in view when the video begins.

Frame 0: Object 4 (cyan rubber sphere) is in view when the video begins.

Frame 0: Object 5 (cyan metal sphere) is in view when the video begins.

Frame 31: Object 0 (yellow rubber cylinder) collides with Object 1 (cyan metal cylinder).

Frame 87: Object 2 (yellow metal sphere) enters the scene.

Frame 98: Object 1 (cyan metal cylinder) collides with Object 2 (yellow metal sphere).

Frame 127: The video ends.

## Video 12082: Motion log

| frame | object | x | y | vx | vy | speed |
|---|---|---|---|---|---|---|
| 0 | 0 yellow rubber cylinder | -3.01 | -2.35 | 2.66 | 0.97 | 2.83 |
| 0 | 1 cyan metal cylinder | -0.58 | -1.48 | 0.00 | 0.00 | 0.00 |
| 0 | 3 brown metal sphere | 1.14 | 1.51 | 0.00 | 0.00 | 0.00 |
| 0 | 4 cyan rubber sphere | 0.18 | -0.23 | 0.00 | 0.00 | 0.00 |
| 0 | 5 cyan metal sphere | -0.54 | 0.64 | 0.00 | 0.00 | 0.00 |
| 8 | 0 yellow rubber cylinder | -2.43 | -2.14 | 2.51 | 0.91 | 2.67 |
| 8 | 1 cyan metal cylinder | -0.58 | -1.48 | 0.00 | 0.00 | 0.00 |
| 8 | 3 brown metal sphere | 1.14 | 1.51 | 0.00 | -0.00 | 0.00 |
| 8 | 4 cyan rubber sphere | 0.18 | -0.23 | 0.00 | -0.00 | 0.00 |
| 8 | 5 cyan metal sphere | -0.54 | 0.64 | -0.00 | -0.00 | 0.00 |
| 16 | 0 yellow rubber cylinder | -1.89 | -1.94 | 2.37 | 0.86 | 2.52 |
| 16 | 1 cyan metal cylinder | -0.58 | -1.48 | 0.00 | -0.00 | 0.00 |
| 16 | 3 brown metal sphere | 1.14 | 1.51 | -0.00 | -0.00 | 0.00 |
| 16 | 4 cyan rubber sphere | 0.18 | -0.23 | 0.00 | -0.00 | 0.00 |
| 16 | 5 cyan metal sphere | -0.54 | 0.64 | -0.00 | -0.00 | 0.00 |
| 24 | 0 yellow rubber cylinder | -1.37 | -1.75 | 2.24 | 0.81 | 2.38 |
| 24 | 1 cyan metal cylinder | -0.58 | -1.48 | -0.00 | -0.00 | 0.00 |
| 24 | 3 brown metal sphere | 1.14 | 1.51 | -0.00 | -0.00 | 0.00 |
| 24 | 4 cyan rubber sphere | 0.18 | -0.23 | 0.00 | -0.00 | 0.00 |
| 24 | 5 cyan metal sphere | -0.54 | 0.64 | -0.00 | -0.00 | 0.00 |
| 31 | 0 yellow rubber cylinder | -0.96 | -1.61 | 0.25 | 0.14 | 0.29 |
| 31 | 1 cyan metal cylinder | -0.57 | -1.47 | 1.17 | 0.39 | 1.23 |
| 31 | 3 brown metal sphere | 1.14 | 1.51 | -0.00 | -0.00 | 0.00 |

| frame | object | x | y | vx | vy | speed |
|---|---|---|---|---|---|---|
| 31 | 4 cyan rubber sphere | 0.18 | -0.23 | 0.00 | -0.00 | 0.00 |
| 31 | 5 cyan metal sphere | -0.54 | 0.64 | -0.00 | -0.00 | 0.00 |
| 32 | 0 yellow rubber cylinder | -0.95 | -1.60 | 0.24 | 0.13 | 0.28 |
| 32 | 1 cyan metal cylinder | -0.54 | -1.46 | 1.16 | 0.39 | 1.22 |
| 32 | 3 brown metal sphere | 1.14 | 1.51 | -0.00 | -0.00 | 0.00 |
| 32 | 4 cyan rubber sphere | 0.18 | -0.23 | 0.00 | -0.00 | 0.00 |
| 32 | 5 cyan metal sphere | -0.54 | 0.64 | -0.00 | -0.00 | 0.00 |
| 40 | 0 yellow rubber cylinder | -0.91 | -1.58 | 0.18 | 0.11 | 0.21 |
| 40 | 1 cyan metal cylinder | -0.29 | -1.38 | 1.08 | 0.36 | 1.13 |
| 40 | 3 brown metal sphere | 1.14 | 1.51 | 0.00 | -0.00 | 0.00 |
| 40 | 4 cyan rubber sphere | 0.18 | -0.23 | 0.00 | -0.00 | 0.00 |
| 40 | 5 cyan metal sphere | -0.54 | 0.64 | -0.00 | -0.00 | 0.00 |
| 48 | 0 yellow rubber cylinder | -0.87 | -1.56 | 0.12 | 0.07 | 0.14 |
| 48 | 1 cyan metal cylinder | -0.06 | -1.30 | 0.99 | 0.33 | 1.05 |
| 48 | 3 brown metal sphere | 1.14 | 1.51 | 0.00 | -0.00 | 0.00 |
| 48 | 4 cyan rubber sphere | 0.18 | -0.23 | 0.00 | -0.00 | 0.00 |
| 48 | 5 cyan metal sphere | -0.54 | 0.64 | 0.00 | -0.00 | 0.00 |
| 56 | 0 yellow rubber cylinder | -0.85 | -1.54 | 0.07 | 0.04 | 0.08 |
| 56 | 1 cyan metal cylinder | 0.15 | -1.23 | 0.92 | 0.30 | 0.96 |
| 56 | 3 brown metal sphere | 1.14 | 1.51 | 0.00 | -0.00 | 0.00 |
| 56 | 4 cyan rubber sphere | 0.18 | -0.23 | 0.00 | -0.00 | 0.00 |
| 56 | 5 cyan metal sphere | -0.54 | 0.64 | 0.00 | -0.00 | 0.00 |
| 64 | 0 yellow rubber cylinder | -0.84 | -1.54 | 0.01 | 0.01 | 0.01 |
| 64 | 1 cyan metal cylinder | 0.35 | -1.17 | 0.84 | 0.28 | 0.88 |
| 64 | 3 brown metal sphere | 1.14 | 1.51 | -0.00 | -0.00 | 0.00 |
| 64 | 4 cyan rubber sphere | 0.18 | -0.23 | 0.00 | -0.00 | 0.00 |
| 64 | 5 cyan metal sphere | -0.54 | 0.64 | 0.00 | -0.00 | 0.00 |
| 72 | 0 yellow rubber cylinder | -0.84 | -1.54 | -0.00 | -0.00 | 0.00 |
| 72 | 1 cyan metal cylinder | 0.53 | -1.11 | 0.76 | 0.25 | 0.80 |
| 72 | 3 brown metal sphere | 1.14 | 1.51 | 0.00 | -0.00 | 0.00 |
| 72 | 4 cyan rubber sphere | 0.18 | -0.23 | 0.00 | -0.00 | 0.00 |
| 72 | 5 cyan metal sphere | -0.54 | 0.64 | 0.00 | -0.00 | 0.00 |
| 80 | 0 yellow rubber cylinder | -0.84 | -1.54 | -0.00 | -0.00 | 0.00 |
| 80 | 1 cyan metal cylinder | 0.69 | -1.05 | 0.69 | 0.23 | 0.73 |
| 80 | 3 brown metal sphere | 1.14 | 1.51 | -0.00 | -0.00 | 0.00 |
| 80 | 4 cyan rubber sphere | 0.18 | -0.23 | 0.00 | -0.00 | 0.00 |
| 80 | 5 cyan metal sphere | -0.54 | 0.64 | 0.00 | -0.00 | 0.00 |
| 87 | 0 yellow rubber cylinder | -0.84 | -1.54 | -0.00 | -0.00 | 0.00 |
| 87 | 1 cyan metal cylinder | 0.82 | -1.01 | 0.63 | 0.21 | 0.66 |
| 87 | 2 yellow metal sphere | 1.17 | -1.95 | -0.15 | 2.09 | 2.10 |
| 87 | 3 brown metal sphere | 1.14 | 1.51 | -0.00 | -0.00 | 0.00 |
| 87 | 4 cyan rubber sphere | 0.18 | -0.23 | 0.00 | -0.00 | 0.00 |
| 87 | 5 cyan metal sphere | -0.54 | 0.64 | 0.00 | -0.00 | 0.00 |
| 88 | 0 yellow rubber cylinder | -0.84 | -1.54 | -0.00 | -0.00 | 0.00 |
| 88 | 1 cyan metal cylinder | 0.84 | -1.01 | 0.62 | 0.20 | 0.65 |
| 88 | 2 yellow metal sphere | 1.17 | -1.89 | -0.15 | 2.09 | 2.09 |
| 88 | 3 brown metal sphere | 1.14 | 1.51 | -0.00 | -0.00 | 0.00 |
| 88 | 4 cyan rubber sphere | 0.18 | -0.23 | 0.00 | -0.00 | 0.00 |
| 88 | 5 cyan metal sphere | -0.54 | 0.64 | 0.00 | -0.00 | 0.00 |
| 96 | 0 yellow rubber cylinder | -0.84 | -1.54 | -0.00 | -0.00 | 0.00 |
| 96 | 1 cyan metal cylinder | 0.97 | -0.96 | 0.55 | 0.18 | 0.58 |
| 96 | 2 yellow metal sphere | 1.14 | -1.43 | -0.14 | 2.03 | 2.03 |
| 96 | 3 brown metal sphere | 1.14 | 1.51 | -0.00 | -0.00 | 0.00 |
| 96 | 4 cyan rubber sphere | 0.18 | -0.23 | 0.00 | -0.00 | 0.00 |
| 96 | 5 cyan metal sphere | -0.54 | 0.64 | 0.00 | -0.00 | 0.00 |
| 98 | 0 yellow rubber cylinder | -0.84 | -1.54 | -0.00 | -0.00 | 0.00 |
| 98 | 1 cyan metal cylinder | 1.00 | -0.94 | 0.04 | 1.63 | 1.63 |

| frame | object | x | y | vx | vy | speed |
|---|---|---|---|---|---|---|
| 98 | 2 yellow metal sphere | 1.13 | -1.33 | 0.28 | 0.76 | 0.81 |
| 98 | 3 brown metal sphere | 1.14 | 1.51 | -0.00 | -0.00 | 0.00 |
| 98 | 4 cyan rubber sphere | 0.18 | -0.23 | 0.00 | -0.00 | 0.00 |
| 98 | 5 cyan metal sphere | -0.54 | 0.64 | 0.00 | -0.00 | 0.00 |
| 104 | 0 yellow rubber cylinder | -0.84 | -1.54 | -0.00 | -0.00 | 0.00 |
| 104 | 1 cyan metal cylinder | 1.00 | -0.67 | 0.04 | 1.56 | 1.56 |
| 104 | 2 yellow metal sphere | 1.18 | -1.20 | 0.28 | 0.75 | 0.80 |
| 104 | 3 brown metal sphere | 1.14 | 1.51 | -0.00 | -0.00 | 0.00 |
| 104 | 4 cyan rubber sphere | 0.18 | -0.23 | 0.00 | -0.00 | 0.00 |
| 104 | 5 cyan metal sphere | -0.54 | 0.64 | 0.00 | -0.00 | 0.00 |
| 112 | 0 yellow rubber cylinder | -0.84 | -1.54 | -0.00 | 0.00 | 0.00 |
| 112 | 1 cyan metal cylinder | 1.01 | -0.34 | 0.04 | 1.46 | 1.46 |
| 112 | 2 yellow metal sphere | 1.24 | -1.03 | 0.27 | 0.74 | 0.79 |
| 112 | 3 brown metal sphere | 1.14 | 1.51 | -0.00 | -0.00 | 0.00 |
| 112 | 4 cyan rubber sphere | 0.18 | -0.23 | 0.00 | -0.00 | 0.00 |
| 112 | 5 cyan metal sphere | -0.54 | 0.64 | 0.00 | -0.00 | 0.00 |
| 120 | 0 yellow rubber cylinder | -0.84 | -1.54 | -0.00 | -0.00 | 0.00 |
| 120 | 1 cyan metal cylinder | 1.02 | -0.02 | 0.04 | 1.36 | 1.36 |
| 120 | 2 yellow metal sphere | 1.30 | -0.87 | 0.27 | 0.73 | 0.78 |
| 120 | 3 brown metal sphere | 1.14 | 1.51 | -0.00 | -0.00 | 0.00 |
| 120 | 4 cyan rubber sphere | 0.18 | -0.23 | 0.00 | -0.00 | 0.00 |
| 120 | 5 cyan metal sphere | -0.54 | 0.64 | 0.00 | -0.00 | 0.00 |
| 127 | 0 yellow rubber cylinder | -0.84 | -1.54 | -0.00 | 0.00 | 0.00 |
| 127 | 1 cyan metal cylinder | 1.03 | 0.24 | 0.03 | 1.28 | 1.28 |
| 127 | 2 yellow metal sphere | 1.35 | -0.73 | 0.26 | 0.72 | 0.77 |
| 127 | 3 brown metal sphere | 1.14 | 1.51 | -0.00 | -0.00 | 0.00 |
| 127 | 4 cyan rubber sphere | 0.18 | -0.23 | 0.00 | -0.00 | 0.00 |
| 127 | 5 cyan metal sphere | -0.54 | 0.64 | 0.00 | -0.00 | 0.00 |

# Video 12123

## Video 12123: Objects

Object 0: purple rubber cube (color=purple, material=rubber, shape=cube)

Object 1: gray rubber cylinder (color=gray, material=rubber, shape=cylinder)

Object 2: cyan metal cylinder (color=cyan, material=metal, shape=cylinder)

Object 3: green metal sphere (color=green, material=metal, shape=sphere)

Object 4: blue rubber sphere (color=blue, material=rubber, shape=sphere)

Object 5: blue rubber cylinder (color=blue, material=rubber, shape=cylinder)

## Video 12123: Events

Frame 0: Object 1 (gray rubber cylinder) is in view when the video begins.

Frame 0: Object 4 (blue rubber sphere) is in view when the video begins.

Frame 0: Object 5 (blue rubber cylinder) is in view when the video begins.

Frame 18: Object 2 (cyan metal cylinder) enters the scene.

Frame 28: Object 0 (purple rubber cube) enters the scene.

Frame 50: Object 0 (purple rubber cube) collides with Object 1 (gray rubber cylinder).

Frame 81: Object 1 (gray rubber cylinder) collides with Object 2 (cyan metal cylinder).

Frame 96: Object 3 (green metal sphere) enters the scene.

Frame 127: The video ends.

## Video 12123: Motion log

| frame | object | x | y | vx | vy | speed |
|---|---|---|---|---|---|---|
| 0 | 1 gray rubber cylinder | -1.72 | -1.55 | 0.00 | 0.00 | 0.00 |
| 0 | 4 blue rubber sphere | -0.58 | -0.76 | 0.00 | 0.00 | 0.00 |
| 0 | 5 blue rubber cylinder | -0.88 | 0.01 | 0.00 | 0.00 | 0.00 |
| 8 | 1 gray rubber cylinder | -1.72 | -1.55 | 0.00 | 0.00 | 0.00 |
| 8 | 4 blue rubber sphere | -0.58 | -0.76 | 0.00 | -0.00 | 0.00 |
| 8 | 5 blue rubber cylinder | -0.88 | 0.01 | 0.00 | 0.00 | 0.00 |
| 16 | 1 gray rubber cylinder | -1.72 | -1.55 | 0.00 | 0.00 | 0.00 |
| 16 | 4 blue rubber sphere | -0.58 | -0.76 | 0.00 | -0.00 | 0.00 |
| 16 | 5 blue rubber cylinder | -0.88 | 0.01 | 0.00 | 0.00 | 0.00 |
| 18 | 1 gray rubber cylinder | -1.72 | -1.55 | 0.00 | 0.00 | 0.00 |
| 18 | 2 cyan metal cylinder | -4.37 | -4.19 | 1.30 | 2.10 | 2.47 |
| 18 | 4 blue rubber sphere | -0.58 | -0.76 | 0.00 | -0.00 | 0.00 |
| 18 | 5 blue rubber cylinder | -0.88 | 0.01 | 0.00 | 0.00 | 0.00 |
| 24 | 1 gray rubber cylinder | -1.72 | -1.55 | 0.00 | 0.00 | 0.00 |
| 24 | 2 cyan metal cylinder | -4.15 | -3.85 | 1.25 | 2.01 | 2.37 |
| 24 | 4 blue rubber sphere | -0.58 | -0.76 | 0.00 | -0.00 | 0.00 |
| 24 | 5 blue rubber cylinder | -0.88 | 0.01 | 0.00 | 0.00 | 0.00 |
| 28 | 0 purple rubber cube | -0.33 | -2.61 | -1.85 | 1.32 | 2.27 |
| 28 | 1 gray rubber cylinder | -1.72 | -1.55 | 0.00 | 0.00 | 0.00 |
| 28 | 2 cyan metal cylinder | -4.02 | -3.63 | 1.21 | 1.96 | 2.30 |
| 28 | 4 blue rubber sphere | -0.58 | -0.76 | 0.00 | -0.00 | 0.00 |
| 28 | 5 blue rubber cylinder | -0.88 | 0.01 | 0.00 | 0.00 | 0.00 |
| 32 | 0 purple rubber cube | -0.53 | -2.47 | -1.80 | 1.28 | 2.21 |

| frame | object | x | y | vx | vy | speed |
|---|---|---|---|---|---|---|
| 32 | 1 gray rubber cylinder | -1.72 | -1.55 | 0.00 | 0.00 | 0.00 |
| 32 | 2 cyan metal cylinder | -3.88 | -3.41 | 1.18 | 1.90 | 2.24 |
| 32 | 4 blue rubber sphere | -0.58 | -0.76 | 0.00 | -0.00 | 0.00 |
| 32 | 5 blue rubber cylinder | -0.88 | 0.01 | 0.00 | 0.00 | 0.00 |
| 40 | 0 purple rubber cube | -0.92 | -2.19 | -1.70 | 1.21 | 2.08 |
| 40 | 1 gray rubber cylinder | -1.72 | -1.55 | 0.00 | 0.00 | 0.00 |
| 40 | 2 cyan metal cylinder | -3.62 | -3.00 | 1.11 | 1.79 | 2.11 |
| 40 | 4 blue rubber sphere | -0.58 | -0.76 | 0.00 | -0.00 | 0.00 |
| 40 | 5 blue rubber cylinder | -0.88 | 0.01 | 0.00 | 0.00 | 0.00 |
| 48 | 0 purple rubber cube | -1.29 | -1.93 | -1.60 | 1.14 | 1.96 |
| 48 | 1 gray rubber cylinder | -1.72 | -1.55 | 0.00 | 0.00 | 0.00 |
| 48 | 2 cyan metal cylinder | -3.38 | -2.61 | 1.05 | 1.69 | 1.99 |
| 48 | 4 blue rubber sphere | -0.58 | -0.76 | 0.00 | -0.00 | 0.00 |
| 48 | 5 blue rubber cylinder | -0.88 | 0.01 | 0.00 | 0.00 | 0.00 |
| 50 | 0 purple rubber cube | -1.38 | -1.87 | -0.82 | 0.34 | 0.89 |
| 50 | 1 gray rubber cylinder | -1.72 | -1.55 | -0.43 | 0.45 | 0.62 |
| 50 | 2 cyan metal cylinder | -3.32 | -2.51 | 1.03 | 1.66 | 1.96 |
| 50 | 4 blue rubber sphere | -0.58 | -0.76 | 0.00 | -0.00 | 0.00 |
| 50 | 5 blue rubber cylinder | -0.88 | 0.01 | 0.00 | -0.00 | 0.00 |
| 56 | 0 purple rubber cube | -1.45 | -1.87 | -0.41 | -0.04 | 0.42 |
| 56 | 1 gray rubber cylinder | -1.82 | -1.44 | -0.60 | 0.63 | 0.87 |
| 56 | 2 cyan metal cylinder | -3.15 | -2.24 | 0.99 | 1.59 | 1.87 |
| 56 | 4 blue rubber sphere | -0.58 | -0.76 | 0.00 | -0.00 | 0.00 |
| 56 | 5 blue rubber cylinder | -0.88 | 0.01 | 0.00 | 0.00 | 0.00 |
| 64 | 0 purple rubber cube | -1.53 | -1.88 | -0.35 | -0.03 | 0.35 |
| 64 | 1 gray rubber cylinder | -1.95 | -1.30 | -0.55 | 0.57 | 0.79 |
| 64 | 2 cyan metal cylinder | -2.94 | -1.90 | 0.93 | 1.49 | 1.76 |
| 64 | 4 blue rubber sphere | -0.58 | -0.76 | 0.00 | -0.00 | 0.00 |
| 64 | 5 blue rubber cylinder | -0.88 | 0.01 | 0.00 | 0.00 | 0.00 |
| 72 | 0 purple rubber cube | -1.60 | -1.89 | -0.28 | -0.02 | 0.28 |
| 72 | 1 gray rubber cylinder | -2.07 | -1.18 | -0.50 | 0.52 | 0.72 |
| 72 | 2 cyan metal cylinder | -2.74 | -1.57 | 0.88 | 1.40 | 1.65 |
| 72 | 4 blue rubber sphere | -0.58 | -0.76 | 0.00 | -0.00 | 0.00 |
| 72 | 5 blue rubber cylinder | -0.88 | 0.01 | 0.00 | 0.00 | 0.00 |
| 80 | 0 purple rubber cube | -1.66 | -1.89 | -0.21 | -0.02 | 0.21 |
| 80 | 1 gray rubber cylinder | -2.17 | -1.07 | -0.44 | 0.46 | 0.64 |
| 80 | 2 cyan metal cylinder | -2.55 | -1.27 | 0.82 | 1.32 | 1.55 |
| 80 | 4 blue rubber sphere | -0.58 | -0.76 | 0.00 | -0.00 | 0.00 |
| 80 | 5 blue rubber cylinder | -0.88 | 0.01 | 0.00 | -0.00 | 0.00 |
| 81 | 0 purple rubber cube | -1.66 | -1.89 | -0.20 | -0.02 | 0.20 |
| 81 | 1 gray rubber cylinder | -2.18 | -1.05 | 0.31 | 0.84 | 0.89 |
| 81 | 2 cyan metal cylinder | -2.54 | -1.24 | -0.36 | 0.70 | 0.79 |
| 81 | 4 blue rubber sphere | -0.58 | -0.76 | 0.00 | -0.00 | 0.00 |
| 81 | 5 blue rubber cylinder | -0.88 | 0.01 | -0.00 | 0.00 | 0.00 |
| 88 | 0 purple rubber cube | -1.70 | -1.89 | -0.14 | -0.01 | 0.14 |
| 88 | 1 gray rubber cylinder | -2.12 | -0.89 | 0.29 | 0.77 | 0.83 |
| 88 | 2 cyan metal cylinder | -2.61 | -1.11 | -0.33 | 0.65 | 0.72 |
| 88 | 4 blue rubber sphere | -0.58 | -0.76 | 0.00 | -0.00 | 0.00 |
| 88 | 5 blue rubber cylinder | -0.88 | 0.01 | 0.00 | 0.00 | 0.00 |
| 96 | 0 purple rubber cube | -1.72 | -1.90 | -0.08 | -0.01 | 0.08 |
| 96 | 1 gray rubber cylinder | -2.06 | -0.73 | 0.26 | 0.70 | 0.75 |
| 96 | 2 cyan metal cylinder | -2.68 | -0.97 | -0.30 | 0.58 | 0.65 |
| 96 | 3 green metal sphere | -0.88 | 2.76 | -0.70 | -1.78 | 1.92 |
| 96 | 4 blue rubber sphere | -0.58 | -0.76 | 0.00 | -0.00 | 0.00 |
| 96 | 5 blue rubber cylinder | -0.88 | 0.01 | 0.00 | 0.00 | 0.00 |
| 104 | 0 purple rubber cube | -1.73 | -1.90 | -0.01 | -0.00 | 0.01 |
| 104 | 1 gray rubber cylinder | -2.00 | -0.58 | 0.23 | 0.63 | 0.67 |

| frame | object | x | y | vx | vy | speed |
|---|---|---|---|---|---|---|
| 104 | 2 cyan metal cylinder | -2.74 | -0.85 | -0.26 | 0.51 | 0.58 |
| 104 | 3 green metal sphere | -1.04 | 2.37 | -0.68 | -1.74 | 1.87 |
| 104 | 4 blue rubber sphere | -0.58 | -0.76 | 0.00 | -0.00 | 0.00 |
| 104 | 5 blue rubber cylinder | -0.88 | 0.01 | -0.00 | 0.00 | 0.00 |
| 112 | 0 purple rubber cube | -1.73 | -1.90 | -0.00 | 0.00 | 0.00 |
| 112 | 1 gray rubber cylinder | -1.95 | -0.45 | 0.21 | 0.56 | 0.60 |
| 112 | 2 cyan metal cylinder | -2.80 | -0.74 | -0.23 | 0.45 | 0.50 |
| 112 | 3 green metal sphere | -1.19 | 1.98 | -0.67 | -1.69 | 1.82 |
| 112 | 4 blue rubber sphere | -0.58 | -0.76 | 0.00 | -0.00 | 0.00 |
| 112 | 5 blue rubber cylinder | -0.88 | 0.01 | 0.00 | 0.00 | 0.00 |
| 120 | 0 purple rubber cube | -1.73 | -1.90 | -0.00 | 0.00 | 0.00 |
| 120 | 1 gray rubber cylinder | -1.91 | -0.33 | 0.18 | 0.49 | 0.52 |
| 120 | 2 cyan metal cylinder | -2.84 | -0.65 | -0.20 | 0.39 | 0.43 |
| 120 | 3 green metal sphere | -1.34 | 1.61 | -0.65 | -1.65 | 1.77 |
| 120 | 4 blue rubber sphere | -0.58 | -0.76 | 0.00 | -0.00 | 0.00 |
| 120 | 5 blue rubber cylinder | -0.88 | 0.01 | 0.00 | 0.00 | 0.00 |
| 127 | 0 purple rubber cube | -1.73 | -1.90 | -0.00 | 0.00 | 0.00 |
| 127 | 1 gray rubber cylinder | -1.88 | -0.24 | 0.16 | 0.43 | 0.46 |
| 127 | 2 cyan metal cylinder | -2.88 | -0.58 | -0.17 | 0.33 | 0.37 |
| 127 | 3 green metal sphere | -1.46 | 1.29 | -0.64 | -1.62 | 1.74 |
| 127 | 4 blue rubber sphere | -0.58 | -0.76 | 0.00 | -0.00 | 0.00 |
| 127 | 5 blue rubber cylinder | -0.88 | 0.01 | 0.00 | 0.00 | 0.00 |

# Video 12564

## Video 12564: Objects

Object 0: red metal cube (color=red, material=metal, shape=cube)

Object 1: gray rubber cylinder (color=gray, material=rubber, shape=cylinder)

Object 2: blue rubber cube (color=blue, material=rubber, shape=cube)

Object 3: red rubber sphere (color=red, material=rubber, shape=sphere)

## Video 12564: Events

Frame 0: Object 1 (gray rubber cylinder) is in view when the video begins.

Frame 2: Object 0 (red metal cube) enters the scene.

Frame 46: Object 0 (red metal cube) collides with Object 1 (gray rubber cylinder).

Frame 63: Object 2 (blue rubber cube) enters the scene.

Frame 89: Object 1 (gray rubber cylinder) collides with Object 2 (blue rubber cube).

Frame 99: Object 3 (red rubber sphere) enters the scene.

Frame 119: Object 0 (red metal cube) collides with Object 3 (red rubber sphere).

Frame 127: The video ends.

## Video 12564: Motion log

| frame | object | x | y | vx | vy | speed |
|---|---|---|---|---|---|---|
| 0 | 1 gray rubber cylinder | 0.23 | 0.60 | 0.00 | 0.00 | 0.00 |
| 2 | 0 red metal cube | -1.77 | 3.17 | 1.57 | -2.26 | 2.75 |
| 2 | 1 gray rubber cylinder | 0.23 | 0.60 | -0.00 | 0.00 | 0.00 |
| 8 | 0 red metal cube | -1.51 | 2.80 | 1.51 | -2.16 | 2.63 |
| 8 | 1 gray rubber cylinder | 0.23 | 0.60 | 0.00 | 0.00 | 0.00 |
| 16 | 0 red metal cube | -1.18 | 2.33 | 1.42 | -2.04 | 2.49 |
| 16 | 1 gray rubber cylinder | 0.23 | 0.60 | 0.00 | -0.00 | 0.00 |
| 24 | 0 red metal cube | -0.87 | 1.88 | 1.34 | -1.93 | 2.35 |
| 24 | 1 gray rubber cylinder | 0.23 | 0.60 | 0.00 | -0.00 | 0.00 |
| 32 | 0 red metal cube | -0.58 | 1.47 | 1.27 | -1.82 | 2.22 |
| 32 | 1 gray rubber cylinder | 0.23 | 0.60 | 0.00 | -0.00 | 0.00 |
| 40 | 0 red metal cube | -0.30 | 1.07 | 1.19 | -1.72 | 2.09 |
| 40 | 1 gray rubber cylinder | 0.23 | 0.60 | 0.00 | -0.00 | 0.00 |
| 46 | 0 red metal cube | -0.12 | 0.79 | 0.24 | -1.20 | 1.22 |
| 46 | 1 gray rubber cylinder | 0.24 | 0.60 | 0.84 | -0.40 | 0.93 |
| 48 | 0 red metal cube | -0.11 | 0.73 | 0.23 | -1.18 | 1.20 |
| 48 | 1 gray rubber cylinder | 0.28 | 0.57 | 0.83 | -0.40 | 0.92 |
| 56 | 0 red metal cube | -0.06 | 0.47 | 0.22 | -1.09 | 1.12 |
| 56 | 1 gray rubber cylinder | 0.46 | 0.49 | 0.75 | -0.36 | 0.84 |
| 63 | 0 red metal cube | -0.02 | 0.26 | 0.20 | -1.02 | 1.04 |
| 63 | 1 gray rubber cylinder | 0.60 | 0.42 | 0.69 | -0.33 | 0.77 |
| 63 | 2 blue rubber cube | 2.55 | 0.86 | -1.70 | -0.83 | 1.89 |
| 64 | 0 red metal cube | -0.01 | 0.24 | 0.20 | -1.01 | 1.03 |
| 64 | 1 gray rubber cylinder | 0.62 | 0.41 | 0.68 | -0.33 | 0.76 |
| 64 | 2 blue rubber cube | 2.51 | 0.84 | -1.69 | -0.82 | 1.88 |
| 72 | 0 red metal cube | 0.03 | 0.02 | 0.18 | -0.93 | 0.95 |
| 72 | 1 gray rubber cylinder | 0.77 | 0.34 | 0.61 | -0.29 | 0.68 |
| 72 | 2 blue rubber cube | 2.14 | 0.66 | -1.59 | -0.77 | 1.77 |

| frame | object | x | y | vx | vy | speed |
|---|---|---|---|---|---|---|
| 80 | 0 red metal cube | 0.07 | -0.18 | 0.16 | -0.85 | 0.87 |
| 80 | 1 gray rubber cylinder | 0.90 | 0.28 | 0.55 | -0.26 | 0.61 |
| 80 | 2 blue rubber cube | 1.80 | 0.49 | -1.49 | -0.73 | 1.66 |
| 88 | 0 red metal cube | 0.11 | -0.36 | 0.15 | -0.77 | 0.79 |
| 88 | 1 gray rubber cylinder | 1.01 | 0.22 | 0.48 | -0.23 | 0.53 |
| 88 | 2 blue rubber cube | 1.47 | 0.33 | -1.40 | -0.68 | 1.56 |
| 89 | 0 red metal cube | 0.11 | -0.38 | 0.15 | -0.76 | 0.78 |
| 89 | 1 gray rubber cylinder | 1.02 | 0.22 | 0.47 | -0.23 | 0.53 |
| 89 | 2 blue rubber cube | 1.43 | 0.31 | -1.39 | -0.68 | 1.55 |
| 96 | 0 red metal cube | 0.14 | -0.53 | 0.13 | -0.70 | 0.71 |
| 96 | 1 gray rubber cylinder | 0.94 | 0.12 | -0.42 | -0.50 | 0.65 |
| 96 | 2 blue rubber cube | 1.36 | 0.25 | -0.30 | -0.27 | 0.40 |
| 99 | 0 red metal cube | 0.15 | -0.58 | 0.13 | -0.67 | 0.68 |
| 99 | 1 gray rubber cylinder | 0.91 | 0.08 | -0.40 | -0.48 | 0.62 |
| 99 | 2 blue rubber cube | 1.34 | 0.23 | -0.28 | -0.25 | 0.38 |
| 99 | 3 red rubber sphere | 0.40 | -2.26 | 0.21 | 2.01 | 2.02 |
| 104 | 0 red metal cube | 0.17 | -0.67 | 0.12 | -0.62 | 0.64 |
| 104 | 1 gray rubber cylinder | 0.86 | 0.01 | -0.37 | -0.44 | 0.58 |
| 104 | 2 blue rubber cube | 1.30 | 0.20 | -0.25 | -0.22 | 0.33 |
| 104 | 3 red rubber sphere | 0.43 | -1.98 | 0.21 | 1.98 | 1.99 |
| 112 | 0 red metal cube | 0.19 | -0.81 | 0.10 | -0.55 | 0.56 |
| 112 | 1 gray rubber cylinder | 0.78 | -0.08 | -0.32 | -0.39 | 0.50 |
| 112 | 2 blue rubber cube | 1.25 | 0.16 | -0.20 | -0.17 | 0.26 |
| 112 | 3 red rubber sphere | 0.48 | -1.54 | 0.20 | 1.93 | 1.94 |
| 119 | 0 red metal cube | 0.21 | -0.91 | -0.21 | -0.25 | 0.33 |
| 119 | 1 gray rubber cylinder | 0.72 | -0.15 | -0.28 | -0.34 | 0.44 |
| 119 | 2 blue rubber cube | 1.22 | 0.13 | -0.15 | -0.14 | 0.20 |
| 119 | 3 red rubber sphere | 0.52 | -1.17 | 0.44 | 1.69 | 1.75 |
| 120 | 0 red metal cube | 0.19 | -0.91 | -0.51 | -0.01 | 0.51 |
| 120 | 1 gray rubber cylinder | 0.71 | -0.16 | -0.28 | -0.33 | 0.43 |
| 120 | 2 blue rubber cube | 1.21 | 0.12 | -0.15 | -0.13 | 0.20 |
| 120 | 3 red rubber sphere | 0.53 | -1.13 | 0.69 | 1.49 | 1.64 |
| 127 | 0 red metal cube | 0.10 | -0.91 | -0.45 | -0.00 | 0.45 |
| 127 | 1 gray rubber cylinder | 0.66 | -0.22 | -0.24 | -0.29 | 0.37 |
| 127 | 2 blue rubber cube | 1.19 | 0.10 | -0.10 | -0.09 | 0.14 |
| 127 | 3 red rubber sphere | 0.67 | -0.84 | 0.68 | 1.45 | 1.60 |

# Video 12821

## Video 12821: Objects

Object 0: yellow metal cylinder (color=yellow, material=metal, shape=cylinder)

Object 1: cyan rubber cube (color=cyan, material=rubber, shape=cube)

Object 2: green rubber sphere (color=green, material=rubber, shape=sphere)

Object 3: gray metal sphere (color=gray, material=metal, shape=sphere)

Object 4: red metal sphere (color=red, material=metal, shape=sphere)

## Video 12821: Events

Frame 0: Object 0 (yellow metal cylinder) is in view when the video begins.

Frame 0: Object 1 (cyan rubber cube) is in view when the video begins.

Frame 0: Object 3 (gray metal sphere) is in view when the video begins.

Frame 0: Object 4 (red metal sphere) is in view when the video begins.

Frame 38: Object 0 (yellow metal cylinder) collides with Object 1 (cyan rubber cube).

Frame 92: Object 0 (yellow metal cylinder) collides with Object 3 (gray metal sphere).

Frame 99: Object 2 (green rubber sphere) enters the scene.

Frame 118: Object 1 (cyan rubber cube) collides with Object 2 (green rubber sphere).

Frame 127: The video ends.

## Video 12821: Motion log

| frame | object | x | y | vx | vy | speed |
|---|---|---|---|---|---|---|
| 0 | 0 yellow metal cylinder | -2.57 | 2.11 | 2.22 | -1.82 | 2.87 |
| 0 | 1 cyan rubber cube | -0.05 | 0.33 | 0.00 | 0.00 | 0.00 |
| 0 | 3 gray metal sphere | 0.28 | -1.02 | 0.00 | 0.00 | 0.00 |
| 0 | 4 red metal sphere | -2.23 | 0.17 | 0.00 | 0.00 | 0.00 |
| 8 | 0 yellow metal cylinder | -2.08 | 1.71 | 2.10 | -1.72 | 2.71 |
| 8 | 1 cyan rubber cube | -0.05 | 0.33 | -0.00 | 0.00 | 0.00 |
| 8 | 3 gray metal sphere | 0.28 | -1.02 | -0.00 | 0.00 | 0.00 |
| 8 | 4 red metal sphere | -2.23 | 0.17 | -0.00 | 0.00 | 0.00 |
| 16 | 0 yellow metal cylinder | -1.63 | 1.34 | 1.98 | -1.62 | 2.56 |
| 16 | 1 cyan rubber cube | -0.05 | 0.33 | -0.00 | 0.00 | 0.00 |
| 16 | 3 gray metal sphere | 0.28 | -1.02 | 0.00 | -0.00 | 0.00 |
| 16 | 4 red metal sphere | -2.23 | 0.17 | 0.00 | 0.00 | 0.00 |
| 24 | 0 yellow metal cylinder | -1.19 | 0.99 | 1.88 | -1.52 | 2.42 |
| 24 | 1 cyan rubber cube | -0.05 | 0.33 | -0.00 | 0.00 | 0.00 |
| 24 | 3 gray metal sphere | 0.28 | -1.02 | 0.00 | -0.00 | 0.00 |
| 24 | 4 red metal sphere | -2.23 | 0.17 | 0.00 | 0.00 | 0.00 |
| 32 | 0 yellow metal cylinder | -0.79 | 0.66 | 1.77 | -1.44 | 2.28 |
| 32 | 1 cyan rubber cube | -0.05 | 0.33 | -0.00 | 0.00 | 0.00 |
| 32 | 3 gray metal sphere | 0.28 | -1.02 | 0.00 | -0.00 | 0.00 |
| 32 | 4 red metal sphere | -2.23 | 0.17 | 0.00 | 0.00 | 0.00 |
| 38 | 0 yellow metal cylinder | -0.50 | 0.42 | 0.64 | -0.91 | 1.11 |
| 38 | 1 cyan rubber cube | -0.05 | 0.33 | 1.12 | -0.50 | 1.23 |
| 38 | 3 gray metal sphere | 0.28 | -1.02 | 0.00 | -0.00 | 0.00 |
| 38 | 4 red metal sphere | -2.23 | 0.17 | 0.00 | 0.00 | 0.00 |
| 40 | 0 yellow metal cylinder | -0.47 | 0.37 | 0.63 | -0.89 | 1.09 |

| frame | object | x | y | vx | vy | speed |
|---|---|---|---|---|---|---|
| 40 | 1 cyan rubber cube | 0.01 | 0.30 | 1.11 | -0.50 | 1.21 |
| 40 | 3 gray metal sphere | 0.28 | -1.02 | 0.00 | -0.00 | 0.00 |
| 40 | 4 red metal sphere | -2.23 | 0.17 | 0.00 | 0.00 | 0.00 |
| 48 | 0 yellow metal cylinder | -0.33 | 0.18 | 0.58 | -0.82 | 1.00 |
| 48 | 1 cyan rubber cube | 0.25 | 0.19 | 1.03 | -0.47 | 1.13 |
| 48 | 3 gray metal sphere | 0.28 | -1.02 | 0.00 | -0.00 | 0.00 |
| 48 | 4 red metal sphere | -2.23 | 0.17 | 0.00 | 0.00 | 0.00 |
| 56 | 0 yellow metal cylinder | -0.21 | 0.00 | 0.53 | -0.76 | 0.92 |
| 56 | 1 cyan rubber cube | 0.48 | 0.09 | 0.95 | -0.43 | 1.05 |
| 56 | 3 gray metal sphere | 0.28 | -1.02 | 0.00 | -0.00 | 0.00 |
| 56 | 4 red metal sphere | -2.23 | 0.17 | 0.00 | 0.00 | 0.00 |
| 64 | 0 yellow metal cylinder | -0.09 | -0.16 | 0.48 | -0.69 | 0.84 |
| 64 | 1 cyan rubber cube | 0.68 | 0.00 | 0.88 | -0.39 | 0.96 |
| 64 | 3 gray metal sphere | 0.28 | -1.02 | 0.00 | -0.00 | 0.00 |
| 64 | 4 red metal sphere | -2.23 | 0.17 | 0.00 | 0.00 | 0.00 |
| 72 | 0 yellow metal cylinder | 0.01 | -0.30 | 0.44 | -0.63 | 0.77 |
| 72 | 1 cyan rubber cube | 0.87 | -0.08 | 0.81 | -0.36 | 0.88 |
| 72 | 3 gray metal sphere | 0.28 | -1.02 | 0.00 | -0.00 | 0.00 |
| 72 | 4 red metal sphere | -2.23 | 0.17 | 0.00 | 0.00 | 0.00 |
| 80 | 0 yellow metal cylinder | 0.10 | -0.44 | 0.39 | -0.56 | 0.69 |
| 80 | 1 cyan rubber cube | 1.04 | -0.16 | 0.74 | -0.33 | 0.80 |
| 80 | 3 gray metal sphere | 0.28 | -1.02 | 0.00 | -0.00 | 0.00 |
| 80 | 4 red metal sphere | -2.23 | 0.17 | 0.00 | 0.00 | 0.00 |
| 88 | 0 yellow metal cylinder | 0.19 | -0.56 | 0.35 | -0.50 | 0.61 |
| 88 | 1 cyan rubber cube | 1.20 | -0.23 | 0.67 | -0.29 | 0.73 |
| 88 | 3 gray metal sphere | 0.28 | -1.02 | 0.00 | -0.00 | 0.00 |
| 88 | 4 red metal sphere | -2.23 | 0.17 | 0.00 | 0.00 | 0.00 |
| 92 | 0 yellow metal cylinder | 0.22 | -0.61 | 0.33 | -0.47 | 0.58 |
| 92 | 1 cyan rubber cube | 1.27 | -0.26 | 0.63 | -0.28 | 0.69 |
| 92 | 3 gray metal sphere | 0.28 | -1.02 | 0.00 | -0.00 | 0.00 |
| 92 | 4 red metal sphere | -2.23 | 0.17 | 0.00 | 0.00 | 0.00 |
| 96 | 0 yellow metal cylinder | 0.26 | -0.64 | 0.26 | -0.14 | 0.30 |
| 96 | 1 cyan rubber cube | 1.34 | -0.29 | 0.60 | -0.26 | 0.65 |
| 96 | 3 gray metal sphere | 0.29 | -1.04 | 0.02 | -0.18 | 0.18 |
| 96 | 4 red metal sphere | -2.23 | 0.17 | 0.00 | 0.00 | 0.00 |
| 99 | 0 yellow metal cylinder | 0.28 | -0.65 | 0.24 | -0.13 | 0.27 |
| 99 | 1 cyan rubber cube | 1.39 | -0.31 | 0.57 | -0.25 | 0.62 |
| 99 | 2 green rubber sphere | 2.43 | 0.64 | -1.43 | -1.26 | 1.90 |
| 99 | 3 gray metal sphere | 0.29 | -1.05 | 0.02 | -0.18 | 0.18 |
| 99 | 4 red metal sphere | -2.23 | 0.17 | 0.00 | 0.00 | 0.00 |
| 104 | 0 yellow metal cylinder | 0.31 | -0.66 | 0.20 | -0.11 | 0.23 |
| 104 | 1 cyan rubber cube | 1.47 | -0.35 | 0.53 | -0.23 | 0.58 |
| 104 | 2 green rubber sphere | 2.23 | 0.46 | -1.40 | -1.24 | 1.87 |
| 104 | 3 gray metal sphere | 0.29 | -1.08 | 0.02 | -0.18 | 0.18 |
| 104 | 4 red metal sphere | -2.23 | 0.17 | 0.00 | 0.00 | 0.00 |
| 112 | 0 yellow metal cylinder | 0.35 | -0.68 | 0.14 | -0.08 | 0.16 |
| 112 | 1 cyan rubber cube | 1.58 | -0.40 | 0.46 | -0.20 | 0.51 |
| 112 | 2 green rubber sphere | 1.92 | 0.19 | -1.37 | -1.21 | 1.83 |
| 112 | 3 gray metal sphere | 0.30 | -1.12 | 0.02 | -0.18 | 0.18 |
| 112 | 4 red metal sphere | -2.23 | 0.17 | 0.00 | 0.00 | 0.00 |
| 118 | 0 yellow metal cylinder | 0.37 | -0.69 | 0.10 | -0.05 | 0.11 |
| 118 | 1 cyan rubber cube | 1.65 | -0.44 | 0.09 | -0.76 | 0.76 |
| 118 | 2 green rubber sphere | 1.70 | -0.01 | -1.11 | -0.78 | 1.36 |
| 118 | 3 gray metal sphere | 0.30 | -1.15 | 0.02 | -0.18 | 0.18 |
| 118 | 4 red metal sphere | -2.23 | 0.17 | 0.00 | 0.00 | 0.00 |
| 120 | 0 yellow metal cylinder | 0.37 | -0.70 | 0.09 | -0.04 | 0.10 |
| 120 | 1 cyan rubber cube | 1.65 | -0.48 | 0.09 | -0.74 | 0.75 |

| frame | object | x | y | vx | vy | speed |
|---|---|---|---|---|---|---|
| 120 | 2 green rubber sphere | 1.64 | -0.05 | -1.10 | -0.77 | 1.35 |
| 120 | 3 gray metal sphere | 0.30 | -1.16 | 0.02 | -0.18 | 0.18 |
| 120 | 4 red metal sphere | -2.23 | 0.17 | 0.00 | 0.00 | 0.00 |
| 127 | 0 yellow metal cylinder | 0.39 | -0.70 | 0.03 | -0.02 | 0.04 |
| 127 | 1 cyan rubber cube | 1.67 | -0.62 | 0.09 | -0.69 | 0.69 |
| 127 | 2 green rubber sphere | 1.42 | -0.20 | -1.08 | -0.76 | 1.32 |
| 127 | 3 gray metal sphere | 0.30 | -1.19 | 0.02 | -0.18 | 0.18 |
| 127 | 4 red metal sphere | -2.23 | 0.17 | 0.00 | 0.00 | 0.00 |

# Video 13177

## Video 13177: Objects

Object 0: gray metal cube (color=gray, material=metal, shape=cube)

Object 1: cyan rubber cube (color=cyan, material=rubber, shape=cube)

Object 2: purple rubber cylinder (color=purple, material=rubber, shape=cylinder)

Object 3: green rubber cylinder (color=green, material=rubber, shape=cylinder)

## Video 13177: Events

Frame 0: Object 1 (cyan rubber cube) is in view when the video begins.

Frame 1: Object 0 (gray metal cube) enters the scene.

Frame 31: Object 2 (purple rubber cylinder) enters the scene.

Frame 43: Object 0 (gray metal cube) collides with Object 1 (cyan rubber cube).

Frame 53: Object 0 (gray metal cube) collides with Object 2 (purple rubber cylinder).

Frame 110: Object 3 (green rubber cylinder) enters the scene.

Frame 127: The video ends.

## Video 13177: Motion log

| frame | object | x | y | vx | vy | speed |
|---|---|---|---|---|---|---|
| 0 | 1 cyan rubber cube | -0.04 | -0.71 | 0.00 | 0.00 | 0.00 |
| 1 | 0 gray metal cube | -2.02 | -3.32 | 1.66 | 2.23 | 2.78 |
| 1 | 1 cyan rubber cube | -0.04 | -0.71 | 0.00 | 0.00 | 0.00 |
| 8 | 0 gray metal cube | -1.71 | -2.90 | 1.57 | 2.13 | 2.65 |
| 8 | 1 cyan rubber cube | -0.04 | -0.71 | 0.00 | 0.00 | 0.00 |
| 16 | 0 gray metal cube | -1.36 | -2.43 | 1.49 | 2.01 | 2.50 |
| 16 | 1 cyan rubber cube | -0.04 | -0.71 | 0.00 | -0.00 | 0.00 |
| 24 | 0 gray metal cube | -1.04 | -2.00 | 1.40 | 1.89 | 2.36 |
| 24 | 1 cyan rubber cube | -0.04 | -0.71 | 0.00 | -0.00 | 0.00 |
| 31 | 0 gray metal cube | -0.77 | -1.63 | 1.33 | 1.80 | 2.24 |
| 31 | 1 cyan rubber cube | -0.04 | -0.71 | 0.00 | -0.00 | 0.00 |
| 31 | 2 purple rubber cylinder | 0.89 | -2.11 | -1.26 | 1.82 | 2.22 |
| 32 | 0 gray metal cube | -0.73 | -1.58 | 1.32 | 1.79 | 2.22 |
| 32 | 1 cyan rubber cube | -0.04 | -0.71 | 0.00 | -0.00 | 0.00 |
| 32 | 2 purple rubber cylinder | 0.85 | -2.06 | -1.26 | 1.81 | 2.20 |
| 40 | 0 gray metal cube | -0.45 | -1.20 | 1.25 | 1.69 | 2.10 |
| 40 | 1 cyan rubber cube | -0.04 | -0.71 | 0.00 | -0.00 | 0.00 |
| 40 | 2 purple rubber cylinder | 0.58 | -1.67 | -1.18 | 1.71 | 2.08 |
| 43 | 0 gray metal cube | -0.34 | -1.06 | 1.22 | 1.65 | 2.05 |
| 43 | 1 cyan rubber cube | -0.04 | -0.71 | 0.00 | -0.00 | 0.00 |
| 43 | 2 purple rubber cylinder | 0.48 | -1.52 | -1.16 | 1.67 | 2.03 |
| 48 | 0 gray metal cube | -0.28 | -0.95 | 0.42 | 0.70 | 0.82 |
| 48 | 1 cyan rubber cube | 0.06 | -0.59 | 0.70 | 0.83 | 1.08 |
| 48 | 2 purple rubber cylinder | 0.32 | -1.29 | -1.11 | 1.61 | 1.96 |
| 53 | 0 gray metal cube | -0.23 | -0.86 | 0.39 | 0.67 | 0.77 |
| 53 | 1 cyan rubber cube | 0.15 | -0.48 | 0.67 | 0.79 | 1.03 |
| 53 | 2 purple rubber cylinder | 0.17 | -1.07 | -1.07 | 1.55 | 1.89 |
| 56 | 0 gray metal cube | -0.24 | -0.75 | -0.21 | 1.27 | 1.28 |
| 56 | 1 cyan rubber cube | 0.21 | -0.41 | 0.65 | 0.76 | 1.00 |

| frame | object | x | y | vx | vy | speed |
|---|---|---|---|---|---|---|
| 56 | 2 purple rubber cylinder | 0.13 | -1.00 | -0.44 | 0.88 | 0.98 |
| 64 | 0 gray metal cube | -0.29 | -0.48 | -0.19 | 1.18 | 1.20 |
| 64 | 1 cyan rubber cube | 0.35 | -0.25 | 0.59 | 0.70 | 0.92 |
| 64 | 2 purple rubber cylinder | 0.04 | -0.81 | -0.41 | 0.81 | 0.90 |
| 72 | 0 gray metal cube | -0.33 | -0.22 | -0.18 | 1.09 | 1.11 |
| 72 | 1 cyan rubber cube | 0.47 | -0.10 | 0.54 | 0.64 | 0.84 |
| 72 | 2 purple rubber cylinder | -0.05 | -0.64 | -0.37 | 0.73 | 0.82 |
| 80 | 0 gray metal cube | -0.37 | 0.01 | -0.16 | 1.01 | 1.02 |
| 80 | 1 cyan rubber cube | 0.59 | 0.04 | 0.49 | 0.58 | 0.76 |
| 80 | 2 purple rubber cylinder | -0.13 | -0.48 | -0.33 | 0.66 | 0.74 |
| 88 | 0 gray metal cube | -0.40 | 0.23 | -0.15 | 0.93 | 0.94 |
| 88 | 1 cyan rubber cube | 0.69 | 0.16 | 0.44 | 0.53 | 0.69 |
| 88 | 2 purple rubber cylinder | -0.20 | -0.34 | -0.30 | 0.60 | 0.67 |
| 96 | 0 gray metal cube | -0.44 | 0.43 | -0.14 | 0.85 | 0.86 |
| 96 | 1 cyan rubber cube | 0.79 | 0.27 | 0.39 | 0.47 | 0.61 |
| 96 | 2 purple rubber cylinder | -0.26 | -0.21 | -0.27 | 0.53 | 0.59 |
| 104 | 0 gray metal cube | -0.47 | 0.61 | -0.13 | 0.77 | 0.78 |
| 104 | 1 cyan rubber cube | 0.87 | 0.37 | 0.35 | 0.41 | 0.54 |
| 104 | 2 purple rubber cylinder | -0.32 | -0.10 | -0.23 | 0.47 | 0.52 |
| 110 | 0 gray metal cube | -0.49 | 0.73 | -0.12 | 0.71 | 0.72 |
| 110 | 1 cyan rubber cube | 0.92 | 0.44 | 0.31 | 0.37 | 0.48 |
| 110 | 2 purple rubber cylinder | -0.36 | -0.03 | -0.21 | 0.42 | 0.47 |
| 110 | 3 green rubber cylinder | 1.07 | 2.05 | 0.09 | -1.18 | 1.19 |
| 112 | 0 gray metal cube | -0.49 | 0.77 | -0.11 | 0.69 | 0.70 |
| 112 | 1 cyan rubber cube | 0.94 | 0.46 | 0.30 | 0.36 | 0.47 |
| 112 | 2 purple rubber cylinder | -0.37 | -0.00 | -0.20 | 0.40 | 0.45 |
| 112 | 3 green rubber cylinder | 1.07 | 1.99 | 0.09 | -1.16 | 1.16 |
| 120 | 0 gray metal cube | -0.52 | 0.92 | -0.10 | 0.62 | 0.63 |
| 120 | 1 cyan rubber cube | 1.00 | 0.53 | 0.25 | 0.30 | 0.40 |
| 120 | 2 purple rubber cylinder | -0.41 | 0.08 | -0.17 | 0.34 | 0.38 |
| 120 | 3 green rubber cylinder | 1.09 | 1.74 | 0.08 | -1.07 | 1.08 |
| 127 | 0 gray metal cube | -0.53 | 1.03 | -0.09 | 0.55 | 0.56 |
| 127 | 1 cyan rubber cube | 1.05 | 0.59 | 0.22 | 0.26 | 0.34 |
| 127 | 2 purple rubber cylinder | -0.44 | 0.14 | -0.14 | 0.29 | 0.32 |
| 127 | 3 green rubber cylinder | 1.11 | 1.53 | 0.08 | -1.00 | 1.00 |

# Video 13265

## Video 13265: Objects

Object 0: cyan rubber cylinder (color=cyan, material=rubber, shape=cylinder)

Object 1: red rubber cylinder (color=red, material=rubber, shape=cylinder)

Object 2: red rubber cube (color=red, material=rubber, shape=cube)

Object 3: cyan rubber cube (color=cyan, material=rubber, shape=cube)

Object 4: blue rubber cube (color=blue, material=rubber, shape=cube)

## Video 13265: Events

Frame 0: Object 2 (red rubber cube) is in view when the video begins.

Frame 0: Object 4 (blue rubber cube) is in view when the video begins.

Frame 12: Object 0 (cyan rubber cylinder) enters the scene.

Frame 12: Object 1 (red rubber cylinder) enters the scene.

Frame 15: Object 3 (cyan rubber cube) enters the scene.

Frame 39: Object 0 (cyan rubber cylinder) collides with Object 1 (red rubber cylinder).

Frame 55: Object 0 (cyan rubber cylinder) collides with Object 2 (red rubber cube).

Frame 127: The video ends.

## Video 13265: Motion log

| frame | object | x | y | vx | vy | speed |
|---|---|---|---|---|---|---|
| 0 | 2 red rubber cube | -0.68 | -0.18 | 0.00 | 0.00 | 0.00 |
| 0 | 4 blue rubber cube | 0.44 | 1.36 | 0.00 | 0.00 | 0.00 |
| 8 | 2 red rubber cube | -0.68 | -0.18 | -0.00 | 0.00 | 0.00 |
| 8 | 4 blue rubber cube | 0.44 | 1.36 | 0.00 | 0.00 | 0.00 |
| 12 | 0 cyan rubber cylinder | 0.51 | -2.28 | -0.16 | 2.59 | 2.60 |
| 12 | 1 red rubber cylinder | 2.54 | 0.10 | -2.71 | -0.41 | 2.74 |
| 12 | 2 red rubber cube | -0.68 | -0.18 | -0.00 | 0.00 | 0.00 |
| 12 | 4 blue rubber cube | 0.44 | 1.36 | 0.00 | 0.00 | 0.00 |
| 15 | 0 cyan rubber cylinder | 0.50 | -2.06 | -0.16 | 2.54 | 2.54 |
| 15 | 1 red rubber cylinder | 2.31 | 0.07 | -2.65 | -0.40 | 2.69 |
| 15 | 2 red rubber cube | -0.68 | -0.18 | -0.00 | 0.00 | 0.00 |
| 15 | 3 cyan rubber cube | -4.59 | 4.35 | 1.24 | -2.11 | 2.45 |
| 15 | 4 blue rubber cube | 0.44 | 1.36 | 0.00 | 0.00 | 0.00 |
| 16 | 0 cyan rubber cylinder | 0.50 | -1.99 | -0.16 | 2.52 | 2.52 |
| 16 | 1 red rubber cylinder | 2.24 | 0.06 | -2.63 | -0.40 | 2.67 |
| 16 | 2 red rubber cube | -0.68 | -0.18 | 0.00 | 0.00 | 0.00 |
| 16 | 3 cyan rubber cube | -4.55 | 4.29 | 1.24 | -2.09 | 2.43 |
| 16 | 4 blue rubber cube | 0.44 | 1.36 | -0.00 | 0.00 | 0.00 |
| 24 | 0 cyan rubber cylinder | 0.46 | -1.44 | -0.15 | 2.38 | 2.38 |
| 24 | 1 red rubber cylinder | 1.67 | -0.03 | -2.49 | -0.38 | 2.52 |
| 24 | 2 red rubber cube | -0.68 | -0.18 | 0.00 | 0.00 | 0.00 |
| 24 | 3 cyan rubber cube | -4.28 | 3.83 | 1.16 | -1.97 | 2.29 |
| 24 | 4 blue rubber cube | 0.44 | 1.36 | -0.00 | 0.00 | 0.00 |
| 32 | 0 cyan rubber cylinder | 0.43 | -0.93 | -0.14 | 2.24 | 2.25 |
| 32 | 1 red rubber cylinder | 1.13 | -0.11 | -2.35 | -0.36 | 2.37 |
| 32 | 2 red rubber cube | -0.68 | -0.18 | 0.00 | 0.00 | 0.00 |

| frame | object | x | y | vx | vy | speed |
|---|---|---|---|---|---|---|
| 32 | 3 cyan rubber cube | -4.03 | 3.40 | 1.10 | -1.86 | 2.16 |
| 32 | 4 blue rubber cube | 0.44 | 1.36 | -0.00 | 0.00 | 0.00 |
| 39 | 0 cyan rubber cylinder | 0.40 | -0.50 | -0.13 | 2.13 | 2.13 |
| 39 | 1 red rubber cylinder | 0.68 | -0.18 | -2.23 | -0.34 | 2.26 |
| 39 | 2 red rubber cube | -0.68 | -0.18 | -0.00 | 0.00 | 0.00 |
| 39 | 3 cyan rubber cube | -3.82 | 3.05 | 1.04 | -1.77 | 2.06 |
| 39 | 4 blue rubber cube | 0.44 | 1.36 | 0.00 | 0.00 | 0.00 |
| 40 | 0 cyan rubber cylinder | 0.37 | -0.48 | -1.70 | 0.27 | 1.72 |
| 40 | 1 red rubber cylinder | 0.65 | -0.15 | -0.81 | 1.29 | 1.52 |
| 40 | 2 red rubber cube | -0.68 | -0.18 | -0.00 | 0.00 | 0.00 |
| 40 | 3 cyan rubber cube | -3.79 | 3.00 | 1.03 | -1.76 | 2.04 |
| 40 | 4 blue rubber cube | 0.44 | 1.36 | 0.00 | 0.00 | 0.00 |
| 48 | 0 cyan rubber cylinder | -0.01 | -0.42 | -1.61 | 0.25 | 1.63 |
| 48 | 1 red rubber cylinder | 0.47 | 0.13 | -0.76 | 1.21 | 1.43 |
| 48 | 2 red rubber cube | -0.68 | -0.18 | -0.00 | 0.00 | 0.00 |
| 48 | 3 cyan rubber cube | -3.57 | 2.61 | 0.97 | -1.66 | 1.92 |
| 48 | 4 blue rubber cube | 0.44 | 1.36 | 0.00 | 0.00 | 0.00 |
| 55 | 0 cyan rubber cylinder | -0.31 | -0.37 | -1.52 | 0.24 | 1.54 |
| 55 | 1 red rubber cylinder | 0.32 | 0.36 | -0.72 | 1.14 | 1.35 |
| 55 | 2 red rubber cube | -0.68 | -0.18 | 0.00 | 0.00 | 0.00 |
| 55 | 3 cyan rubber cube | -3.38 | 2.30 | 0.92 | -1.57 | 1.82 |
| 55 | 4 blue rubber cube | 0.44 | 1.36 | -0.00 | 0.00 | 0.00 |
| 56 | 0 cyan rubber cylinder | -0.34 | -0.37 | -0.93 | -0.16 | 0.94 |
| 56 | 1 red rubber cylinder | 0.30 | 0.39 | -0.71 | 1.14 | 1.34 |
| 56 | 2 red rubber cube | -0.70 | -0.17 | -0.73 | 0.51 | 0.89 |
| 56 | 3 cyan rubber cube | -3.36 | 2.25 | 0.92 | -1.56 | 1.81 |
| 56 | 4 blue rubber cube | 0.44 | 1.36 | 0.00 | 0.00 | 0.00 |
| 64 | 0 cyan rubber cylinder | -0.54 | -0.41 | -0.85 | -0.15 | 0.86 |
| 64 | 1 red rubber cylinder | 0.15 | 0.64 | -0.66 | 1.06 | 1.25 |
| 64 | 2 red rubber cube | -0.86 | -0.06 | -0.67 | 0.47 | 0.82 |
| 64 | 3 cyan rubber cube | -3.16 | 1.91 | 0.86 | -1.47 | 1.70 |
| 64 | 4 blue rubber cube | 0.44 | 1.36 | 0.00 | 0.00 | 0.00 |
| 72 | 0 cyan rubber cylinder | -0.73 | -0.44 | -0.77 | -0.13 | 0.78 |
| 72 | 1 red rubber cylinder | 0.01 | 0.86 | -0.61 | 0.98 | 1.16 |
| 72 | 2 red rubber cube | -1.00 | 0.04 | -0.61 | 0.42 | 0.74 |
| 72 | 3 cyan rubber cube | -2.97 | 1.59 | 0.81 | -1.38 | 1.60 |
| 72 | 4 blue rubber cube | 0.44 | 1.36 | 0.00 | 0.00 | 0.00 |
| 80 | 0 cyan rubber cylinder | -0.89 | -0.46 | -0.70 | -0.12 | 0.71 |
| 80 | 1 red rubber cylinder | -0.12 | 1.08 | -0.57 | 0.91 | 1.07 |
| 80 | 2 red rubber cube | -1.13 | 0.13 | -0.55 | 0.38 | 0.67 |
| 80 | 3 cyan rubber cube | -2.80 | 1.30 | 0.75 | -1.30 | 1.50 |
| 80 | 4 blue rubber cube | 0.44 | 1.36 | 0.00 | 0.00 | 0.00 |
| 88 | 0 cyan rubber cylinder | -1.04 | -0.49 | -0.62 | -0.11 | 0.63 |
| 88 | 1 red rubber cylinder | -0.24 | 1.27 | -0.52 | 0.84 | 0.99 |
| 88 | 2 red rubber cube | -1.24 | 0.21 | -0.49 | 0.34 | 0.59 |
| 88 | 3 cyan rubber cube | -2.63 | 1.01 | 0.70 | -1.21 | 1.40 |
| 88 | 4 blue rubber cube | 0.44 | 1.36 | 0.00 | 0.00 | 0.00 |
| 96 | 0 cyan rubber cylinder | -1.17 | -0.51 | -0.55 | -0.09 | 0.56 |
| 96 | 1 red rubber cylinder | -0.36 | 1.45 | -0.48 | 0.77 | 0.91 |
| 96 | 2 red rubber cube | -1.34 | 0.28 | -0.43 | 0.29 | 0.52 |
| 96 | 3 cyan rubber cube | -2.48 | 0.75 | 0.66 | -1.13 | 1.31 |
| 96 | 4 blue rubber cube | 0.44 | 1.36 | 0.00 | 0.00 | 0.00 |
| 104 | 0 cyan rubber cylinder | -1.28 | -0.53 | -0.48 | -0.08 | 0.49 |
| 104 | 1 red rubber cylinder | -0.46 | 1.62 | -0.44 | 0.70 | 0.83 |
| 104 | 2 red rubber cube | -1.43 | 0.34 | -0.37 | 0.25 | 0.45 |
| 104 | 3 cyan rubber cube | -2.34 | 0.51 | 0.61 | -1.06 | 1.22 |
| 104 | 4 blue rubber cube | 0.44 | 1.36 | 0.00 | 0.00 | 0.00 |

| frame | object | x | y | vx | vy | speed |
|---|---|---|---|---|---|---|
| 112 | 0 cyan rubber cylinder | -1.38 | -0.55 | -0.41 | -0.07 | 0.41 |
| 112 | 1 red rubber cylinder | -0.55 | 1.77 | -0.39 | 0.64 | 0.75 |
| 112 | 2 red rubber cube | -1.51 | 0.40 | -0.31 | 0.21 | 0.38 |
| 112 | 3 cyan rubber cube | -2.21 | 0.28 | 0.56 | -0.98 | 1.13 |
| 112 | 4 blue rubber cube | 0.44 | 1.36 | 0.00 | 0.00 | 0.00 |
| 120 | 0 cyan rubber cylinder | -1.47 | -0.56 | -0.34 | -0.06 | 0.34 |
| 120 | 1 red rubber cylinder | -0.63 | 1.90 | -0.35 | 0.57 | 0.67 |
| 120 | 2 red rubber cube | -1.57 | 0.44 | -0.25 | 0.17 | 0.31 |
| 120 | 3 cyan rubber cube | -2.09 | 0.07 | 0.52 | -0.91 | 1.05 |
| 120 | 4 blue rubber cube | 0.44 | 1.36 | 0.00 | 0.00 | 0.00 |
| 127 | 0 cyan rubber cylinder | -1.53 | -0.57 | -0.28 | -0.05 | 0.28 |
| 127 | 1 red rubber cylinder | -0.70 | 2.01 | -0.32 | 0.52 | 0.61 |
| 127 | 2 red rubber cube | -1.62 | 0.47 | -0.20 | 0.14 | 0.25 |
| 127 | 3 cyan rubber cube | -1.99 | -0.11 | 0.48 | -0.85 | 0.98 |
| 127 | 4 blue rubber cube | 0.44 | 1.36 | 0.00 | 0.00 | 0.00 |

# Video 13461

## Video 13461: Objects

Object 0: red metal cylinder (color=red, material=metal, shape=cylinder)

Object 1: purple metal sphere (color=purple, material=metal, shape=sphere)

Object 2: yellow rubber sphere (color=yellow, material=rubber, shape=sphere)

Object 3: brown rubber sphere (color=brown, material=rubber, shape=sphere)

## Video 13461: Events

Frame 0: Object 1 (purple metal sphere) is in view when the video begins.

Frame 16: Object 0 (red metal cylinder) enters the scene.

Frame 18: Object 2 (yellow rubber sphere) enters the scene.

Frame 20: Object 0 (red metal cylinder) collides with Object 1 (purple metal sphere).

Frame 91: Object 3 (brown rubber sphere) enters the scene.

Frame 100: Object 1 (purple metal sphere) collides with Object 3 (brown rubber sphere).

Frame 127: The video ends.

## Video 13461: Motion log

| frame | object | x | y | vx | vy | speed |
|---|---|---|---|---|---|---|
| 0 | 1 purple metal sphere | 1.10 | 1.33 | 0.00 | 0.00 | 0.00 |
| 8 | 1 purple metal sphere | 1.10 | 1.33 | -0.00 | -0.00 | 0.00 |
| 16 | 0 red metal cylinder | 1.22 | 2.00 | -1.00 | -2.25 | 2.46 |
| 16 | 1 purple metal sphere | 1.10 | 1.33 | -0.00 | -0.00 | 0.00 |
| 18 | 0 red metal cylinder | 1.16 | 1.87 | -0.99 | -2.22 | 2.42 |
| 18 | 1 purple metal sphere | 1.10 | 1.33 | -0.00 | -0.00 | 0.00 |
| 18 | 2 yellow rubber sphere | -6.47 | 4.77 | 1.91 | -1.85 | 2.66 |
| 20 | 0 red metal cylinder | 1.11 | 1.75 | -0.97 | -2.18 | 2.39 |
| 20 | 1 purple metal sphere | 1.10 | 1.33 | -0.00 | -0.00 | 0.00 |
| 20 | 2 yellow rubber sphere | -6.36 | 4.67 | 1.90 | -1.84 | 2.64 |
| 24 | 0 red metal cylinder | 1.00 | 1.61 | -0.92 | -1.19 | 1.50 |
| 24 | 1 purple metal sphere | 1.10 | 1.20 | -0.03 | -1.17 | 1.17 |
| 24 | 2 yellow rubber sphere | -6.15 | 4.47 | 1.87 | -1.81 | 2.60 |
| 32 | 0 red metal cylinder | 0.80 | 1.35 | -0.86 | -1.11 | 1.40 |
| 32 | 1 purple metal sphere | 1.09 | 0.94 | -0.02 | -1.14 | 1.14 |
| 32 | 2 yellow rubber sphere | -5.74 | 4.07 | 1.81 | -1.75 | 2.52 |
| 40 | 0 red metal cylinder | 0.62 | 1.11 | -0.80 | -1.04 | 1.31 |
| 40 | 1 purple metal sphere | 1.08 | 0.69 | -0.02 | -1.12 | 1.12 |
| 40 | 2 yellow rubber sphere | -5.34 | 3.68 | 1.75 | -1.70 | 2.44 |
| 48 | 0 red metal cylinder | 0.44 | 0.89 | -0.75 | -0.96 | 1.22 |
| 48 | 1 purple metal sphere | 1.08 | 0.44 | -0.02 | -1.10 | 1.10 |
| 48 | 2 yellow rubber sphere | -4.95 | 3.31 | 1.70 | -1.65 | 2.37 |
| 56 | 0 red metal cylinder | 0.28 | 0.68 | -0.69 | -0.90 | 1.13 |
| 56 | 1 purple metal sphere | 1.07 | 0.20 | -0.02 | -1.08 | 1.08 |
| 56 | 2 yellow rubber sphere | -4.58 | 2.94 | 1.65 | -1.60 | 2.30 |
| 64 | 0 red metal cylinder | 0.13 | 0.49 | -0.64 | -0.83 | 1.05 |
| 64 | 1 purple metal sphere | 1.07 | -0.04 | -0.02 | -1.06 | 1.06 |
| 64 | 2 yellow rubber sphere | -4.21 | 2.59 | 1.60 | -1.55 | 2.23 |
| 72 | 0 red metal cylinder | -0.01 | 0.31 | -0.59 | -0.76 | 0.96 |

| frame | object | x | y | vx | vy | speed |
|---|---|---|---|---|---|---|
| 72 | 1 purple metal sphere | 1.06 | -0.28 | -0.02 | -1.04 | 1.04 |
| 72 | 2 yellow rubber sphere | -3.86 | 2.25 | 1.56 | -1.51 | 2.17 |
| 80 | 0 red metal cylinder | -0.13 | 0.14 | -0.54 | -0.70 | 0.88 |
| 80 | 1 purple metal sphere | 1.06 | -0.51 | -0.02 | -1.02 | 1.02 |
| 80 | 2 yellow rubber sphere | -3.52 | 1.91 | 1.51 | -1.47 | 2.11 |
| 88 | 0 red metal cylinder | -0.25 | -0.00 | -0.49 | -0.64 | 0.80 |
| 88 | 1 purple metal sphere | 1.05 | -0.73 | -0.02 | -1.00 | 1.00 |
| 88 | 2 yellow rubber sphere | -3.18 | 1.59 | 1.47 | -1.43 | 2.05 |
| 91 | 0 red metal cylinder | -0.29 | -0.06 | -0.47 | -0.61 | 0.77 |
| 91 | 1 purple metal sphere | 1.05 | -0.82 | -0.02 | -1.00 | 1.00 |
| 91 | 2 yellow rubber sphere | -3.06 | 1.47 | 1.46 | -1.41 | 2.03 |
| 91 | 3 brown rubber sphere | 1.18 | -1.97 | -0.34 | 1.98 | 2.01 |
| 96 | 0 red metal cylinder | -0.35 | -0.14 | -0.44 | -0.58 | 0.73 |
| 96 | 1 purple metal sphere | 1.05 | -0.96 | -0.02 | -0.99 | 0.99 |
| 96 | 2 yellow rubber sphere | -2.86 | 1.27 | 1.43 | -1.39 | 2.00 |
| 96 | 3 brown rubber sphere | 1.13 | -1.70 | -0.34 | 1.95 | 1.98 |
| 100 | 0 red metal cylinder | -0.40 | -0.20 | -0.42 | -0.55 | 0.69 |
| 100 | 1 purple metal sphere | 1.05 | -1.07 | -0.02 | -0.98 | 0.98 |
| 100 | 2 yellow rubber sphere | -2.70 | 1.12 | 1.41 | -1.37 | 1.97 |
| 100 | 3 brown rubber sphere | 1.09 | -1.48 | -0.33 | 1.92 | 1.95 |
| 104 | 0 red metal cylinder | -0.45 | -0.26 | -0.40 | -0.52 | 0.65 |
| 104 | 1 purple metal sphere | 1.02 | -0.95 | -0.21 | 1.07 | 1.09 |
| 104 | 2 yellow rubber sphere | -2.54 | 0.97 | 1.39 | -1.35 | 1.94 |
| 104 | 3 brown rubber sphere | 1.07 | -1.46 | -0.17 | 0.16 | 0.23 |
| 112 | 0 red metal cylinder | -0.53 | -0.37 | -0.35 | -0.46 | 0.58 |
| 112 | 1 purple metal sphere | 0.98 | -0.72 | -0.21 | 1.05 | 1.07 |
| 112 | 2 yellow rubber sphere | -2.23 | 0.67 | 1.36 | -1.32 | 1.89 |
| 112 | 3 brown rubber sphere | 1.03 | -1.42 | -0.16 | 0.16 | 0.23 |
| 120 | 0 red metal cylinder | -0.60 | -0.47 | -0.31 | -0.40 | 0.50 |
| 120 | 1 purple metal sphere | 0.93 | -0.48 | -0.20 | 1.03 | 1.05 |
| 120 | 2 yellow rubber sphere | -1.93 | 0.38 | 1.32 | -1.28 | 1.84 |
| 120 | 3 brown rubber sphere | 1.00 | -1.38 | -0.16 | 0.16 | 0.23 |
| 127 | 0 red metal cylinder | -0.66 | -0.54 | -0.27 | -0.35 | 0.44 |
| 127 | 1 purple metal sphere | 0.89 | -0.28 | -0.20 | 1.01 | 1.03 |
| 127 | 2 yellow rubber sphere | -1.68 | 0.13 | 1.30 | -1.26 | 1.80 |
| 127 | 3 brown rubber sphere | 0.97 | -1.35 | -0.16 | 0.16 | 0.23 |

# Video 13549

## Video 13549: Objects

Object 0: blue rubber cylinder (color=blue, material=rubber, shape=cylinder)

Object 1: purple rubber cube (color=purple, material=rubber, shape=cube)

Object 2: yellow rubber cylinder (color=yellow, material=rubber, shape=cylinder)

Object 3: blue rubber cube (color=blue, material=rubber, shape=cube)

Object 4: red metal cylinder (color=red, material=metal, shape=cylinder)

Object 5: gray rubber cylinder (color=gray, material=rubber, shape=cylinder)

## Video 13549: Events

Frame 0: Object 1 (purple rubber cube) is in view when the video begins.

Frame 0: Object 4 (red metal cylinder) is in view when the video begins.

Frame 0: Object 5 (gray rubber cylinder) is in view when the video begins.

Frame 12: Object 0 (blue rubber cylinder) enters the scene.

Frame 25: Object 2 (yellow rubber cylinder) enters the scene.

Frame 47: Object 3 (blue rubber cube) enters the scene.

Frame 59: Object 0 (blue rubber cylinder) collides with Object 1 (purple rubber cube).

Frame 107: Object 1 (purple rubber cube) collides with Object 3 (blue rubber cube).

Frame 122: Object 2 (yellow rubber cylinder) collides with Object 3 (blue rubber cube).

Frame 127: The video ends.

## Video 13549: Motion log

| frame | object | x | y | vx | vy | speed |
|---|---|---|---|---|---|---|
| 0 | 1 purple rubber cube | -0.65 | 0.22 | 0.00 | 0.00 | 0.00 |
| 0 | 4 red metal cylinder | -0.29 | -0.64 | 0.00 | 0.00 | 0.00 |
| 0 | 5 gray rubber cylinder | 1.05 | -1.39 | 0.00 | 0.00 | 0.00 |
| 8 | 1 purple rubber cube | -0.65 | 0.22 | 0.00 | 0.00 | 0.00 |
| 8 | 4 red metal cylinder | -0.29 | -0.64 | -0.00 | 0.00 | 0.00 |
| 8 | 5 gray rubber cylinder | 1.05 | -1.39 | 0.00 | 0.00 | 0.00 |
| 12 | 0 blue rubber cylinder | 2.53 | 1.03 | -2.46 | -0.71 | 2.55 |
| 12 | 1 purple rubber cube | -0.65 | 0.22 | -0.00 | 0.00 | 0.00 |
| 12 | 4 red metal cylinder | -0.29 | -0.64 | -0.00 | 0.00 | 0.00 |
| 12 | 5 gray rubber cylinder | 1.05 | -1.39 | 0.00 | 0.00 | 0.00 |
| 16 | 0 blue rubber cylinder | 2.26 | 0.95 | -2.38 | -0.69 | 2.48 |
| 16 | 1 purple rubber cube | -0.65 | 0.22 | -0.00 | 0.00 | 0.00 |
| 16 | 4 red metal cylinder | -0.29 | -0.64 | -0.00 | 0.00 | 0.00 |
| 16 | 5 gray rubber cylinder | 1.05 | -1.39 | 0.00 | 0.00 | 0.00 |
| 24 | 0 blue rubber cylinder | 1.74 | 0.80 | -2.25 | -0.65 | 2.34 |
| 24 | 1 purple rubber cube | -0.65 | 0.22 | -0.00 | 0.00 | 0.00 |
| 24 | 4 red metal cylinder | -0.29 | -0.64 | -0.00 | 0.00 | 0.00 |
| 24 | 5 gray rubber cylinder | 1.05 | -1.39 | 0.00 | -0.00 | 0.00 |
| 25 | 0 blue rubber cylinder | 1.68 | 0.78 | -2.23 | -0.64 | 2.33 |
| 25 | 1 purple rubber cube | -0.65 | 0.22 | -0.00 | 0.00 | 0.00 |
| 25 | 2 yellow rubber cylinder | -6.62 | -2.15 | 2.25 | 0.80 | 2.39 |
| 25 | 4 red metal cylinder | -0.29 | -0.64 | -0.00 | 0.00 | 0.00 |

| frame | object | x | y | vx | vy | speed |
|---|---|---|---|---|---|---|
| 25 | 5 gray rubber cylinder | 1.05 | -1.39 | 0.00 | -0.00 | 0.00 |
| 32 | 0 blue rubber cylinder | 1.25 | 0.66 | -2.12 | -0.61 | 2.21 |
| 32 | 1 purple rubber cube | -0.65 | 0.22 | -0.00 | 0.00 | 0.00 |
| 32 | 2 yellow rubber cylinder | -6.19 | -1.99 | 2.14 | 0.76 | 2.27 |
| 32 | 4 red metal cylinder | -0.29 | -0.64 | -0.00 | 0.00 | 0.00 |
| 32 | 5 gray rubber cylinder | 1.05 | -1.39 | 0.00 | -0.00 | 0.00 |
| 40 | 0 blue rubber cylinder | 0.79 | 0.52 | -2.00 | -0.58 | 2.08 |
| 40 | 1 purple rubber cube | -0.65 | 0.22 | -0.00 | 0.00 | 0.00 |
| 40 | 2 yellow rubber cylinder | -5.73 | -1.83 | 2.02 | 0.71 | 2.15 |
| 40 | 4 red metal cylinder | -0.29 | -0.64 | -0.00 | 0.00 | 0.00 |
| 40 | 5 gray rubber cylinder | 1.05 | -1.39 | 0.00 | -0.00 | 0.00 |
| 47 | 0 blue rubber cylinder | 0.41 | 0.41 | -1.90 | -0.55 | 1.98 |
| 47 | 1 purple rubber cube | -0.65 | 0.22 | -0.00 | 0.00 | 0.00 |
| 47 | 2 yellow rubber cylinder | -5.34 | -1.69 | 1.92 | 0.68 | 2.04 |
| 47 | 3 blue rubber cube | -0.32 | -2.66 | -1.12 | 1.69 | 2.03 |
| 47 | 4 red metal cylinder | -0.29 | -0.64 | -0.00 | 0.00 | 0.00 |
| 47 | 5 gray rubber cylinder | 1.05 | -1.39 | 0.00 | -0.00 | 0.00 |
| 48 | 0 blue rubber cylinder | 0.35 | 0.40 | -1.89 | -0.55 | 1.96 |
| 48 | 1 purple rubber cube | -0.65 | 0.22 | -0.00 | 0.00 | 0.00 |
| 48 | 2 yellow rubber cylinder | -5.29 | -1.67 | 1.91 | 0.67 | 2.02 |
| 48 | 3 blue rubber cube | -0.35 | -2.61 | -1.11 | 1.68 | 2.01 |
| 48 | 4 red metal cylinder | -0.29 | -0.64 | -0.00 | 0.00 | 0.00 |
| 48 | 5 gray rubber cylinder | 1.05 | -1.39 | 0.00 | -0.00 | 0.00 |
| 56 | 0 blue rubber cylinder | -0.06 | 0.28 | -1.77 | -0.52 | 1.85 |
| 56 | 1 purple rubber cube | -0.65 | 0.22 | -0.00 | 0.00 | 0.00 |
| 56 | 2 yellow rubber cylinder | -4.87 | -1.53 | 1.80 | 0.63 | 1.90 |
| 56 | 3 blue rubber cube | -0.59 | -2.25 | -1.04 | 1.58 | 1.89 |
| 56 | 4 red metal cylinder | -0.29 | -0.64 | -0.00 | 0.00 | 0.00 |
| 56 | 5 gray rubber cylinder | 1.05 | -1.39 | 0.00 | -0.00 | 0.00 |
| 59 | 0 blue rubber cylinder | -0.19 | 0.24 | -0.82 | -0.35 | 0.89 |
| 59 | 1 purple rubber cube | -0.66 | 0.22 | -1.12 | -0.20 | 1.14 |
| 59 | 2 yellow rubber cylinder | -4.72 | -1.48 | 1.75 | 0.62 | 1.86 |
| 59 | 3 blue rubber cube | -0.67 | -2.12 | -1.02 | 1.54 | 1.85 |
| 59 | 4 red metal cylinder | -0.29 | -0.64 | -0.00 | 0.00 | 0.00 |
| 59 | 5 gray rubber cylinder | 1.05 | -1.39 | 0.00 | -0.00 | 0.00 |
| 64 | 0 blue rubber cylinder | -0.30 | 0.19 | -0.78 | -0.33 | 0.84 |
| 64 | 1 purple rubber cube | -0.82 | 0.19 | -1.08 | -0.19 | 1.09 |
| 64 | 2 yellow rubber cylinder | -4.48 | -1.39 | 1.69 | 0.60 | 1.79 |
| 64 | 3 blue rubber cube | -0.81 | -1.91 | -0.98 | 1.49 | 1.78 |
| 64 | 4 red metal cylinder | -0.29 | -0.64 | -0.00 | 0.00 | 0.00 |
| 64 | 5 gray rubber cylinder | 1.05 | -1.39 | -0.00 | -0.00 | 0.00 |
| 72 | 0 blue rubber cylinder | -0.47 | 0.12 | -0.70 | -0.30 | 0.76 |
| 72 | 1 purple rubber cube | -1.05 | 0.15 | -0.99 | -0.17 | 1.01 |
| 72 | 2 yellow rubber cylinder | -4.12 | -1.26 | 1.59 | 0.56 | 1.68 |
| 72 | 3 blue rubber cube | -1.03 | -1.58 | -0.92 | 1.40 | 1.67 |
| 72 | 4 red metal cylinder | -0.29 | -0.64 | -0.00 | 0.00 | 0.00 |
| 72 | 5 gray rubber cylinder | 1.05 | -1.39 | 0.00 | -0.00 | 0.00 |
| 80 | 0 blue rubber cylinder | -0.62 | 0.06 | -0.63 | -0.27 | 0.69 |
| 80 | 1 purple rubber cube | -1.26 | 0.12 | -0.91 | -0.16 | 0.92 |
| 80 | 2 yellow rubber cylinder | -3.77 | -1.14 | 1.49 | 0.53 | 1.58 |
| 80 | 3 blue rubber cube | -1.23 | -1.28 | -0.86 | 1.31 | 1.57 |
| 80 | 4 red metal cylinder | -0.29 | -0.64 | -0.00 | 0.00 | 0.00 |
| 80 | 5 gray rubber cylinder | 1.05 | -1.39 | 0.00 | -0.00 | 0.00 |
| 88 | 0 blue rubber cylinder | -0.75 | 0.00 | -0.56 | -0.24 | 0.61 |
| 88 | 1 purple rubber cube | -1.46 | 0.08 | -0.83 | -0.15 | 0.84 |
| 88 | 2 yellow rubber cylinder | -3.45 | -1.02 | 1.39 | 0.49 | 1.48 |
| 88 | 3 blue rubber cube | -1.41 | -1.00 | -0.81 | 1.23 | 1.47 |

| frame | object | x | y | vx | vy | speed |
|---|---|---|---|---|---|---|
| 88 | 4 red metal cylinder | -0.29 | -0.64 | -0.00 | 0.00 | 0.00 |
| 88 | 5 gray rubber cylinder | 1.05 | -1.39 | 0.00 | -0.00 | 0.00 |
| 96 | 0 blue rubber cylinder | -0.87 | -0.05 | -0.50 | -0.21 | 0.54 |
| 96 | 1 purple rubber cube | -1.63 | 0.05 | -0.75 | -0.13 | 0.76 |
| 96 | 2 yellow rubber cylinder | -3.15 | -0.92 | 1.30 | 0.46 | 1.38 |
| 96 | 3 blue rubber cube | -1.59 | -0.73 | -0.76 | 1.14 | 1.37 |
| 96 | 4 red metal cylinder | -0.29 | -0.64 | -0.00 | 0.00 | 0.00 |
| 96 | 5 gray rubber cylinder | 1.05 | -1.39 | 0.00 | -0.00 | 0.00 |
| 104 | 0 blue rubber cylinder | -0.97 | -0.09 | -0.43 | -0.18 | 0.47 |
| 104 | 1 purple rubber cube | -1.79 | 0.02 | -0.68 | -0.12 | 0.69 |
| 104 | 2 yellow rubber cylinder | -2.87 | -0.82 | 1.22 | 0.43 | 1.29 |
| 104 | 3 blue rubber cube | -1.75 | -0.49 | -0.70 | 1.07 | 1.28 |
| 104 | 4 red metal cylinder | -0.29 | -0.64 | -0.00 | 0.00 | 0.00 |
| 104 | 5 gray rubber cylinder | 1.05 | -1.39 | 0.00 | -0.00 | 0.00 |
| 107 | 0 blue rubber cylinder | -1.01 | -0.11 | -0.41 | -0.17 | 0.44 |
| 107 | 1 purple rubber cube | -1.85 | 0.02 | -0.67 | 0.29 | 0.72 |
| 107 | 2 yellow rubber cylinder | -2.77 | -0.78 | 1.18 | 0.42 | 1.25 |
| 107 | 3 blue rubber cube | -1.81 | -0.40 | -0.67 | 0.73 | 0.99 |
| 107 | 4 red metal cylinder | -0.29 | -0.64 | 0.00 | -0.00 | 0.00 |
| 107 | 5 gray rubber cylinder | 1.05 | -1.39 | -0.00 | -0.00 | 0.00 |
| 112 | 0 blue rubber cylinder | -1.06 | -0.13 | -0.37 | -0.15 | 0.40 |
| 112 | 1 purple rubber cube | -1.94 | 0.09 | -0.62 | 0.62 | 0.87 |
| 112 | 2 yellow rubber cylinder | -2.60 | -0.73 | 1.13 | 0.40 | 1.20 |
| 112 | 3 blue rubber cube | -1.90 | -0.33 | -0.64 | 0.42 | 0.77 |
| 112 | 4 red metal cylinder | -0.29 | -0.64 | -0.00 | 0.00 | 0.00 |
| 112 | 5 gray rubber cylinder | 1.05 | -1.39 | 0.00 | -0.00 | 0.00 |
| 120 | 0 blue rubber cylinder | -1.14 | -0.16 | -0.30 | -0.13 | 0.33 |
| 120 | 1 purple rubber cube | -2.07 | 0.23 | -0.56 | 0.56 | 0.79 |
| 120 | 2 yellow rubber cylinder | -2.36 | -0.64 | 1.05 | 0.37 | 1.11 |
| 120 | 3 blue rubber cube | -2.04 | -0.24 | -0.58 | 0.38 | 0.69 |
| 120 | 4 red metal cylinder | -0.29 | -0.64 | -0.00 | 0.00 | 0.00 |
| 120 | 5 gray rubber cylinder | 1.05 | -1.39 | 0.00 | 0.00 | 0.00 |
| 122 | 0 blue rubber cylinder | -1.15 | -0.17 | -0.29 | -0.12 | 0.31 |
| 122 | 1 purple rubber cube | -2.10 | 0.26 | -0.55 | 0.55 | 0.77 |
| 122 | 2 yellow rubber cylinder | -2.30 | -0.62 | 1.03 | 0.36 | 1.09 |
| 122 | 3 blue rubber cube | -2.07 | -0.22 | -0.56 | 0.37 | 0.67 |
| 122 | 4 red metal cylinder | -0.29 | -0.64 | -0.00 | 0.00 | 0.00 |
| 122 | 5 gray rubber cylinder | 1.05 | -1.39 | -0.00 | -0.00 | 0.00 |
| 127 | 0 blue rubber cylinder | -1.19 | -0.18 | -0.25 | -0.10 | 0.27 |
| 127 | 1 purple rubber cube | -2.17 | 0.33 | -0.51 | 0.51 | 0.72 |
| 127 | 2 yellow rubber cylinder | -2.16 | -0.57 | 0.98 | 0.33 | 1.03 |
| 127 | 3 blue rubber cube | -2.15 | -0.17 | -0.52 | 0.36 | 0.63 |
| 127 | 4 red metal cylinder | -0.29 | -0.64 | 0.00 | -0.00 | 0.00 |
| 127 | 5 gray rubber cylinder | 1.05 | -1.39 | -0.00 | -0.00 | 0.00 |

# Video 13894

## Video 13894: Objects

Object 0: gray metal sphere (color=gray, material=metal, shape=sphere)

Object 1: brown metal cube (color=brown, material=metal, shape=cube)

Object 2: red metal cube (color=red, material=metal, shape=cube)

Object 3: purple metal cube (color=purple, material=metal, shape=cube)

Object 4: green rubber sphere (color=green, material=rubber, shape=sphere)

## Video 13894: Events

Frame 0: Object 0 (gray metal sphere) is in view when the video begins.

Frame 0: Object 1 (brown metal cube) is in view when the video begins.

Frame 0: Object 3 (purple metal cube) is in view when the video begins.

Frame 0: Object 4 (green rubber sphere) is in view when the video begins.

Frame 26: Object 0 (gray metal sphere) collides with Object 1 (brown metal cube).

Frame 56: Object 1 (brown metal cube) collides with Object 3 (purple metal cube).

Frame 101: Object 2 (red metal cube) enters the scene.

Frame 127: The video ends.

## Video 13894: Motion log

| frame | object | x | y | vx | vy | speed |
|---|---|---|---|---|---|---|
| 0 | 0 gray metal sphere | -2.83 | 1.10 | 2.60 | -1.39 | 2.95 |
| 0 | 1 brown metal cube | -0.63 | 0.07 | 0.00 | 0.00 | 0.00 |
| 0 | 3 purple metal cube | 0.90 | -0.58 | 0.00 | 0.00 | 0.00 |
| 0 | 4 green rubber sphere | -1.08 | 1.52 | 0.00 | 0.00 | 0.00 |
| 8 | 0 gray metal sphere | -2.25 | 0.80 | 2.51 | -1.35 | 2.85 |
| 8 | 1 brown metal cube | -0.63 | 0.07 | 0.00 | 0.00 | 0.00 |
| 8 | 3 purple metal cube | 0.90 | -0.58 | 0.00 | -0.00 | 0.00 |
| 8 | 4 green rubber sphere | -1.08 | 1.52 | -0.00 | -0.00 | 0.00 |
| 16 | 0 gray metal sphere | -1.70 | 0.50 | 2.43 | -1.30 | 2.76 |
| 16 | 1 brown metal cube | -0.63 | 0.07 | 0.00 | 0.00 | 0.00 |
| 16 | 3 purple metal cube | 0.90 | -0.58 | 0.00 | -0.00 | 0.00 |
| 16 | 4 green rubber sphere | -1.08 | 1.52 | -0.00 | 0.00 | 0.00 |
| 24 | 0 gray metal sphere | -1.17 | 0.21 | 2.35 | -1.26 | 2.67 |
| 24 | 1 brown metal cube | -0.63 | 0.07 | 0.00 | 0.00 | 0.00 |
| 24 | 3 purple metal cube | 0.90 | -0.58 | 0.00 | -0.00 | 0.00 |
| 24 | 4 green rubber sphere | -1.08 | 1.52 | -0.00 | -0.00 | 0.00 |
| 26 | 0 gray metal sphere | -1.03 | 0.14 | 2.33 | -1.25 | 2.64 |
| 26 | 1 brown metal cube | -0.63 | 0.07 | 0.00 | 0.00 | 0.00 |
| 26 | 3 purple metal cube | 0.90 | -0.58 | 0.00 | -0.00 | 0.00 |
| 26 | 4 green rubber sphere | -1.08 | 1.52 | -0.00 | -0.00 | 0.00 |
| 32 | 0 gray metal sphere | -0.98 | 0.10 | 0.32 | -0.28 | 0.42 |
| 32 | 1 brown metal cube | -0.40 | -0.04 | 1.34 | -0.65 | 1.49 |
| 32 | 3 purple metal cube | 0.90 | -0.58 | 0.00 | -0.00 | 0.00 |
| 32 | 4 green rubber sphere | -1.08 | 1.52 | -0.00 | -0.00 | 0.00 |
| 40 | 0 gray metal sphere | -0.91 | 0.03 | 0.31 | -0.28 | 0.42 |
| 40 | 1 brown metal cube | -0.11 | -0.18 | 1.26 | -0.60 | 1.40 |

| frame | object | x | y | vx | vy | speed |
|---|---|---|---|---|---|---|
| 40 | 3 purple metal cube | 0.90 | -0.58 | 0.00 | -0.00 | 0.00 |
| 40 | 4 green rubber sphere | -1.08 | 1.52 | -0.00 | -0.00 | 0.00 |
| 48 | 0 gray metal sphere | -0.84 | -0.03 | 0.31 | -0.27 | 0.41 |
| 48 | 1 brown metal cube | 0.17 | -0.32 | 1.18 | -0.56 | 1.31 |
| 48 | 3 purple metal cube | 0.90 | -0.58 | 0.00 | -0.00 | 0.00 |
| 48 | 4 green rubber sphere | -1.08 | 1.52 | -0.00 | -0.00 | 0.00 |
| 56 | 0 gray metal sphere | -0.77 | -0.09 | 0.31 | -0.27 | 0.41 |
| 56 | 1 brown metal cube | 0.42 | -0.44 | 1.10 | -0.52 | 1.21 |
| 56 | 3 purple metal cube | 0.90 | -0.58 | 0.00 | -0.00 | 0.00 |
| 56 | 4 green rubber sphere | -1.08 | 1.52 | -0.00 | -0.00 | 0.00 |
| 64 | 0 gray metal sphere | -0.70 | -0.15 | 0.30 | -0.27 | 0.40 |
| 64 | 1 brown metal cube | 0.58 | -0.49 | 0.60 | -0.19 | 0.63 |
| 64 | 3 purple metal cube | 1.00 | -0.66 | 0.52 | -0.37 | 0.64 |
| 64 | 4 green rubber sphere | -1.08 | 1.52 | -0.00 | -0.00 | 0.00 |
| 72 | 0 gray metal sphere | -0.64 | -0.21 | 0.30 | -0.26 | 0.40 |
| 72 | 1 brown metal cube | 0.70 | -0.53 | 0.52 | -0.16 | 0.55 |
| 72 | 3 purple metal cube | 1.11 | -0.74 | 0.47 | -0.33 | 0.58 |
| 72 | 4 green rubber sphere | -1.08 | 1.52 | -0.00 | -0.00 | 0.00 |
| 80 | 0 gray metal sphere | -0.57 | -0.26 | 0.29 | -0.26 | 0.39 |
| 80 | 1 brown metal cube | 0.81 | -0.56 | 0.45 | -0.14 | 0.47 |
| 80 | 3 purple metal cube | 1.21 | -0.81 | 0.41 | -0.29 | 0.50 |
| 80 | 4 green rubber sphere | -1.08 | 1.52 | -0.00 | -0.00 | 0.00 |
| 88 | 0 gray metal sphere | -0.50 | -0.32 | 0.29 | -0.26 | 0.39 |
| 88 | 1 brown metal cube | 0.91 | -0.59 | 0.39 | -0.12 | 0.40 |
| 88 | 3 purple metal cube | 1.30 | -0.87 | 0.35 | -0.25 | 0.43 |
| 88 | 4 green rubber sphere | -1.08 | 1.52 | -0.00 | -0.00 | 0.00 |
| 96 | 0 gray metal sphere | -0.44 | -0.38 | 0.29 | -0.25 | 0.38 |
| 96 | 1 brown metal cube | 0.99 | -0.62 | 0.32 | -0.10 | 0.33 |
| 96 | 3 purple metal cube | 1.37 | -0.92 | 0.30 | -0.21 | 0.36 |
| 96 | 4 green rubber sphere | -1.08 | 1.52 | -0.00 | -0.00 | 0.00 |
| 101 | 0 gray metal sphere | -0.40 | -0.41 | 0.28 | -0.25 | 0.38 |
| 101 | 1 brown metal cube | 1.03 | -0.63 | 0.28 | -0.08 | 0.29 |
| 101 | 2 red metal cube | 0.81 | -2.20 | -0.64 | 1.14 | 1.31 |
| 101 | 3 purple metal cube | 1.41 | -0.95 | 0.26 | -0.18 | 0.32 |
| 101 | 4 green rubber sphere | -1.08 | 1.52 | -0.00 | -0.00 | 0.00 |
| 104 | 0 gray metal sphere | -0.38 | -0.43 | 0.28 | -0.25 | 0.38 |
| 104 | 1 brown metal cube | 1.05 | -0.64 | 0.25 | -0.08 | 0.26 |
| 104 | 2 red metal cube | 0.76 | -2.11 | -0.62 | 1.11 | 1.27 |
| 104 | 3 purple metal cube | 1.43 | -0.96 | 0.24 | -0.17 | 0.29 |
| 104 | 4 green rubber sphere | -1.08 | 1.52 | -0.00 | -0.00 | 0.00 |
| 112 | 0 gray metal sphere | -0.31 | -0.49 | 0.28 | -0.25 | 0.37 |
| 112 | 1 brown metal cube | 1.10 | -0.65 | 0.19 | -0.06 | 0.20 |
| 112 | 2 red metal cube | 0.62 | -1.87 | -0.58 | 1.04 | 1.19 |
| 112 | 3 purple metal cube | 1.48 | -0.99 | 0.18 | -0.13 | 0.22 |
| 112 | 4 green rubber sphere | -1.08 | 1.52 | -0.00 | -0.00 | 0.00 |
| 120 | 0 gray metal sphere | -0.25 | -0.54 | 0.28 | -0.24 | 0.37 |
| 120 | 1 brown metal cube | 1.13 | -0.66 | 0.12 | -0.04 | 0.13 |
| 120 | 2 red metal cube | 0.50 | -1.64 | -0.53 | 0.96 | 1.10 |
| 120 | 3 purple metal cube | 1.51 | -1.02 | 0.13 | -0.09 | 0.16 |
| 120 | 4 green rubber sphere | -1.08 | 1.52 | -0.00 | -0.00 | 0.00 |
| 127 | 0 gray metal sphere | -0.20 | -0.59 | 0.27 | -0.24 | 0.36 |
| 127 | 1 brown metal cube | 1.15 | -0.67 | 0.07 | -0.02 | 0.07 |
| 127 | 2 red metal cube | 0.40 | -1.46 | -0.50 | 0.90 | 1.03 |
| 127 | 3 purple metal cube | 1.53 | -1.03 | 0.08 | -0.05 | 0.10 |
| 127 | 4 green rubber sphere | -1.08 | 1.52 | -0.00 | -0.00 | 0.00 |

# Video 14082

## Video 14082: Objects

Object 0: cyan rubber cube (color=cyan, material=rubber, shape=cube)

Object 1: purple rubber cube (color=purple, material=rubber, shape=cube)

Object 2: purple metal cube (color=purple, material=metal, shape=cube)

Object 3: cyan metal cylinder (color=cyan, material=metal, shape=cylinder)

## Video 14082: Events

Frame 0: Object 0 (cyan rubber cube) is in view when the video begins.

Frame 0: Object 1 (purple rubber cube) is in view when the video begins.

Frame 0: Object 2 (purple metal cube) is in view when the video begins.

Frame 13: Object 0 (cyan rubber cube) collides with Object 2 (purple metal cube).

Frame 29: Object 0 (cyan rubber cube) collides with Object 1 (purple rubber cube).

Frame 85: Object 1 (purple rubber cube) collides with Object 2 (purple metal cube).

Frame 114: Object 3 (cyan metal cylinder) enters the scene.

Frame 127: The video ends.

## Video 14082: Motion log

| frame | object | x | y | vx | vy | speed |
|---|---|---|---|---|---|---|
| 0 | 0 cyan rubber cube | -2.88 | 1.55 | 2.75 | -0.65 | 2.83 |
| 0 | 1 purple rubber cube | 1.43 | 1.58 | -2.64 | -0.68 | 2.73 |
| 0 | 2 purple metal cube | -1.82 | 0.87 | 0.00 | 0.00 | 0.00 |
| 8 | 0 cyan rubber cube | -2.28 | 1.41 | 2.60 | -0.61 | 2.67 |
| 8 | 1 purple rubber cube | 0.86 | 1.43 | -2.49 | -0.65 | 2.58 |
| 8 | 2 purple metal cube | -1.82 | 0.87 | 0.00 | 0.00 | 0.00 |
| 13 | 0 cyan rubber cube | -1.93 | 1.32 | 2.51 | -0.59 | 2.58 |
| 13 | 1 purple rubber cube | 0.52 | 1.34 | -2.40 | -0.62 | 2.48 |
| 13 | 2 purple metal cube | -1.82 | 0.87 | -0.00 | 0.00 | 0.00 |
| 16 | 0 cyan rubber cube | -1.73 | 1.32 | 2.26 | 0.05 | 2.26 |
| 16 | 1 purple rubber cube | 0.32 | 1.29 | -2.35 | -0.61 | 2.43 |
| 16 | 2 purple metal cube | -1.80 | 0.82 | 0.20 | -0.63 | 0.66 |
| 24 | 0 cyan rubber cube | -1.24 | 1.33 | 2.13 | 0.05 | 2.13 |
| 24 | 1 purple rubber cube | -0.19 | 1.16 | -2.22 | -0.58 | 2.29 |
| 24 | 2 purple metal cube | -1.76 | 0.69 | 0.18 | -0.56 | 0.59 |
| 29 | 0 cyan rubber cube | -0.95 | 1.34 | 2.05 | 0.05 | 2.05 |
| 29 | 1 purple rubber cube | -0.50 | 1.08 | -2.14 | -0.55 | 2.21 |
| 29 | 2 purple metal cube | -1.74 | 0.61 | 0.16 | -0.52 | 0.54 |
| 32 | 0 cyan rubber cube | -0.96 | 1.35 | -0.24 | 0.17 | 0.29 |
| 32 | 1 purple rubber cube | -0.54 | 1.02 | -0.50 | -0.64 | 0.81 |
| 32 | 2 purple metal cube | -1.73 | 0.57 | 0.15 | -0.49 | 0.52 |
| 40 | 0 cyan rubber cube | -1.02 | 1.39 | -0.23 | 0.17 | 0.28 |
| 40 | 1 purple rubber cube | -0.65 | 0.88 | -0.48 | -0.60 | 0.77 |
| 40 | 2 purple metal cube | -1.70 | 0.47 | 0.13 | -0.42 | 0.44 |
| 48 | 0 cyan rubber cube | -1.07 | 1.42 | -0.21 | 0.15 | 0.26 |
| 48 | 1 purple rubber cube | -0.76 | 0.75 | -0.46 | -0.57 | 0.73 |
| 48 | 2 purple metal cube | -1.67 | 0.38 | 0.11 | -0.36 | 0.37 |
| 56 | 0 cyan rubber cube | -1.11 | 1.46 | -0.20 | 0.13 | 0.24 |

| frame | object | x | y | vx | vy | speed |
|---|---|---|---|---|---|---|
| 56 | 1 purple rubber cube | -0.86 | 0.63 | -0.43 | -0.52 | 0.68 |
| 56 | 2 purple metal cube | -1.65 | 0.31 | 0.09 | -0.29 | 0.30 |
| 64 | 0 cyan rubber cube | -1.15 | 1.48 | -0.16 | 0.10 | 0.19 |
| 64 | 1 purple rubber cube | -0.95 | 0.52 | -0.38 | -0.48 | 0.61 |
| 64 | 2 purple metal cube | -1.63 | 0.25 | 0.07 | -0.22 | 0.23 |
| 72 | 0 cyan rubber cube | -1.18 | 1.50 | -0.10 | 0.06 | 0.12 |
| 72 | 1 purple rubber cube | -1.03 | 0.42 | -0.33 | -0.43 | 0.54 |
| 72 | 2 purple metal cube | -1.62 | 0.21 | 0.05 | -0.16 | 0.17 |
| 80 | 0 cyan rubber cube | -1.20 | 1.51 | -0.05 | 0.03 | 0.05 |
| 80 | 1 purple rubber cube | -1.10 | 0.33 | -0.29 | -0.37 | 0.47 |
| 80 | 2 purple metal cube | -1.61 | 0.18 | 0.03 | -0.10 | 0.10 |
| 85 | 0 cyan rubber cube | -1.20 | 1.51 | -0.01 | 0.01 | 0.01 |
| 85 | 1 purple rubber cube | -1.13 | 0.28 | -0.19 | -0.30 | 0.36 |
| 85 | 2 purple metal cube | -1.60 | 0.17 | -0.08 | -0.10 | 0.13 |
| 88 | 0 cyan rubber cube | -1.20 | 1.51 | 0.00 | -0.00 | 0.00 |
| 88 | 1 purple rubber cube | -1.15 | 0.26 | -0.14 | -0.26 | 0.29 |
| 88 | 2 purple metal cube | -1.61 | 0.16 | -0.12 | -0.12 | 0.17 |
| 96 | 0 cyan rubber cube | -1.20 | 1.51 | 0.00 | -0.00 | 0.00 |
| 96 | 1 purple rubber cube | -1.17 | 0.21 | -0.08 | -0.18 | 0.20 |
| 96 | 2 purple metal cube | -1.64 | 0.14 | -0.12 | -0.10 | 0.15 |
| 104 | 0 cyan rubber cube | -1.20 | 1.51 | 0.00 | -0.00 | 0.00 |
| 104 | 1 purple rubber cube | -1.19 | 0.17 | -0.05 | -0.12 | 0.13 |
| 104 | 2 purple metal cube | -1.66 | 0.12 | -0.07 | -0.06 | 0.09 |
| 112 | 0 cyan rubber cube | -1.20 | 1.51 | 0.00 | -0.00 | 0.00 |
| 112 | 1 purple rubber cube | -1.19 | 0.15 | -0.03 | -0.06 | 0.06 |
| 112 | 2 purple metal cube | -1.67 | 0.11 | -0.02 | -0.01 | 0.02 |
| 114 | 0 cyan rubber cube | -1.20 | 1.51 | 0.00 | 0.00 | 0.00 |
| 114 | 1 purple rubber cube | -1.19 | 0.15 | -0.02 | -0.04 | 0.05 |
| 114 | 2 purple metal cube | -1.67 | 0.11 | -0.00 | -0.00 | 0.00 |
| 114 | 3 cyan metal cylinder | -0.94 | 2.85 | -0.23 | -1.17 | 1.19 |
| 120 | 0 cyan rubber cube | -1.20 | 1.51 | 0.00 | -0.00 | 0.00 |
| 120 | 1 purple rubber cube | -1.20 | 0.15 | 0.00 | 0.00 | 0.00 |
| 120 | 2 purple metal cube | -1.67 | 0.11 | 0.00 | -0.00 | 0.00 |
| 120 | 3 cyan metal cylinder | -0.97 | 2.66 | -0.21 | -1.11 | 1.13 |
| 127 | 0 cyan rubber cube | -1.20 | 1.51 | 0.00 | -0.00 | 0.00 |
| 127 | 1 purple rubber cube | -1.20 | 0.15 | 0.00 | 0.00 | 0.00 |
| 127 | 2 purple metal cube | -1.67 | 0.11 | 0.00 | -0.00 | 0.00 |
| 127 | 3 cyan metal cylinder | -1.01 | 2.45 | -0.20 | -1.03 | 1.05 |

# Video 14195

## Video 14195: Objects

Object 0: red metal cylinder (color=red, material=metal, shape=cylinder)

Object 1: green metal cylinder (color=green, material=metal, shape=cylinder)

Object 2: blue rubber cube (color=blue, material=rubber, shape=cube)

Object 3: gray rubber cube (color=gray, material=rubber, shape=cube)

## Video 14195: Events

Frame 0: Object 1 (green metal cylinder) is in view when the video begins.

Frame 0: Object 2 (blue rubber cube) is in view when the video begins.

Frame 8: Object 0 (red metal cylinder) enters the scene.

Frame 41: Object 0 (red metal cylinder) collides with Object 1 (green metal cylinder).

Frame 69: Object 1 (green metal cylinder) collides with Object 2 (blue rubber cube).

Frame 85: Object 3 (gray rubber cube) enters the scene.

Frame 127: The video ends.

## Video 14195: Motion log

| frame | object | x | y | vx | vy | speed |
|---|---|---|---|---|---|---|
| 0 | 1 green metal cylinder | -0.17 | -0.54 | 0.00 | 0.00 | 0.00 |
| 0 | 2 blue rubber cube | -5.34 | -3.24 | 2.37 | 1.78 | 2.97 |
| 8 | 0 red metal cylinder | 2.34 | -1.46 | -2.56 | 1.08 | 2.78 |
| 8 | 1 green metal cylinder | -0.17 | -0.54 | -0.00 | 0.00 | 0.00 |
| 8 | 2 blue rubber cube | -4.82 | -2.85 | 2.24 | 1.68 | 2.80 |
| 16 | 0 red metal cylinder | 1.78 | -1.22 | -2.42 | 1.02 | 2.62 |
| 16 | 1 green metal cylinder | -0.17 | -0.54 | -0.00 | 0.00 | 0.00 |
| 16 | 2 blue rubber cube | -4.33 | -2.48 | 2.11 | 1.59 | 2.64 |
| 24 | 0 red metal cylinder | 1.26 | -1.00 | -2.28 | 0.96 | 2.48 |
| 24 | 1 green metal cylinder | -0.17 | -0.54 | -0.00 | -0.00 | 0.00 |
| 24 | 2 blue rubber cube | -3.87 | -2.14 | 2.00 | 1.50 | 2.50 |
| 32 | 0 red metal cylinder | 0.76 | -0.79 | -2.15 | 0.91 | 2.34 |
| 32 | 1 green metal cylinder | -0.17 | -0.54 | -0.00 | -0.00 | 0.00 |
| 32 | 2 blue rubber cube | -3.44 | -1.81 | 1.88 | 1.42 | 2.36 |
| 40 | 0 red metal cylinder | 0.29 | -0.59 | -2.03 | 0.86 | 2.20 |
| 40 | 1 green metal cylinder | -0.17 | -0.54 | -0.00 | -0.00 | 0.00 |
| 40 | 2 blue rubber cube | -3.03 | -1.50 | 1.78 | 1.34 | 2.22 |
| 41 | 0 red metal cylinder | 0.24 | -0.57 | -2.02 | 0.85 | 2.19 |
| 41 | 1 green metal cylinder | -0.17 | -0.54 | -0.00 | -0.00 | 0.00 |
| 41 | 2 blue rubber cube | -2.98 | -1.47 | 1.76 | 1.33 | 2.21 |
| 48 | 0 red metal cylinder | 0.11 | -0.42 | -0.58 | 0.72 | 0.93 |
| 48 | 1 green metal cylinder | -0.48 | -0.53 | -1.60 | 0.09 | 1.61 |
| 48 | 2 blue rubber cube | -2.64 | -1.21 | 1.68 | 1.26 | 2.10 |
| 56 | 0 red metal cylinder | -0.01 | -0.27 | -0.53 | 0.66 | 0.85 |
| 56 | 1 green metal cylinder | -0.83 | -0.51 | -1.50 | 0.08 | 1.50 |
| 56 | 2 blue rubber cube | -2.28 | -0.94 | 1.58 | 1.19 | 1.98 |
| 64 | 0 red metal cylinder | -0.13 | -0.13 | -0.48 | 0.60 | 0.77 |
| 64 | 1 green metal cylinder | -1.16 | -0.49 | -1.40 | 0.08 | 1.41 |
| 64 | 2 blue rubber cube | -1.94 | -0.68 | 1.49 | 1.12 | 1.86 |

| frame | object | x | y | vx | vy | speed |
|---|---|---|---|---|---|---|
| 69 | 0 red metal cylinder | -0.19 | -0.05 | -0.45 | 0.56 | 0.72 |
| 69 | 1 green metal cylinder | -1.34 | -0.48 | 0.94 | 0.37 | 1.01 |
| 69 | 2 blue rubber cube | -1.74 | -0.53 | 0.09 | 0.89 | 0.90 |
| 72 | 0 red metal cylinder | -0.23 | 0.00 | -0.43 | 0.54 | 0.69 |
| 72 | 1 green metal cylinder | -1.26 | -0.44 | 0.91 | 0.36 | 0.98 |
| 72 | 2 blue rubber cube | -1.73 | -0.45 | 0.09 | 0.88 | 0.88 |
| 80 | 0 red metal cylinder | -0.32 | 0.11 | -0.39 | 0.48 | 0.62 |
| 80 | 1 green metal cylinder | -1.06 | -0.37 | 0.83 | 0.33 | 0.90 |
| 80 | 2 blue rubber cube | -1.72 | -0.27 | 0.08 | 0.79 | 0.80 |
| 85 | 0 red metal cylinder | -0.38 | 0.18 | -0.36 | 0.45 | 0.57 |
| 85 | 1 green metal cylinder | -0.95 | -0.32 | 0.79 | 0.31 | 0.85 |
| 85 | 2 blue rubber cube | -1.70 | -0.16 | 0.07 | 0.74 | 0.75 |
| 85 | 3 gray rubber cube | 0.60 | -2.23 | -0.32 | 1.39 | 1.42 |
| 88 | 0 red metal cylinder | -0.40 | 0.22 | -0.34 | 0.42 | 0.54 |
| 88 | 1 green metal cylinder | -0.88 | -0.30 | 0.76 | 0.30 | 0.82 |
| 88 | 2 blue rubber cube | -1.70 | -0.10 | 0.07 | 0.71 | 0.72 |
| 88 | 3 gray rubber cube | 0.57 | -2.12 | -0.31 | 1.35 | 1.39 |
| 96 | 0 red metal cylinder | -0.48 | 0.30 | -0.30 | 0.37 | 0.47 |
| 96 | 1 green metal cylinder | -0.72 | -0.23 | 0.69 | 0.27 | 0.74 |
| 96 | 2 blue rubber cube | -1.68 | 0.05 | 0.06 | 0.64 | 0.64 |
| 96 | 3 gray rubber cube | 0.50 | -1.83 | -0.29 | 1.26 | 1.29 |
| 104 | 0 red metal cylinder | -0.54 | 0.38 | -0.25 | 0.31 | 0.40 |
| 104 | 1 green metal cylinder | -0.58 | -0.17 | 0.62 | 0.24 | 0.66 |
| 104 | 2 blue rubber cube | -1.67 | 0.19 | 0.06 | 0.56 | 0.57 |
| 104 | 3 gray rubber cube | 0.44 | -1.55 | -0.27 | 1.17 | 1.20 |
| 112 | 0 red metal cylinder | -0.59 | 0.44 | -0.21 | 0.26 | 0.33 |
| 112 | 1 green metal cylinder | -0.45 | -0.12 | 0.55 | 0.22 | 0.59 |
| 112 | 2 blue rubber cube | -1.66 | 0.30 | 0.05 | 0.49 | 0.49 |
| 112 | 3 gray rubber cube | 0.38 | -1.30 | -0.25 | 1.09 | 1.12 |
| 120 | 0 red metal cylinder | -0.63 | 0.50 | -0.16 | 0.21 | 0.26 |
| 120 | 1 green metal cylinder | -0.33 | -0.08 | 0.48 | 0.19 | 0.52 |
| 120 | 2 blue rubber cube | -1.65 | 0.41 | 0.04 | 0.42 | 0.42 |
| 120 | 3 gray rubber cube | 0.33 | -1.07 | -0.23 | 1.00 | 1.03 |
| 127 | 0 red metal cylinder | -0.66 | 0.53 | -0.13 | 0.16 | 0.20 |
| 127 | 1 green metal cylinder | -0.24 | -0.04 | 0.42 | 0.17 | 0.45 |
| 127 | 2 blue rubber cube | -1.64 | 0.48 | 0.04 | 0.36 | 0.36 |
| 127 | 3 gray rubber cube | 0.28 | -0.88 | -0.21 | 0.93 | 0.96 |

# Video 14981

## Video 14981: Objects

Object 0: brown rubber sphere (color=brown, material=rubber, shape=sphere)

Object 1: green metal cylinder (color=green, material=metal, shape=cylinder)

Object 2: yellow rubber cube (color=yellow, material=rubber, shape=cube)

Object 3: gray rubber cube (color=gray, material=rubber, shape=cube)

## Video 14981: Events

Frame 0: Object 1 (green metal cylinder) is in view when the video begins.

Frame 0: Object 2 (yellow rubber cube) is in view when the video begins.

Frame 10: Object 3 (gray rubber cube) enters the scene.

Frame 19: Object 0 (brown rubber sphere) enters the scene.

Frame 40: Object 0 (brown rubber sphere) collides with Object 1 (green metal cylinder).

Frame 74: Object 0 (brown rubber sphere) collides with Object 2 (yellow rubber cube).

Frame 91: Object 1 (green metal cylinder) exits the scene.

Frame 127: The video ends.

## Video 14981: Motion log

| frame | object | x | y | vx | vy | speed |
|---|---|---|---|---|---|---|
| 0 | 1 green metal cylinder | 0.93 | 0.87 | 0.00 | 0.00 | 0.00 |
| 0 | 2 yellow rubber cube | -0.03 | 0.08 | 0.00 | 0.00 | 0.00 |
| 8 | 1 green metal cylinder | 0.93 | 0.87 | 0.00 | 0.00 | 0.00 |
| 8 | 2 yellow rubber cube | -0.03 | 0.08 | -0.00 | 0.00 | 0.00 |
| 10 | 1 green metal cylinder | 0.93 | 0.87 | 0.00 | 0.00 | 0.00 |
| 10 | 2 yellow rubber cube | -0.03 | 0.08 | -0.00 | 0.00 | 0.00 |
| 10 | 3 gray rubber cube | -6.63 | 2.00 | 2.51 | -0.95 | 2.68 |
| 16 | 1 green metal cylinder | 0.93 | 0.87 | 0.00 | -0.00 | 0.00 |
| 16 | 2 yellow rubber cube | -0.03 | 0.08 | 0.00 | 0.00 | 0.00 |
| 16 | 3 gray rubber cube | -6.22 | 1.85 | 2.40 | -0.91 | 2.57 |
| 19 | 0 brown rubber sphere | 2.46 | -0.26 | -2.32 | 1.48 | 2.75 |
| 19 | 1 green metal cylinder | 0.93 | 0.87 | -0.00 | 0.00 | 0.00 |
| 19 | 2 yellow rubber cube | -0.03 | 0.08 | -0.00 | -0.00 | 0.00 |
| 19 | 3 gray rubber cube | -6.02 | 1.77 | 2.35 | -0.89 | 2.51 |
| 24 | 0 brown rubber sphere | 2.13 | -0.05 | -2.28 | 1.45 | 2.70 |
| 24 | 1 green metal cylinder | 0.93 | 0.87 | 0.00 | -0.00 | 0.00 |
| 24 | 2 yellow rubber cube | -0.03 | 0.08 | 0.00 | 0.00 | 0.00 |
| 24 | 3 gray rubber cube | -5.70 | 1.65 | 2.27 | -0.85 | 2.42 |
| 32 | 0 brown rubber sphere | 1.63 | 0.26 | -2.20 | 1.40 | 2.61 |
| 32 | 1 green metal cylinder | 0.93 | 0.87 | 0.00 | -0.00 | 0.00 |
| 32 | 2 yellow rubber cube | -0.03 | 0.08 | 0.00 | 0.00 | 0.00 |
| 32 | 3 gray rubber cube | -5.20 | 1.47 | 2.14 | -0.80 | 2.29 |
| 40 | 0 brown rubber sphere | 1.16 | 0.55 | -0.90 | -0.23 | 0.93 |
| 40 | 1 green metal cylinder | 0.92 | 0.89 | -1.19 | 1.53 | 1.93 |
| 40 | 2 yellow rubber cube | -0.03 | 0.08 | -0.00 | 0.00 | 0.00 |
| 40 | 3 gray rubber cube | -4.74 | 1.29 | 2.02 | -0.76 | 2.16 |
| 48 | 0 brown rubber sphere | 0.96 | 0.50 | -0.88 | -0.23 | 0.91 |
| 48 | 1 green metal cylinder | 0.66 | 1.22 | -1.11 | 1.44 | 1.82 |

| frame | object | x | y | vx | vy | speed |
|---|---|---|---|---|---|---|
| 48 | 2 yellow rubber cube | -0.03 | 0.08 | -0.00 | 0.00 | 0.00 |
| 48 | 3 gray rubber cube | -4.30 | 1.13 | 1.91 | -0.71 | 2.03 |
| 56 | 0 brown rubber sphere | 0.77 | 0.45 | -0.87 | -0.22 | 0.90 |
| 56 | 1 green metal cylinder | 0.42 | 1.54 | -1.05 | 1.35 | 1.71 |
| 56 | 2 yellow rubber cube | -0.03 | 0.08 | -0.00 | 0.00 | 0.00 |
| 56 | 3 gray rubber cube | -3.88 | 0.97 | 1.80 | -0.67 | 1.92 |
| 64 | 0 brown rubber sphere | 0.57 | 0.40 | -0.86 | -0.22 | 0.88 |
| 64 | 1 green metal cylinder | 0.19 | 1.83 | -0.98 | 1.27 | 1.60 |
| 64 | 2 yellow rubber cube | -0.03 | 0.08 | -0.00 | 0.00 | 0.00 |
| 64 | 3 gray rubber cube | -3.49 | 0.83 | 1.69 | -0.63 | 1.80 |
| 72 | 0 brown rubber sphere | 0.38 | 0.35 | -0.84 | -0.22 | 0.87 |
| 72 | 1 green metal cylinder | -0.02 | 2.11 | -0.92 | 1.19 | 1.50 |
| 72 | 2 yellow rubber cube | -0.03 | 0.08 | -0.00 | 0.00 | 0.00 |
| 72 | 3 gray rubber cube | -3.13 | 0.69 | 1.59 | -0.59 | 1.70 |
| 74 | 0 brown rubber sphere | 0.33 | 0.34 | -0.84 | -0.22 | 0.86 |
| 74 | 1 green metal cylinder | -0.07 | 2.17 | -0.91 | 1.17 | 1.48 |
| 74 | 2 yellow rubber cube | -0.03 | 0.08 | -0.00 | 0.00 | 0.00 |
| 74 | 3 gray rubber cube | -3.04 | 0.66 | 1.57 | -0.58 | 1.67 |
| 80 | 0 brown rubber sphere | 0.23 | 0.32 | -0.55 | -0.12 | 0.56 |
| 80 | 1 green metal cylinder | -0.22 | 2.36 | -0.86 | 1.11 | 1.41 |
| 80 | 2 yellow rubber cube | -0.09 | 0.06 | -0.40 | -0.13 | 0.42 |
| 80 | 3 gray rubber cube | -2.78 | 0.57 | 1.49 | -0.55 | 1.59 |
| 88 | 0 brown rubber sphere | 0.12 | 0.29 | -0.48 | -0.09 | 0.49 |
| 88 | 1 green metal cylinder | -0.41 | 2.60 | -0.80 | 1.04 | 1.31 |
| 88 | 2 yellow rubber cube | -0.18 | 0.03 | -0.42 | -0.15 | 0.45 |
| 88 | 3 gray rubber cube | -2.46 | 0.45 | 1.40 | -0.51 | 1.49 |
| 91 | 0 brown rubber sphere | 0.08 | 0.29 | -0.48 | -0.08 | 0.48 |
| 91 | 2 yellow rubber cube | -0.22 | 0.02 | -0.41 | -0.15 | 0.44 |
| 91 | 3 gray rubber cube | -2.34 | 0.41 | 1.37 | -0.50 | 1.46 |
| 96 | 0 brown rubber sphere | 0.01 | 0.28 | -0.46 | -0.07 | 0.47 |
| 96 | 2 yellow rubber cube | -0.27 | -0.00 | -0.39 | -0.15 | 0.42 |
| 96 | 3 gray rubber cube | -2.15 | 0.34 | 1.31 | -0.48 | 1.40 |
| 104 | 0 brown rubber sphere | -0.09 | 0.26 | -0.44 | -0.05 | 0.44 |
| 104 | 2 yellow rubber cube | -0.35 | -0.04 | -0.35 | -0.15 | 0.38 |
| 104 | 3 gray rubber cube | -1.87 | 0.24 | 1.22 | -0.44 | 1.30 |
| 112 | 0 brown rubber sphere | -0.18 | 0.25 | -0.41 | -0.04 | 0.42 |
| 112 | 2 yellow rubber cube | -0.43 | -0.07 | -0.31 | -0.14 | 0.35 |
| 112 | 3 gray rubber cube | -1.60 | 0.14 | 1.14 | -0.41 | 1.21 |
| 120 | 0 brown rubber sphere | -0.27 | 0.25 | -0.39 | -0.02 | 0.39 |
| 120 | 2 yellow rubber cube | -0.49 | -0.10 | -0.28 | -0.14 | 0.31 |
| 120 | 3 gray rubber cube | -1.36 | 0.05 | 1.06 | -0.38 | 1.12 |
| 127 | 0 brown rubber sphere | -0.35 | 0.24 | -0.37 | -0.00 | 0.37 |
| 127 | 2 yellow rubber cube | -0.55 | -0.13 | -0.25 | -0.13 | 0.29 |
| 127 | 3 gray rubber cube | -1.16 | -0.02 | 0.99 | -0.35 | 1.05 |